INS 2222
FLUID MECHANICS
LECTURE NOTES

Prof. Dr. Yalçın Yüksel and Prof. Dr. Esin Çevik

İstanbul, March 2021

1
Fluid Properties

1.1 Introduction

Fluid mechanics is encountered in almost every area of our lives. Blood flows through our veins and arteries, a ship moves through water, water flows through rives, airplanes fly in the air, air flows around wind machines, dam holds water behind, air is heated and cooled in our homes. All engineering disciplines require some expertise in the area of fluid mechanics.

Civil Engineering education is based on two main subjects which are solid and fluid mechanics. However, they are branches of continuum mechanics.

Fluid mechanics is that discipline within the broad field of applied mechanics concerned with the behavior of liquids and gases at rest or in motion. The scope of fluid mechanics is the understanding and knowledge of the basic fluid properties and the laws of fluid mechanics are essential for the analysis and design of a system in which fluid is working medium. Such systems are used in aircraft, rockets, automobiles, trains, ships, submarines, hydraulic structures (dams, bridges, culvers, etc.), hydraulic machines (water pumps, turbines), and many others. It will be encountered the key conservation principles on which fluid mechanics is based, namely, conservation of mass, momentum, and energy. Also Newton's second law will be applied in fluid mechanics.

The term "fluid" is applied to certain substances, which, whether at rest, or under conditions of static equilibrium, cannot resist a shear force- no matter how small that force is. Thus, a fluid deforms continuously under the influence of shear force- i.e. the fluid particles continuously change their position relative to one another when subjected to a shear force (Fig.1.1). In the case of a solid, the deformation is small; whereas in the case of a fluid, the deformation is large.

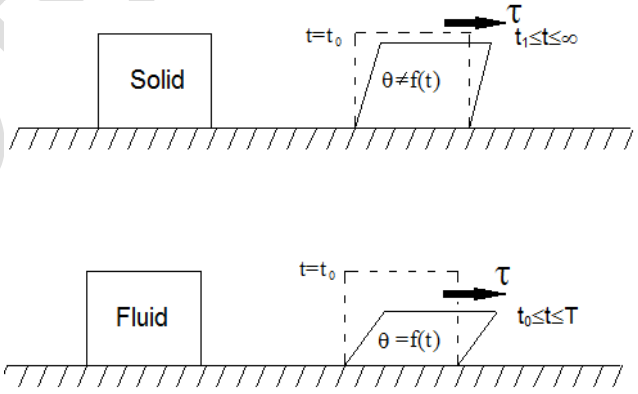


Figure 1.1 Effect of shear stress on a solid and a fluid

Fluids may be divided into two classes: liquids and gases. Liquids are highly resistant to compression while gases are easily compressed. Also, liquids are not greatly affected by changes in temperature, while gases are.

DRAFT COPY

1.2 Molecular Structure of Fluids

All substances in nature are made up of molecules separated by empty spaces. The differences between solids, liquids and gases may be distinguished with their molecular structures. In solids, the molecules are much closer together than in a liquid; in a liquid the molecules are much closer together than in a gas. The force of attraction between the molecules is inversely proportional to the square of the distance between them. Liquid is not a rigid substance, however, the force of attraction between molecules is large enough to keep the liquid in a definite volume. Therefore, solids resist the external forces than fluids. The molecules may change their positions under the external forces in fluids and they don't back to their original positions (Fig.1.2). However, solids only deformed beyond their elastic limit.

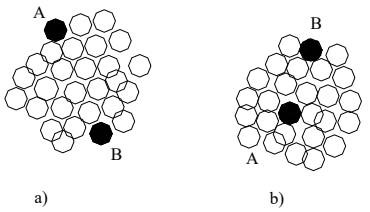


Figure 1.2 Molecular positions under external forces in fluids

In gases, the forces of attraction between the molecules are negligible because of the large distance between molecules. When a gas is enclosed in a container, it will expand until it completely fills the container, but not liquids, they take their definite shape. In other words, in a liquid, the forces of attraction between molecules are only large enough to hold them together to give the liquid a definite shape.

1.3 Fluid as a Continuum

All fluids are made up of molecules, which are separated from each other by spaces. The properties of a fluid cannot be defined in these spaces due to the nonexistence of mass. Therefore a fluid is regarded as hypothetical continuous medium, called continuum. The mean free path of fluid molecules at standard temperature and pressure is very small, and hence the assumption of a continuum may be valid.

Fluid properties such as density, temperature, velocity, etc., are considered to have definite value at each point in the fluid, and are defined with respect to the *representative elementary volume* (REV) (Fig 1.3).

$$\rho_E\,(x,t)=\frac{E}{\forall(x)} \tag{1.1}$$

where ρ_E concentration, $\forall(x)$ volume of a spatial domain, E is a property of the fluid.

$$\rho_E(x,t)=\lim_{\forall_{(x)}\to\forall_{min}}\frac{E}{\forall(x)} \tag{1.2}$$

For all practical purposes, the mean density $\bar{\rho}_E$ of property E for a fluid of volume $\forall$ may be determined using the relationship.

$$\bar{\rho}_E=\frac{E}{\forall} \tag{1.3}$$

DRAFT COPY

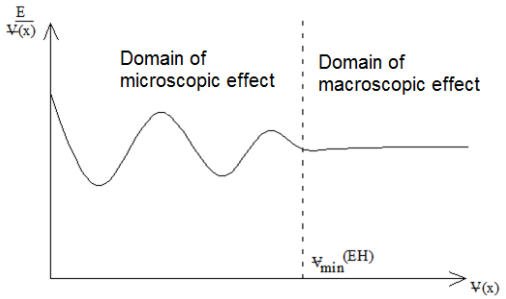


Figure 1.3 Property of a fluid as a function of its volume (Khan,1987)

1.4 Density (ρ)

The density of a fluid is defined as its mass per unit volume. Density is used to characterize the mass of a fluid system. The value of density can vary widely between different fluids, but variation of density for fluids is small under the variation of pressure and temperature.

$$\rho_o = \lim_{\Delta\forall_{(x)} \to \Delta\forall_{min}} \frac{\Delta m}{\Delta\forall} \tag{1.4}$$

$$\rho = \frac{\Delta m}{\Delta\forall} \tag{1.5}$$

The density of a gas is calculated using the perfect gas law.

$$p = \rho RT \tag{1.6}$$

where p is absolute pressure, R gas constant, T absolute temperature

$$R = \frac{8312}{M} \text{ mN/kg K}$$

where M is the molecular weight of gas.

1.5 Specific Weight (γ)

Specific weight is weight per unit volume.

$$\gamma = \frac{\Delta W}{\Delta\forall} \tag{1.7}$$

weight is the product of mass and local acceleration due to gravity

$$\Delta W = \Delta m g$$

$$\gamma = \frac{\Delta W}{\Delta\forall} = \frac{\Delta m g}{\Delta\forall} = \rho g$$

$$\gamma = \rho g \tag{1.8}$$

1.6 Specific Gravity (d)

Specific gravity is the ratio of fluid density to the density of water at a certain temperature. It is a dimensionless quantity. Usually this specified temperature is considered as 4 °C.

$$d = \frac{\rho}{\rho_{water}} \tag{1.9}$$

DRAFT COPY

1.7 Compressibility of Fluids (Bulk Modulus of Elasticity)

The fluids are elastic media, and it is customary in engineering to summarize this property by defining a modulus of elasticity, as is done for solid elastic materials such as steel. Since fluids do not posses rigidity of form, however, the modulus of elasticity must be defined on the basis of volume and is termed a bulk modulus.

The bulk modulus of elasticity (K) of a fluid is a measure of its elasticity and is defined as ratio of stress (pressure) to the volumetric strain. Liquids are usually considered to be incompressible, whereas gases are generally considered compressible.

$$-\frac{d\forall}{\forall} = \frac{dp}{K} \tag{1.10}$$

where dp change in pressure, d$\forall$ change in volume, and $\forall$ initial volume. The minus sign indicates that an increase in pressure results in a decrease in volume and vice versa. Since a decrease in volume of a given mass will result in an increase in density.

$m=\rho\forall=$constant

If we take the derivative of above equation and since mass is constant,

$dm=\rho d\forall+\forall d\rho=0$

$\forall d\rho=-\rho d\forall$

hence we may write the bulk modulus is a function of the density

$$\frac{d\rho}{\rho} = -\frac{d\forall}{\forall}$$

we get another equation for bulk modulus

$$\frac{d\rho}{\rho} = \frac{dp}{K} \tag{1.11}$$

The bulk modulus of elasticity (K) of gas depends on its mode of compression. For isothermal compression;

$$\frac{dp}{d\rho} = R\,T \tag{1.12}$$

and

$$K = \rho\frac{dp}{d\rho} = \rho\,R\,T = p \tag{1.13}$$

$$K = kp \tag{1.14}$$

where k= Cp/cv

Bulk modulus has dimensions of pressure. The unit is N/m^2 (Pa) in SI units. Large values for the bulk modulus the fluid indicates that the fluid is rather incompressible which means under a large pressure change resuls a small change in volume.

DRAFT COPY

1.8 Viscosity

Fluid motion can cause shearing stress. This resistance is attributed to a property of the fluid called viscosity. Therefore, viscosity is the property of fluid, which is a measure of its internal resistance to relative motion between adjacent layers.

The phenomenon of viscosity can be further investigated by following experiment (Fig.1.4). Consider two plates, one lying on top of another. The lower plate is fixed while the upper plate can be moved by applying a tangential force (F) with a velocity (u). Let the area of the upper plate be A.

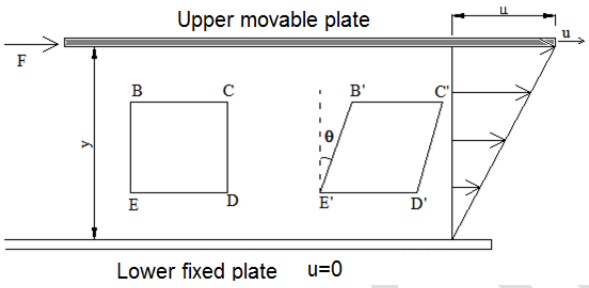


Figure 1.4 Deformation of fluid under the influence of a shear stress

Newton (1642-1727) observed the following relationships among the various variables:

1-Keeping the area A and the distance y constant, the velocity u attained by the plate is directly proportional to the applied force F:

$F \alpha u$

2-Keeping the velocity u and the distance y constant, the force required to move the plate with a velocity u, is directly proportional to the area of the plate:

$F \alpha A$

3-Keeping the velocity u and the area A constant, the force required is inversely proportional to distance between the plates:

$$F \alpha \frac{1}{y}$$

Combining these equations, following equation may be obtained:

$$F \alpha \frac{u A}{y} \qquad \text{or} \qquad \frac{F}{A} \propto \frac{u}{y}$$

$$\tau \propto \frac{u}{y} \qquad \text{or} \qquad \tau = \mu \frac{u}{y} \tag{1.15}$$

In Fig. 1.4 a linear velocity variation across the plate is shown. In actual practice, such a linear variation is uncommon. A nonlinear variation, such as shown in Fig 1.5, is more common. Since the slope of a nonlinear velocity gradient varies from point to point across the plate, it is convenient to write as follows in differential form:

DRAFT COPY

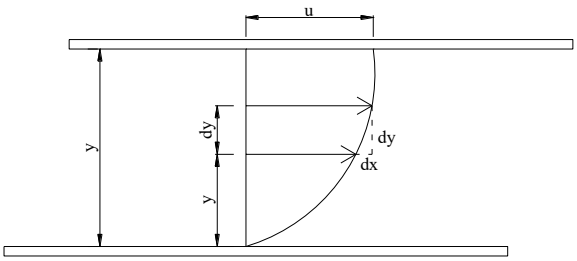


Figure 1.5 A fluid with a nonlinear velocity variation between the plates

$$\tau = \mu \frac{du}{dy} \tag{1.16}$$

This equation is known as Newton's Law of Viscosity.

The Causes of Viscosity:

There are two major causes of viscosity in fluids, which are:

1-Molecular attraction; this is major cause of viscosity in liquids, since the molecular attraction decreases with the increase in temperature, the viscosity of a liquid also decreases with increase in temperature,

2-Transfer of molecular momentum; this is major cause of viscosity in gases, since the molecular movement increases with the increase in temperature, the viscosity of a gas increases with the increase in temperature (Fig.1.6).

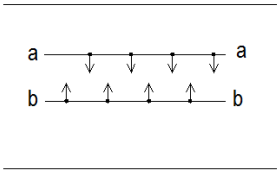


Figure 1.6 Transfer of momentum between two fluid layers

Units of Viscosity:

Dynamic viscosity (μ) is the fluid property that relates shearing stress and fluid motion. Kinematic viscosity is defined as the ratio of the dynamic viscosity to the fluid density (ρ).

$$\nu = \frac{\mu}{\rho} \tag{1.17}$$

μ; $N.s/m^2$, $dyn.s/cm^2$, $kgf.s/m^2$

ν; m^2/s, cm^2/s(stoke)

Newtonian and Non-Newtonian Fluids:

If the relationship between shear stress and velocity gradient is linear, such a fluid is called a Newtonian fluid since viscosity is independent of the velocity gradient. If this relationship is non-linear, fluid is called Non-Newtonian fluid (Fig 1.7). A hypothetical fluid of zero viscosity is termed an ideal fluid. This term is used because it gives a mathematical simplicity.

DRAFT COPY

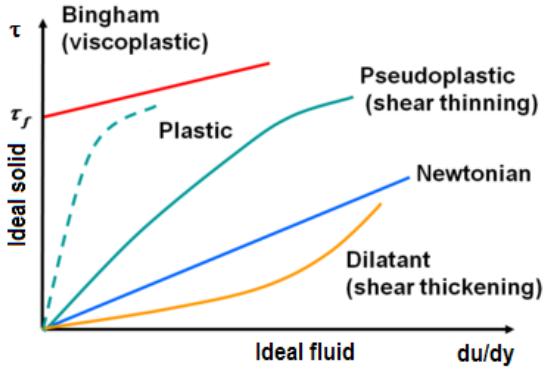


Figure 1.7 Relation between shear stress and velocity gradient for various types of fluids and a plastic

1.9 Surface Tension

The intensity of the molecular attraction per unit length along any line in the surface is called *the surface tension.* At the interface between a liquid and a gas or between immiscible liquids, forces develop in the liquid surface which causes the surface to behave as if it were a skin or membrane although such a skin is not actually present. The physical consequence of this unbalanced force along the surface is to create the hypothetical skin. Consider a container full of water (Fig. 1.8). The surface of the water is in contact with air. A molecule A of water located beneath the surface is acted upon by equal forces in all directions. But a molecule B at the surface is acted by unbalanced forces which have the tendency to drawn by molecule. This causes the liquid surface to seek a minimum possible area by developing a force at the surface whose effect is to pull the surface molecules upwards.

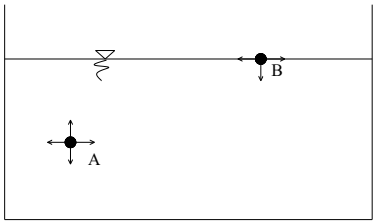


Figure.1.8 A vessel containing water

Surface tension (σ) is measured in units of force per unit length. The magnitude of the surface tension force depends on the type of fluid and substance it is in contact with. For example, surface tension for a water-air surface is 0.073 N/m at room temperature. Surface tension decreases with increasing temperature.

1.10 Capillarity

Capillarity action in small tubes, which involves a liquid-gas-solid interface, is caused by surface tension and by the relative value of adhesion between liquid and solid to cohesion of the fluid. A rise or drop of a liquid in a solid tube is known as capillarity. Adhesion between fluids and solids is expressed by the contact angle, θ.

DRAFT COPY

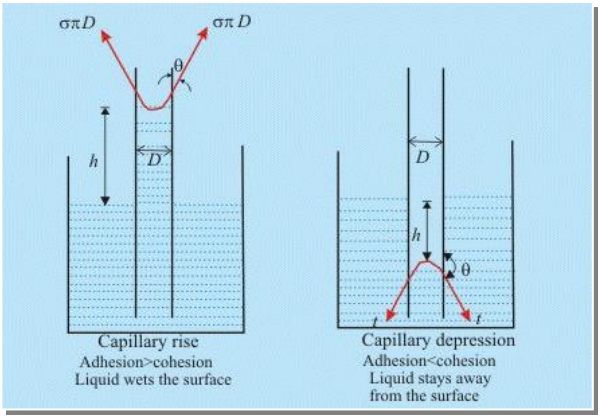


Figure 1.9 Capillarity in a small tube immersed in water (adhesion>cohesion) and mercury (adhesion<cohesion)

The height of fluid raised in a tube can be calculated as described below:

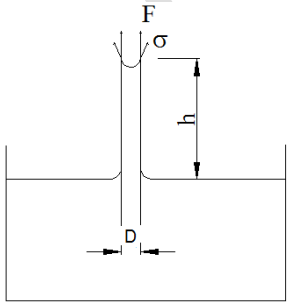


Figure1.10- Forces in capillarity

$F=\sigma\cos\theta\pi D$

$W=\gamma\frac{\pi D^2}{4}h$

$W=F$

$$\gamma\frac{\pi D^2}{4}h=\sigma\cos\theta\pi D$$

$$h=\frac{4\sigma\cos\theta}{\gamma D} \qquad (1.18)$$

generally $\cos\theta = 1$

$$h=\frac{4\sigma}{\gamma D} \qquad (1.19)$$

1.11 Vapor Pressure

The molecules of a liquid are in constant motion. Liquids will evaporate if they are simply placed in a container open to the atmosphere. Since some liquid molecules at the surface have

DRAFT COPY

sufficient momentum to overcome the intermolecular cohesive forces and may escape into the atmosphere as vapors. This vapor exerts a partial pressure in space just like any other gaseous. This partial pressure due to the vapor is known as *the vapor pressure*. Not only are vapors escaping from the liquid to space; some are returning from the space to the liquid. If the space above the liquid is confined and a small air space left above the surface, after a sufficient time, an equilibrium condition is reached when the number of molecules leaving the surface is equal to the number of molecules entering. If equilibrium is reached, vapor pressure in this stage is called *the saturation vapor pressure* (C_v).This phenomenon depends upon molecular activity which is function of temperature.

Vapor pressure is said to be saturated and the pressure is saturated vapor pressure.

A liquid boils when the pressure above a liquid equals the vapor pressure of the liquid. The molecules of a liquid are in constant motion. For example, water boils at room temperature if the pressure is reduced sufficiently. Water at standard atmospheric pressure will boil when the temperature reaches 100 °C. Boiling can be induced at a given pressure acting on the fluid by raising temperature, or at a given temperature by lowering the pressure.

There are many situations involving the flow of liquid, at certain locations in the system pressure decreases to very low values. If pressure is lowered to the vapor pressure boiling will occur. The dissolved gases due to boiling in a liquid may come out in the form of bubbles if the pressure of the liquid is sufficiently lowered. These bubbles, if subsequently transferred to a region of high pressure, may collapse producing noise and high impact forces and cause structural damage. This phenomenon is known as *cavitation.*

This can occur when liquid flows through valves, elbows, or turbine blades.

DRAFT COPY

2
Fluid Statics

2.1. Introduction

Fluid static is that branch of fluid mechanics, which deals with fluids at rest. It can be concluded from the definition of a fluid that at rest cannot have any shear force acting on it; otherwise it will not be at rest. Therefore, all forces due to the pressure of a fluid must be normal to the surfaces on which they act.

2.2 Pressure

The average pressure $\bar{p}$ over a plane δA in a fluid is the ratio of the normal force δF to the area, or

$$\bar{p} = \frac{\delta F}{\delta A} \tag{2.1}$$

The pressure p at a point is the limit of Eq. 2.1 as the area is indefinitely reduced to subject to the restriction of a continuum, or

$$p = \lim_{\delta A \to \delta A_{min}} \frac{\delta F}{\delta A} = \frac{dF}{dA} \tag{2.2}$$

Since force and area are vectors, it is evident from Eq.(2.1) that pressure is a scalar. Just like temperature and density, pressure is the property of a fluid at a point. The unit of pressure in the SI systems is N/m^2. The unit N/m^2 is called a *Pascal* with the abbreviation Pa. A pressure of 100 kpa is called a *bar*.

Due to molecular collision, any part of a fluid experiences pressure due to the adjoining fluid or the solid boundaries containing it (Fig.2.1). Therefore, if we divide a fluid into two parts by an imaginary plane passing through it, the plane will experience pressure. At a point in a fluid with no relative motion between the adjacent layers, the magnitude of the pressure is independent of the orientation of the plane passing through that point.

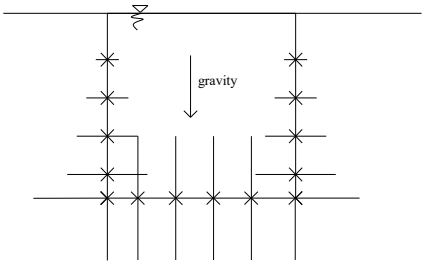


Figure 2.1 Pressure distributions around the fluid volume

DRAFT COPY

Prof. Dr. Yalçın Yüksel and Prof. Dr. Esin Çevik

2.3 Basic Principles

1) If the surface of the fluid is free surface, than the free surface is horizontal

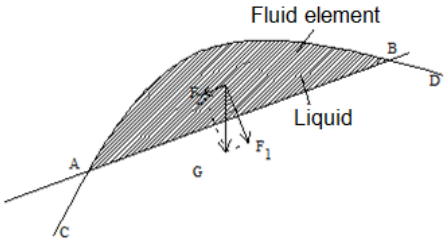


Figure 2.2

Since the fluid is in static case F_2=0. Hence there exists only gravity force. The direction of F_1 is the same as G in this case G should be perpendicular AB (Fig.2.2). This shows that AB should be horizontal

2) *The magnitude of pressure is independent of the orientation of the plane passing through that point*. Consider a wedge-shaped body of unit width with plane faces surrounding the point P, the point under consideration (Fig. 2.3). Since there is no relative motion, there are no shear forces and the only forces are the normal surface forces and the gravity.

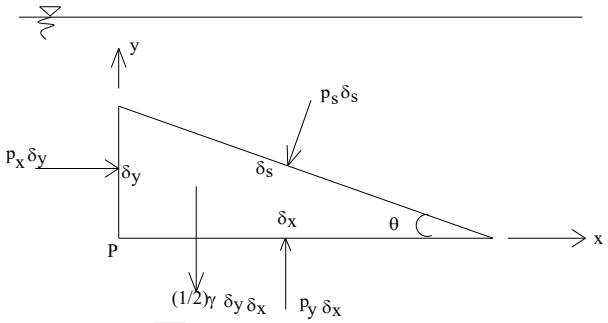


Figure 2.3 A wedge shaped fluid element with forces acting on it

The equation of motion in x-direction,

$$\sum F_x = 0 \qquad p_x \delta y - p_s \delta s \sin\theta = 0$$

Since $\delta s \sin\theta = \delta y$

$$p_x \delta y - p_s \delta y = 0 \qquad (2.2a)$$

$$p_x = p_s$$

The equation of motion in the y-direction

$$\sum F_y = 0 \qquad p_y \delta x - p_s \delta s \cos\theta - \frac{1}{2}\gamma \delta x \delta y = 0$$

At the limit when the wedge is reduced to the point P, then $\delta x \to 0$ and $\delta y \to 0$

Since $\delta s \cos\theta = \delta x$

$$p_x \delta x - p_s \delta x = 0 \qquad (2.2b)$$

$$p_y = p_s$$

Combining equations 2.2a and 2.2b, below equality is obtained

$$p_x = p_y = p_s \qquad (2.2c)$$

DRAFT COPY

This result is valid for any value of the angle θ and any orientation of the wedge. *This means that at any point within a static fluid, the pressure is the same in all directions*. This result is known as *Pascal's Law*.

If there is a relative motion between the adjacent fluid layers, shear forces are present and Eq.2.2c is not valid. However, if the shear forces are small as compared to the normal forces, Pascal's Law may still be approximately true.

3) To further elaborate Pascal's Law, consider a vessel as shown in Fig.2.4. The vessel has a unit thickness normal to the plane of the paper and is equipped with four circular openings of different diameters. Each opening is fitted with a piston, which can move freely through the mouth of the opening. The areas of the four openings are shown in Fig. 2.4

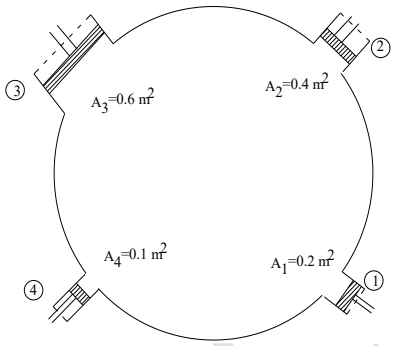


Figure 2.4 A vessel with four opening filled with a fluid.

Let us assume that the system is in equilibrium and under pressure. Now, if we increase the pressure force at the piston (1) 1 N, then, *according to Pascal's Law, this pressure change will be transmitted equally in all directions*. To keep the system in equilibrium, the same pressure must be applied on pistons 2, 3, and 4, respectively. Hence, forces of 2N, 3N, and 0.5N will be needed on pistons 2, 3, and 4 respectively, to keep the system in equilibrium.

Pascal's Law has wide application in the development of hydraulic controls for heavy earthmoving equipment, aircraft, and hydraulic presses.

4) In a fluid that is in equilibrium, the pressure difference between any two points is equal to the multiplication of the vertical difference between two points and the specific weight of the fluid

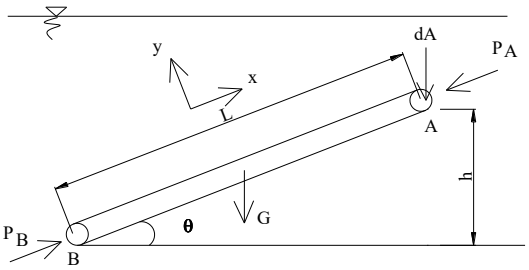


Figure 2.5

For equilibrium of the elementary volume

$$\sum F_x = 0 \qquad p_B\,dA - p_A\,dA - G\sin\theta = 0$$

and

DRAFT COPY

$$G \sin\theta = \gamma \forall \sin\theta = \gamma \, dA \, L \sin\theta$$

$$p_B \, dA - p_A \, dA - \gamma \, dA \, L \sin\theta = 0$$

$$p_B - p_A = \gamma \, L \sin\theta$$

Since

$$\sin\theta = \frac{h}{L} \quad \text{or} \quad L = \frac{h}{\sin\theta}$$

$$p_B - p_A = \gamma \frac{h}{\sin\theta} \sin\theta$$

$$p_B - p_A = \gamma \, h \tag{2.3}$$

5) In a fluid that is in equilibrium, all the points at the same level have pressure equal and constant.

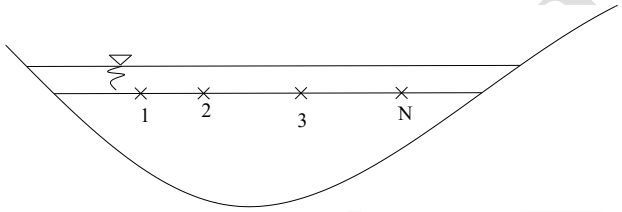


Figure 2.6 Equal pressure surface (Nivo plane or Equipressure plane)

p_1-p_2=γh

and h=0, $\gamma \neq \Rightarrow p_1=p_2=p_3$

From 4th and 5th principals in a static fluid, the pressure does not vary from point to point in a horizontal plane, but it does vary from point to point in a vertical plane.

6) The interface surface of two different liquids having different specific weights is horizontal

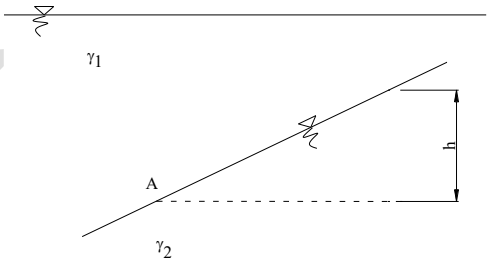


Figure 2.7

p_A–p_B=γ_1h or p_A–p_B=γ_2h

so γ_1h = γ_2h or h(γ_1-γ_2)=0

If $\gamma_1 \neq \gamma_2$ is, so h=0 this shows that interface should be horizontal

DRAFT COPY

2.4 Pressure Measurement

Since pressure is a very important characteristic of a fluid field, it is not surprising that numerous devices and techniques are used in its measurement. The pressure at point within a fluid mass will be designated as either an *absolute* pressure or a *gage (relative)* pressure (Fig.2.8). Absolute pressure

$$p_{abs} = p_{atm} + p \tag{2.4}$$

$$p = \rho g y \tag{2.5}$$

Pressure can also be expressed as a column height (Fig.2.9).

$$h = \frac{p}{\gamma} \tag{2.6}$$

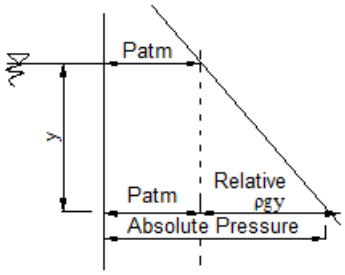


Figure 2.8 Absolute and gage pressure

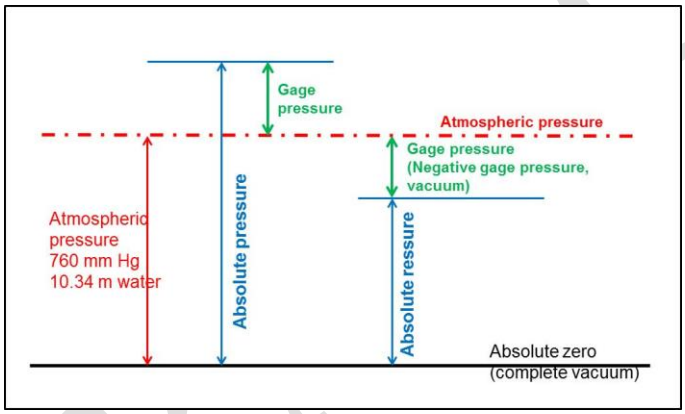


Figure 2.9 Graphical representations of gage and absolute pressure

For example at sea level atmospheric pressure is P= 101.3 kN/m^2

$$h = \frac{101.3 \times 10^3}{9810} = 10.33\text{m} \quad \text{meter of water or (water column)}$$

or

$$h = \frac{101.3 \times 10^3}{132926} = 0.762\,\text{mHgs (meter of mercury) or (mercury column)}$$

DRAFT COPY

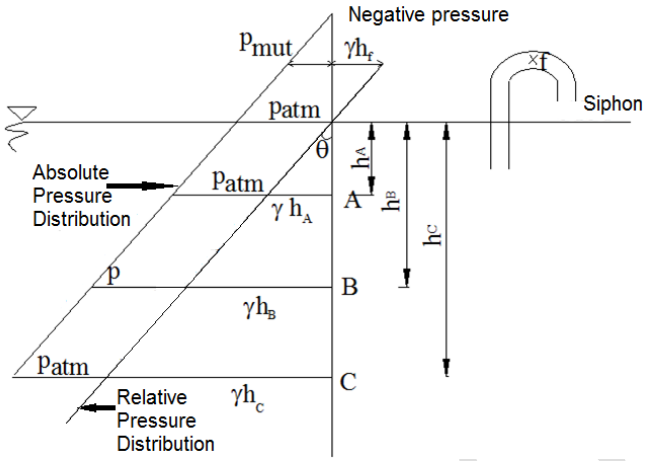


Figure 2.10 Distribution of hydrostatic pressure

Fig.2.10 shows that;

$$tg\theta = \frac{\gamma h_A}{h_A} = \frac{\gamma h_B}{h_B} = \frac{\gamma h_C}{h_C} = \gamma = \text{constant}$$

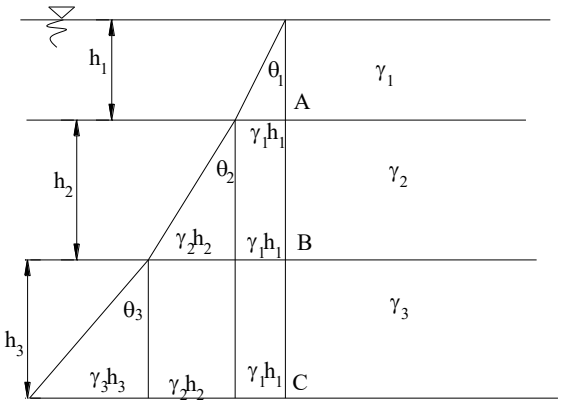


Figure 2.11 Distribution of hydrostatic pressure for different specific weight fluids

Fig. 2.11 indicates the hydrostatic pressure distribution for different fluids;

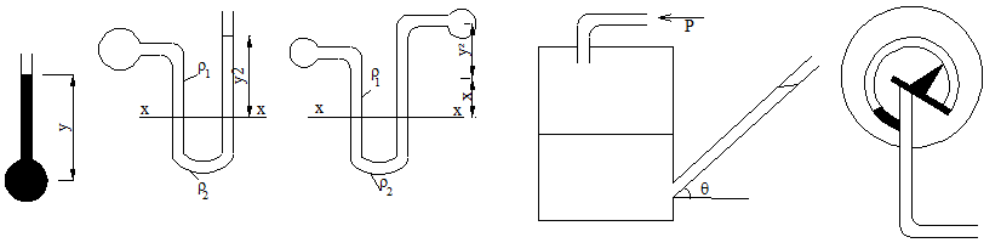
$$tg\theta_1 = \gamma_1$$

$$tg\theta_2 = \gamma_2$$

$$tg\theta_3 = \gamma_3$$

$$p_C = \gamma_1 h_1 + \gamma_2 h_2 + \gamma_3 h_3$$

DRAFT COPY



a) Piezometer tube b) Manometer c)Differential manometer d) Inclined manometer e) Bourden manometer

Figure 2.12 Pressure measurement devices

A manometer is a device used to measure the pressure of a fluid in a pipe. In its simplest form, it consists of a clear plastic or glass tube, one end of which is inserted in the pipe where the pressure is to be measured and the other end are left open to the atmosphere, as shown in Fig. 2.12 a, b. The height to which the fluid rises in the tube gives the pressure at the point of insertion in terms of the height of the fluid column. *A manometer can only measure the difference of pressure between its two ends*. The pressure at one end can only be found if the pressure at the other end is known. That is why one end of the manometer is left open to the atmosphere. In this way, the pressure at the open end is known to be zero. If the two ends of a manometer are connected to two different points of unknown pressures, then only the difference of pressure between the two points can be determined. The individual pressures at the two points cannot be determined in this way. A manometer may be in the form of a U-tube.

The unknown pressure at a point can be determined by moving along the length of the manometer from the point of unknown pressure to the point of known pressure while adding the column heights when moving down and subtracting them when moving up.

The pressure is the same within the same continuous fluid at the same level.

All heights must be expressed in terms of equivalent heights. This is done by multiplying the fluid heights by the respective specific gravities.

Add the height when moving down and subtract it when moving up

2.5 Pressure Variation in a Fluid at Rest

Although we have answered the question of how the pressure at a point varies with direction, we are now faced with an equally important question-how does the pressure in a fluid in which there are no shearing stresses vary from point to point? To answer this question lets consider a small rectangular element of fluid removed from some arbitrary position within the mass of fluid of interest as illustrated in Fig.2.13. There are two types of forces acting on this element: surface forces due to the pressure, and a body force equal to the weight of the element. Other possible types of body forces, such as those due to magnetic fields, will not be considered in this text.

If we let the pressure at the center of the element be designated as p, then the average pressure on the various faces can be expressed in terms of p and its derivatives as shown in Fig. 2.13. We are actually using a Taylor series expansion of the pressure at the element center to

approximate the pressures a short distance away and neglecting higher order terms that will vanish as we let dx, dy, and dz approach zero. Pressure increases in the +x, +y, +z directions and decreases in the –x, -y, -z directions.

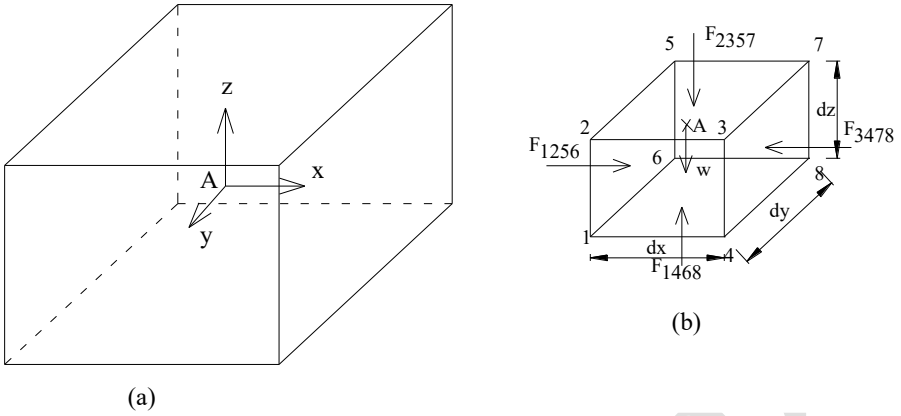


Figure 2.13

The pressure on 1256 surface

$$p_{1256} = p - \frac{\partial p}{\partial x}\frac{dx}{2}$$

Similarly

$$p_{3478} = p + \frac{\partial p}{\partial x}\frac{dx}{2}$$

The forces in x-direction are

$$F_{1256} = (p - \frac{\partial p}{\partial x}\frac{dx}{2})\,dy\,dz$$

$$F_{3478} = (p + \frac{\partial p}{\partial x}\frac{dx}{2})\,dy\,dz$$

The total surface forces on x-direction;

$$\sum F_x = F_{1256} - F_{3478}$$

Apart from hydrostatic pressure forces, if there are other body forces such as magnetic field, the components acting on unit mass are shown as X,Y, and Z and if the elemental mass, dm=ρdxdydz, the forces in x-direction.

Since the fluid is at rest

$$\sum F_x = 0 \qquad F_{1256} - F_{3478} + X\,dm = 0$$

$$(p - \frac{\partial p}{\partial x}\frac{dx}{2})dy\,dz - (p + \frac{\partial p}{\partial x}\frac{dx}{2})dy\,dz + X\rho\,dx\,dy\,dz = 0$$

$$p\,dy\,dz - \frac{\partial p}{\partial x}\frac{dx\,dy\,dz}{2} - p\,dy\,dz - \frac{\partial p}{\partial x}\frac{dx\,dy\,dz}{2} + X\rho\,dx\,dy\,dz = 0$$

$$-\frac{\partial p}{\partial x}dx\,dy\,dz + X\rho\,dx\,dy\,dz = 0 \qquad (2.7)$$

DRAFT COPY

if there are no body forces in this direction

$$\frac{\partial p}{\partial x} dx\, dy\, dz = 0$$

is obtained. In this equation, either $\frac{\partial p}{\partial x}$ or $dx\, dy\, dz$ should be equal to zero, but since $dx\, dy\, dz$ is the volume of this element cannot be zero

$$\frac{\partial p}{\partial x} = 0 \tag{2.8}$$

This shows that the pressure in the x-direction is constant.

Similarly, for y direction

$$-\frac{\partial p}{\partial y} dx\, dy\, dz + Y \rho\, dx\, dy\, dz = 0 \tag{2.9}$$

is obtained and if there is no body force in this direction

$$\frac{\partial p}{\partial y} = 0 \tag{2.10}$$

because of the equations (2.8) and (2.10) when combined together suggest that the *pressure is constant every point in the x-y plane (this is the basic principle)*

The surface where the pressure is constant is called as Nivo plane or Equipotential plane.

Again, equations similar to 2.8 and 2.10 may be written in the z-direction. But in addition to these forces, there is also a force W equal to the weight of the fluid in the elemental volume.

$$-\frac{\partial p}{\partial z} dx\, dy\, dz + Z \rho\, dx\, dy\, dz = 0 \tag{2.11}$$

in this direction the body force is the gravity force, hence W=mg and Z = -g

$$-\frac{\partial p}{\partial z} dx\, dy\, dz - \rho g\, dx\, dy\, dz = 0$$

or

$$\frac{\partial p}{\partial z} + \rho g = 0$$

is obtained.

Since pressure is a function of z only, we may change partial to total derivatives. Hence,

$$\frac{dp}{dz} + \rho g = 0$$

or

$$\frac{dp}{dz} = -\rho g \tag{2.12}$$

This is the differential equation of the variation of pressure in a static fluid. The equation is valid for a compressible, as well as an incompressible fluid.

DRAFT COPY

The results in the form of Eqs. (2.8), (2.10), (2.12) show that in a static fluid, the pressure does not vary from point to point in a horizontal plane, but it does vary from point to point in a vertical plane.

General Equations

$$\begin{bmatrix} \frac{\partial p}{\partial x} = \rho X \\ \frac{\partial p}{\partial y} = \rho Y \\ \frac{\partial p}{\partial z} = \rho Z \end{bmatrix} \tag{2.13}$$

These are called as Euler equations for static case

$$X\,dx + Y\,dy + Z\,dz = \frac{1}{\rho}\left(\frac{\partial p}{\partial x}dx + \frac{\partial p}{\partial y}dy + \frac{\partial p}{\partial z}dz\right) \tag{2.14}$$

$$X\,dx + Y\,dy + Z\,dz = \frac{1}{\rho}dp \tag{2.15}$$

Pressure Variation in an Incompressible Fluid

If a fluid is incompressible, its density ρ may be assumed to remain constant throughout the fluid. Then integrating Eq. (2.12) which respect to z, and replacing ρg with γ, we obtain

$$\frac{dp}{dz} = -\rho g$$

$$p = -\gamma z + C$$

where C is a constant of integration. If we change the z-axis is positive downwards by convention, then

$$p = \gamma z + C$$

if $p = p_0$ for $z = 0$ then $C = p_0$

$$p = p_0 + \gamma z$$

As a matter of convention, z is replaced by h which signifies depth below the liquid surface. Then the above equation becomes:

$$p = p_0 + \gamma h$$

If the pressure on the surface of the liquid is atmospheric, $p_0=0$ and the above equation simplifies to

$$p = \gamma h$$

Comments:

The equation $p = \gamma h$ assumes that γ is constant. Therefore, it is applicable only to incompressible fluids.

DRAFT COPY

The equation p=γh or $p = p_0 + \gamma h$ are valid only if the fluid is at rest or all shear stresses are negligibly small.

2.6 Hydrostatic Force on Immersed Bodies

2.6.1 Hydrostatic Force on Immersed Plane Surfaces

When a surface is submerged in a fluid, forces develop on the surface due to the fluid. The determination of these forces is important in the design of hydraulic structures such as storage tanks, ships, dams. For fluids at rest we know that the force must be perpendicular to the surface since there is no shearing stress present. We also know that the pressure will vary linearly with depth if the fluid is incompressible

It was shown in Section 2.3 that every particle below the surface of a fluid is under pressure which is given by the relationship p=γh. This pressure causes a force to act on any surface, which comes into contact with the fluid. This force, caused by the fluid at rest on surfaces in contact with it, is called the *fluid static* force. If the fluid is water, the force is known as the *hydrostatic* force. The point of application of the hydrostatic force is called the *center of pressure*.

Consider a plane area AB making an angle θ with the fluid surface as shown in Fig. 2.14. Let dA be a infinitesimally small area on the surface at a distance h vertically below the fluid surface. Considering the fluid to be incompressible, the pressure on the small areas is, p=γh

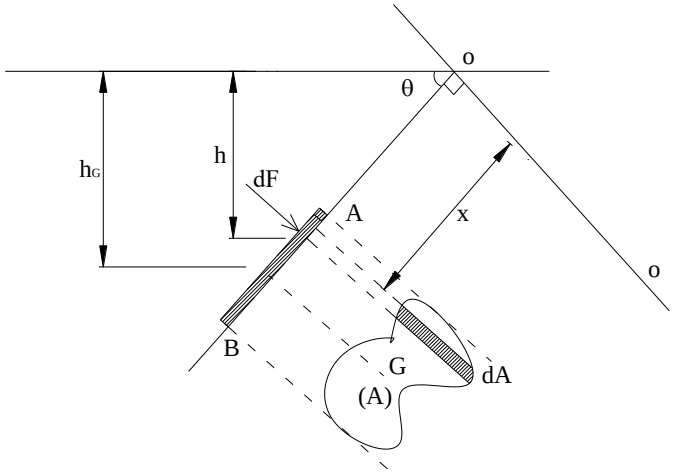


Figure 2.14 Hydrostatic force on an inclined plane surface of arbitrary shape.

The force on the small area is

$$dF = p\,dA$$

The total force on the entire area AB is

$$F = \int_A \gamma h\,dA = \gamma \int_A h\,dA$$

Here, $\int_A h\,dA$ is the static moment and that is equal to $h_G\,A$

Hence

$$F = \gamma h_G A \tag{2.16}$$

where A is the total area of the surface and h_G is the vertical depth of the centroid of the area below the fluid surface.

$$h_G = \frac{1}{A}\int_A h\,dA \tag{2.17}$$

It is seen from Eq. (2.16) that the hydrostatic force acting on a plane area is equal to its area multiplied by the pressure acting on its centroid. Since shear forces are not present in a static fluid, the hydrostatic force always acts perpendicular to the plane area.

Since θ does not appear in the expression for the force, the magnitude of the hydrostatic force does not change with the orientation of the area as long as the centroid remains at the same depth below the fluid surface. However, the direction of the force will change with the change in the orientation of the area in such a way that it always perpendicular to the area (see Fig. 2.15).

The hydrostatic force acts at the center of the pressure distribution, which is not the same point as the centroid of the area (Fig. 2.16).

$$F x_P = \int_A dF\,x$$

$$F x_P = \int_A p\,dA\,x$$

$$F x_P = \int_A \gamma h\,dA\,x$$

$$\gamma h_G A x_P = \gamma \int_A h\,dA\,x$$

$$\sin\theta = \frac{h}{x} = \frac{h_G}{x_G} = \frac{h_P}{x_P}$$

$$h_G = x_G \sin\theta \quad \text{ve} \quad h = x\sin\theta$$

$$x_G \sin\theta A x_P = \int_A x \sin\theta\,dA\,x$$

$$A x_G x_P = \int_A dA\,x^2$$

where $\int_A dA\,x^2$ moment of inertia of the plane area about an axis passing through 0-0

$$A x_G x_P = I_{0-0}$$

According to the Stainer Law

$$I_{0-0} = I_G + A x_G^2$$

$$A x_G x_P = I_G + A x_G^2$$

I_G is the moment of inertia of the plane area about an axis passing through the centroid and parallel to the fluid surface.

DRAFT COPY

or

$$x_P = \frac{I_G}{A\,x_G} + x_G$$

or

$$\frac{h_P}{\sin\theta} = \frac{I_G}{A\dfrac{h_G}{\sin\theta}} + \frac{h_G}{\sin\theta}$$

$$h_p = \frac{I_G \sin^2\theta}{A\,h_G} + h_G \qquad (2.18)$$

In Eq.2.18 the term $\dfrac{I_G \sin^2\theta}{A\,h_G}$ is always positive. Therefore, the center of pressure is always located below the centroid of surface. The only exception is when θ =0. In this case, the center of pressure coincides with the centroid.

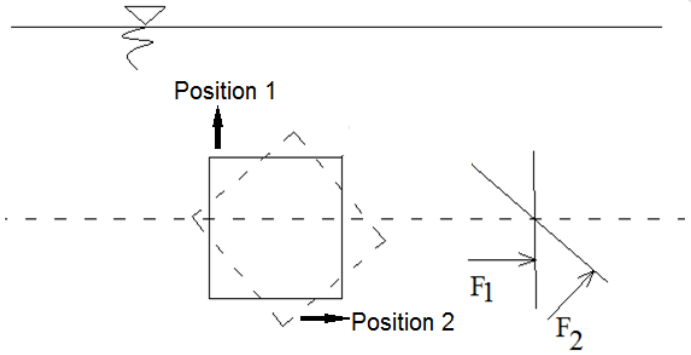


Figure 2.15 A plane area immersed in a fluid in two different positions.

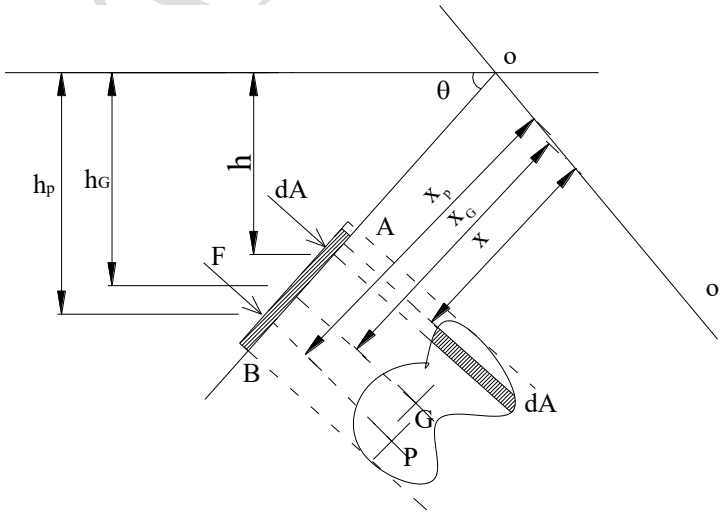


Figure 2.16 Point of application of the hydrostatic force.

2.6.1.1 Pressure Prism

An informative and useful graphical interpretation can be made for the force developed by a fluid on a plane area. Consider the pressure distribution along a vertical wall of a tank of

width b, which contains a liquid having a specific weight, γ. The resultant force must pass through the centroid of the pressure of the prism. For the volume under consideration the centroid is located along the vertical axis of symmetry of the surface, and at a distance of h/3 above the base (since the centroid of a triangle is located at h/3 above its base). The result can readily be shown to be consistent with that obtained from Eq.2.16.

This same graphical approach can be used for plane surfaces that do not extend up to the fluid surface as Fig.2.17a, b. In this instance, the cross section of pressure prism is trapezoidal. However, the resultant force is still equal in magnitude to the volume of the pressure prism, and it passes through the centroid of the volume. Specific values can be obtained by decomposing the pressure prism into two parts.

ΣF = hydrostatic force × b

$$\Sigma F = \left[\frac{\gamma h_A + \gamma h_B}{2}\overline{AB}\right] b$$

Note: The pressure prism method is only applicable when the width of the body is constant.

Imaginary Free Surface



(a)Inclined surface

(b) Vertical surface

Figure 2.17

All the equations obtained to this section are based on the assumption that the pressure on the free surface of the fluid is atmospheric. If the pressure on the surface is greater than the atmospheric pressure, additional force due to this pressure must also act on the immersed area. This additional force may be calculated in one of two ways, as follows:

1. Let p′ be the gauge pressure on the surface. Then additional force F′ due to this pressure on an immersed plane area A would be given by:

$$F' = \pm p' A \tag{2.19}$$

The total force on the plane area will be

$$\sum F = F \pm F' \tag{2.20}$$

The additional force F′ acts on the centroid of the area and, therefore, will shift the position of the center of pressure. The new location of the center of pressure, h′, is given by:

$$h' = \frac{F h_p + F' h_G}{F + F'} \tag{2.21}$$

2. Another method of dealing with the excess pressure is to convert the pressure into an equivalent height of the fluid column and then create an imaginary fluid surface at this level.

DRAFT COPY

The force and the location of the center of pressure may now be calculated as usual using Eqs.(2.16) and (2.18).

Summary

The force F always acts perpendicular to the plane area and is independent of the orientation of the area.

The center of pressure is always below the centroid of the plane area unless the area is horizontal.

While calculating I_G care must be exercised so that it is always about an axis, which passes through the centroid of the area and is parallel to the fluid surface.

Equations $F = \gamma h_G A$ and $h_P = \frac{I_G \sin^2\theta}{A h_G} + h_G$ are based on the assumption that the pressure on the surface of the fluid is atmospheric. If the pressure is not atmospheric, adjustment must be made in the force and its location as explained.

Since $F = \gamma h_G A$ and $h_P = \frac{I_G \sin^2\theta}{A h_G} + h_G$ were derived using a constant value of density, they are valid only for a single homogeneous fluid.

2.6.2 Hydrostatic Force on Curved Areas

The equations developed in the previous section for the magnitude and the location of the resultant force acting on a submerged surface can only be applied to plane surfaces. They cannot be used if the area is curved. However, many surfaces of interest (such as those associated with dams, pipes, and tanks) are non-planar.

Consider the curved surface AB shown in Fig.2.18. The direction of the forces shown is due to the reaction of the surface. The hydrostatic forces will have directions opposite to the ones shown here. The hydrostatic forces dF acting on individual infinitesimal area dA do not act parallel to each other due to the curvature of the surface. Hence, simple summation of these forces cannot yield the resultant force. However, these forces can be resolved into their components which can then be summed up vectorially using a convenient coordinate system.

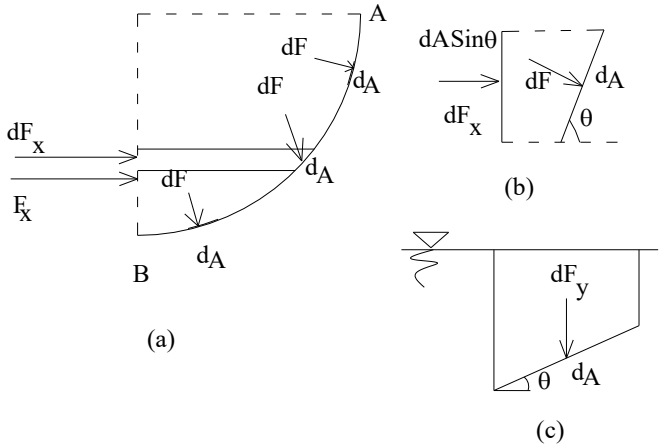


Figure 2.18 Hydrostatic forces on a curved immersed area.

DRAFT COPY

a) Horizontal hydrostatic pressure force;

Consider an elemental area dA shown exaggerated in Fig.2.18b. The hydrostatic force acting on the area is dF=pdA, where p is the pressure at the centroid of the elemental area. The x-component of this force is given by

$$dF = p\,dA$$

$$dF_x = p\,dA\sin\theta = p\,dA_x$$

where dA_x is the projection area of dA on a vertical plane. The x-component of the total force acting on the curved area is then given by

$$F_x = \int_A p\,dA_x = \int_A \gamma\, y\, dA_x = \gamma \int_A y\, dA_x$$

or

$$F_x = \gamma\, A_x\, y_G$$

$$F_x = \gamma\, h_G\, A_x \qquad (2.22)$$

where A_x is the area of projection of the curved surface on a vertical plane. Since this formula is identical to the one for a plane area, the location of F_x can be found using Eq.2.16.

b) Vertical hydrostatic pressure force;

The y-component of the small force dF acting on the elemental area dA can be found by considering Fig.2.18c. For equilibrium in the y-direction, dF_y must be equal in magnitude and same in direction to W, which is equal to the weight of the fluid overlying the elemental area. The y-component of the total force acting on the entire curved area will be equal to the sum of all such elemental fluid weights. Hence:

$$dF_y = pdA\cos\theta = pdA_y$$

$$F_y = \int_A p\,dA_y = \gamma \int_A y\,dA_y$$

where $dA_y=dA\cos\theta$

$$y\,dA_y = d\forall$$

$$F_y = \gamma \int_{\forall} d\forall$$

$$F_y = \gamma\, \forall \qquad (2.23)$$

F_y is the total volume of the fluid overlying the curved surface. The vertical component acts through the center of gravity of the overlying volume of the fluid.

The x-component and the y-component can then be combined vectorially to the resultant force acting on the curved surface.

$$F = \sqrt{F_x^2 + F_y^2} \qquad (2.24)$$

DRAFT COPY

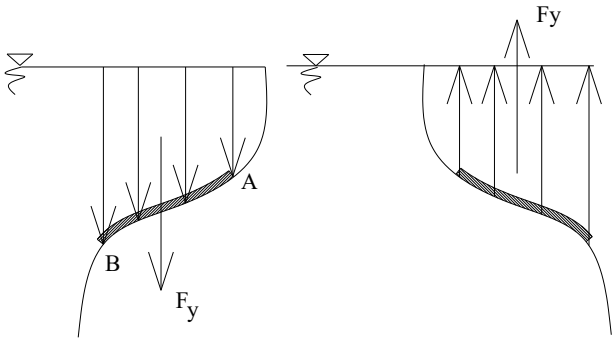


Figure 2.19 Obtaining of the direction of the vertical component acting on the curved area.

Example: Find the resultant force acting on the cylindrical surface AB.

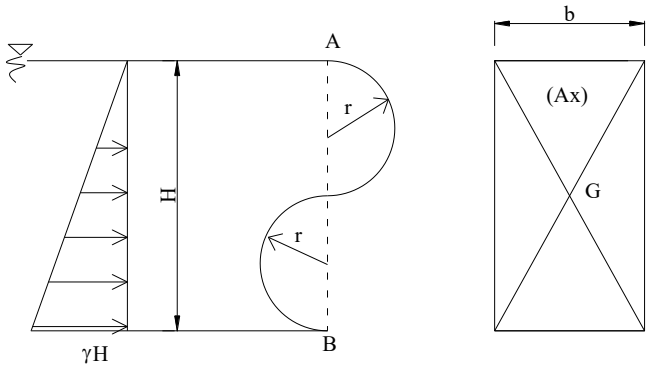


Figure 2.20

a) Horizontal component

$$F_x = \gamma h_G A_x = \gamma \frac{H}{2} H b = \frac{\gamma H^2}{2} b$$

$$F_x = \frac{\gamma H^2}{2} b$$

b) Vertical component

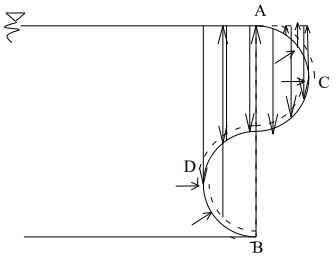


Figure 2.21

DRAFT COPY

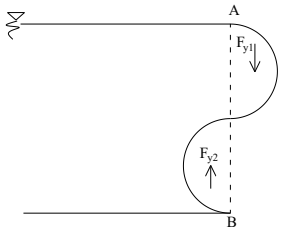


Figure 2.22

$$F_y = F_{1y} - F_{2y} = \gamma \frac{\pi r^2}{2} b - \gamma \frac{\pi r^2}{2} b = 0$$

The resultant force $F = F_x$

2.7 Relative Equilibrium

If a body of fluid is subjected to motion such that no layer moves relative to an adjacent layer, shear stresses do not exist within the fluid. In other words, in a moving mass if the fluid particles do not move relative to each other, they are said to be in static condition and a relative or dynamic equilibrium exists between them under the action of accelerating force and fluid pressures are everywhere normal to the surfaces on which they act. Here two cases when the methods of fluid statics can be used to study behavior of fluids in motion. One is the case of uniform linear acceleration when the entire fluid is moving as a solid with no relative movement of one layer with respect to another. The other is the case when a fluid is rotated uniformly about a vertical axis. Fluids in such motion are said to be in relative motion. In these cases, the planes of equal pressure are not horizontal but parallel to the fluid surface whose slope and shape are determined by the magnitude and the nature of the uniform acceleration.

2.7.1 Uniform Linear Acceleration

A liquid in an open vessel subjected to a uniform acceleration adjusts to the acceleration after some time so that it moves as a solid and the whole mass of fluid will be in relative equilibrium.

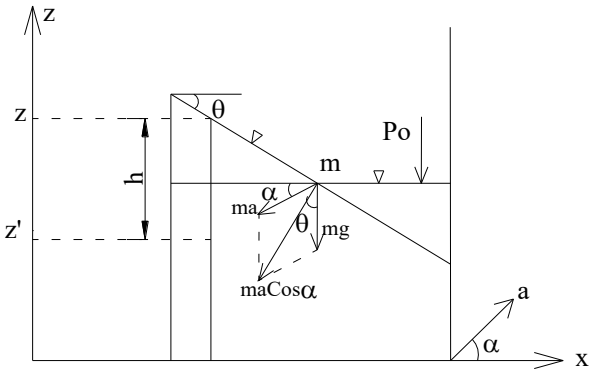


Figure 2.23 Linear acceleration with free surface

$$tg\theta = \frac{a\cos\alpha}{g + a\sin\alpha} \qquad (2.25)$$

where θ is the angle between horizontal and the free surface.

$$dp = \rho(X\,dx + Y\,dy + Z\,dz)$$

DRAFT COPY

From the general equation of hydrostatic, if there is no change in y direction and if you examine the pressure variation at a fixed x position then dx = 0, Y =0

$$dp = \rho Z\, dz$$

here $Z = -(g + a\sin\alpha)$

$$\int dp = -\rho(g + a\sin\alpha)\int dz$$

$$p = -\rho(g + a\sin\alpha)z + C$$

For $z = z$, $p = p_0$ ve $C = p_0 + \rho(g + a\sin\alpha)z$

for $z = z'$ pressure at a point A inside the fluid

$$p_A = p_0 + \rho(g + a\sin\alpha)(z - z')$$

$$p_A = p_0 + \rho(g + a\sin\alpha)h$$

or

$$p_A = p_0 + \gamma(1 + \frac{a}{g}\sin\alpha)h \qquad (2.26)$$

If one wants to obtain pressure at different x and z locations, the variation in the x direction should be taken account and a more general equation is obtained as below:

$$p = p_0 - \gamma\frac{a\sin\theta}{g}x - \gamma\left(1 + \frac{a\cos\theta}{g}\right)y$$

Here h is the vertical distance from the new free surface to the point A

For $\alpha = 0$ $p = p_0 + \gamma h$ hydrostatic pressure distribution

For $\alpha = 90°$ and for $a = g$, $p = p_0 + 2\gamma h$

For $\alpha = 270°$ and $a = g$ freely falling container $p = p_0$

2.7.2 Uniform Rotation about a Vertical Axis

In the case of uniform rotation about a vertical axis (also known as forced vortex motion), every particle of the fluid has the same angular velocity. Therefore there is no relative movement between the fluid particles and no shear forces. Since there are no shear forces, the laws of fluid statics apply.

Let a jar containing a fluid be rotated about its vertical axis with a constant angular velocity of ω rad/s (Fig. 2.24). Liquid in the container moves like a solid after some time. Pressure is constant for all points on the equipotential lines. Equipotential surface is a surface on which the pressure is constant.

DRAFT COPY

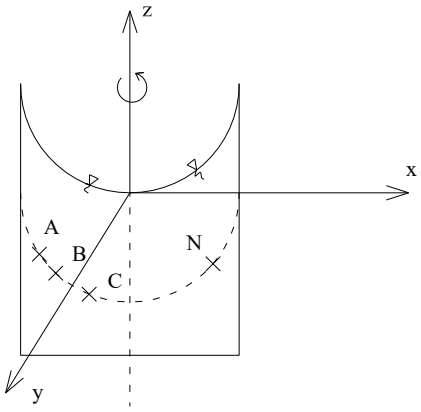


Figure 2.24 Rotation of a fluid about a vertical axis

In Fig. 2.24, pressures at the points on equipotential lines are;

$$p_A = p_B = p_C = ... = p_N$$

Since atmospheric pressure is effective to all points at the free surface, the free surface is equipotential surface.

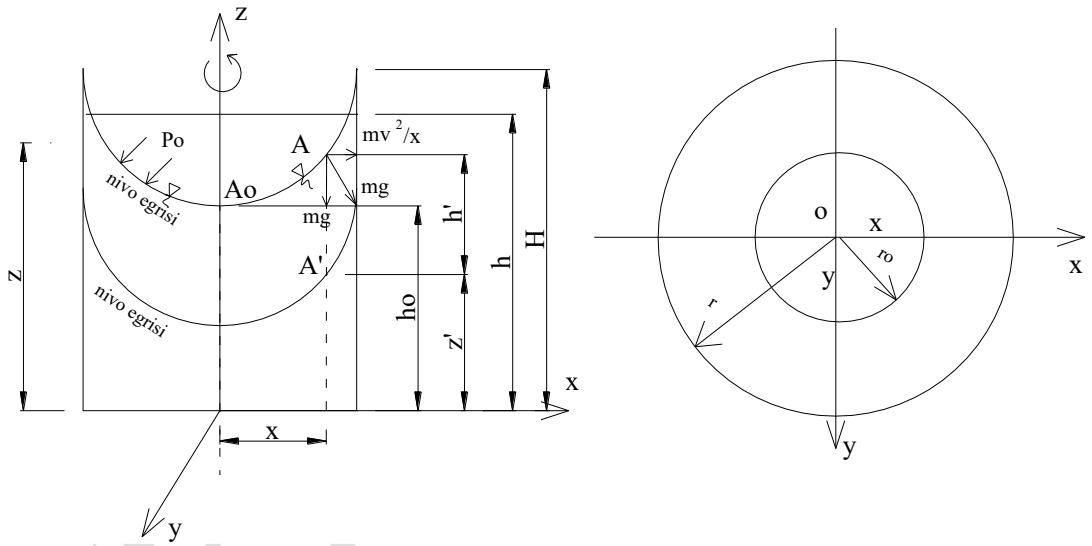


Figure 2.25 Forces acting on fluid particles in a uniformly rotating fluid

In equipotential surface p= constant, dp=0

$$dp = \rho(X\,dx + Y\,dy + Z\,dz)$$

$$X\,dx + Y\,dy + Z\,dz = 0$$

The centrifugal force on a unit mass in the x-direction;

$$X = \frac{v^2}{x} = \frac{\omega^2 x^2}{x} = \omega^2 x$$

The centrifugal force on a unit mass in the y-direction;

$$Y = \omega^2 y$$

The gravity force on a unit mass;

$$Z = -g$$

$$\omega^2 x dx + \omega^2 y dy - g dz = 0$$

by integrating

$$\frac{\omega^2 x^2}{2} + \frac{\omega^2 y}{2} - gz = C$$

$$\frac{\omega^2}{2}(x^2 + y^2) - gz = C$$

$$\frac{\omega^2 r_0^2}{2} - gz = C$$

integral constant is obtained by using the boundary condition;

for $z = h_0$ and $r_0 = 0$ $C = -gh_0$

$$\frac{\omega^2 r_0^2}{2} - gz = -gh_0$$

$$\frac{\omega^2 r_0^2}{2} = g(z - h_0)$$

$$\omega^2 r_0^2 = 2g(z - h_0) \qquad (2.27)$$

The equation of equipotential surface is obtained, this is the equation of paraboloid $x^2 = 2px + C$.

In order to find the pressure at any point inside the fluid, A′

$$dp = \rho(X dx + Y dy + Z dz)$$

$$dp = \rho(\omega^2 x dx + \omega^2 y dy - g dz)$$

by integrating

$$p_{A'} = \rho\frac{\omega^2}{2}(x'^2 + y'^2) - \rho g z' + C$$

since A (is on the free surface) and A′ are in the same vertical x′ = x and y′ = y. In this case

$$p_{A^1} = \rho\frac{\omega^2}{2}(x^2 + y^2) - \rho g z' + C = \rho\frac{\omega^2}{2} r_0^2 - \rho g z' + C$$

pressure at point A

$$p_A = p_0 = \rho\frac{\omega^2}{2}(x^2 + y^2) - \rho g z = \rho\frac{\omega^2}{2} r_0^2 - \rho g z$$

The pressure difference between the two points

$$p_{A^1} - p_0 = \rho g(z - z') = \gamma h'$$

$$p_{A^1} = p_0 + \gamma h' \qquad (2.28)$$

In a rotating jar the pressure distribution is hydrostatic and on the bottom of the jar pressure increases from center to the edges.

DRAFT COPY

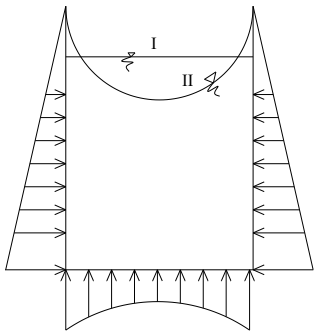


Figure 2.26 Pressure distributions in a uniformly rotating jar.

In order to find h_0 it is assumed that the volume of the fluid doesn't change

$$\pi r^2 h = \pi r^2 H - \frac{1}{2}\pi r^2 (H - h_0) \quad (2.29)$$

(original volume) (volume after rotation) (volume of paraboloid)

$$2h = H + h_0 \quad (2.30)$$

Equation of the equipotential surface

$$\omega^2 r_0^2 = 2g(z - h_0)$$

for $r_0 = r$; $z = H$

$$\omega^2 r^2 = 2g(H - h_0)$$

if the equation is substituted in Eq.(2.30)

$$\frac{\omega^2 r^2}{2g} = H - h_0 = (2h - h_0) - h_0 = 2h - 2h_0$$

or

$$\frac{\omega^2 r^2}{4g} = h - h_0$$

or

$$h_0 = h - \frac{\omega^2 r^2}{4g} \quad (2.31)$$

is obtained.

2.8 Stability of Floating Body

Let ABCD (Fig.2.27a) be a floating in a liquid. Let it be heeled through a small angle θ and acquire a new position A′B′C′D′ (Fig 2.27 b).

Also let

B=position of the center of buoyancy before heeling

B1=position of the center of buoyancy after heeling

G=position of the center of gravity

W=weight of body

DRAFT COPY

M= the point where a vertical line passing through B1 meets a line drawn through B and G.

Since the body is floating, the magnitude of the magnitude of the buoyant force is equal to the weight of the body, or

$F_B=W$

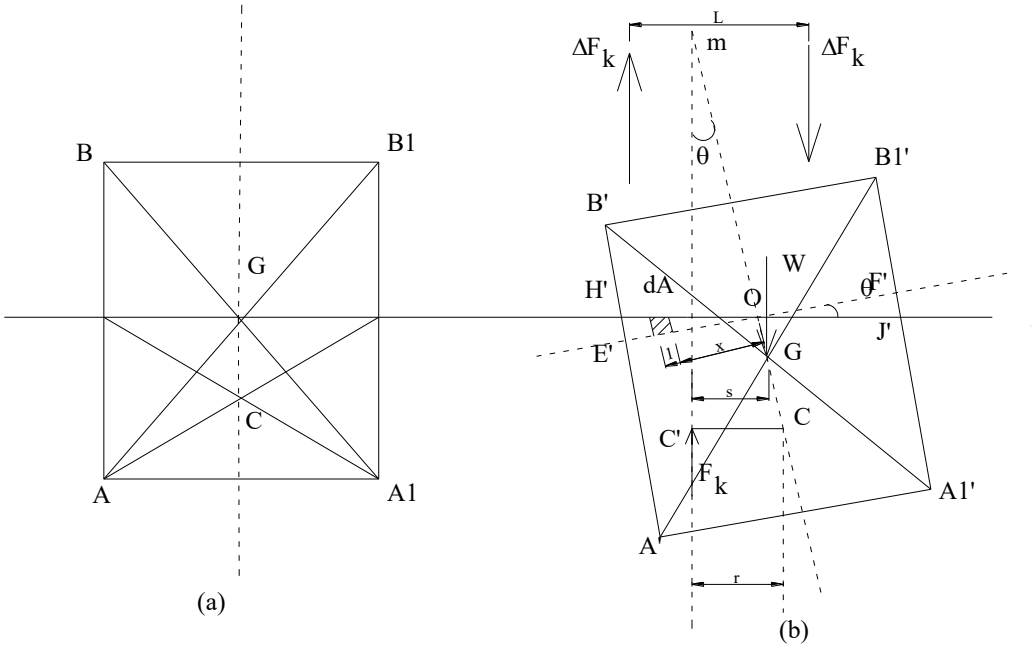


Figure 2.27 Stability of a prismatic floating body

This creates a couple which has a tendency to bring the body back to its original position. This couple is called *the righting couple.*

Let S= the length of a perpendicular drawn from G to MB_1

Then:

Moment of the couple= $W \times S$

$$= W \times MG \sin\theta \quad (2.32)$$

$$= W \times MG \tan\theta (\text{as } \theta \text{ is small}) \quad (2.33)$$

$$= W \times MG\, \theta \quad (2.34)$$

The point M is called the *metacenter.*

The distance MG is known as the *metacentric height*.

If M is above G, the equilibrium is *stable.*

If M is below G, the equilibrium is *unstable.*

If M coincides with G, the equilibrium is *neutral.*

A formula is now derived to determine the metacentric height for a very small angle of rotation, θ. Notice from Fig. 2.27b that as a result of rotation, a small wedge OH′E′ of the body is submerged in the liquid causing an upward force ΔF_B, and an identical wedge OF′J′ of the body comes out of the liquid causing a downward force ΔF_B . The horizontal shift in the center of buoyancy r is determined by change in buoyant forces caused by the wedges OH′E′ and OF′J′, or

$\Delta F_B l = Wr$ and

$\Delta F_B l = \gamma \forall r$ (2.35)

where $\forall$ is the total volume of the liquid displaced.

The term $\Delta F_B l$ can be evaluated by considering an elemental volume of the wedge having a base area equal to dA (Fig2.27b) at a distance x from the point O. Then volume of the elemental wedge is $\gamma x \theta dA$ and its moment about O is $\gamma x^2 \theta dA$.

Then:

$$\Delta F_B l = \int_A \gamma \theta x^2 dA = \gamma \theta \int_A x^2 dA$$

or

$\Delta F_B l = \gamma \theta I_0$

where I_0 is the moment of inertia of the area at the waterline about the longitudinal axis passing through O.

Substituting for $\Delta F_B l$ in Eq. 2.35

$\gamma \theta I_0 = \gamma \forall r$ or $\theta I_0 = \forall r$

also

$r = MB \times \sin \theta$

$= MB \times \theta$ (since θ is small)

Hence

$$\theta I_0 = \forall MB\theta \quad \text{or} \quad MG = \frac{I_0}{\forall}$$

The metacentric height MG is now given by

$$MG = MB \pm GB \quad \text{or} \quad MG = \frac{I_0}{\forall} \pm GB$$

DRAFT COPY

3
Fluid Kinematics

Introduction

So far we considered the equilibrium of a mass of fluid, which was at rest or in uniform motion as solid with no relative movement between various fluid layers. Now, we consider a mass of fluid in motion by examining the geometry of the motion of its particles. No attention, however, will be paid to the forces, which cause the motion of fluid mass.

The rules in fluid statics directly apply to the practices because we only need an experimental study for the fluid density. However, the problem is very complex if fluid moves. Kinematics involves position, velocity, and acceleration, not force.

Analyzing of Fluid Mechanics

The motion of a fluid may be described using two methods.

The first is the Lagrangian Method in which attention is focused on motion of a single particle as it occupies various positions in flow field (Fig.3.1).

$$V = V\left[x(t), y(t), z(t), t\right] \tag{3.1}$$

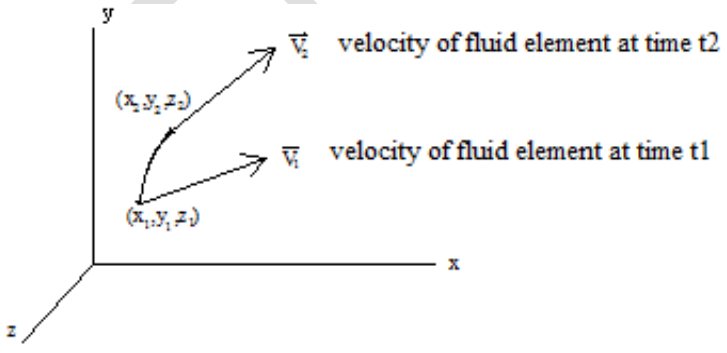


Figure 3.1 Lagrangian approach to the movement of a fluid particle

The second is the Eulerian method in which attention is focused on fixed point in the flow field and the velocities of different particles are examined as they pass through that point (Fig.3.2).

$$V = (x, y, z, t) \tag{3.2}$$

Either Eulerian or Lagrangian method can be used to describe flow fields. Most fluid mechanics considerations involve the Eulerian method because mathematics is easy to define the motion of fluid particles.

DRAFT COPY

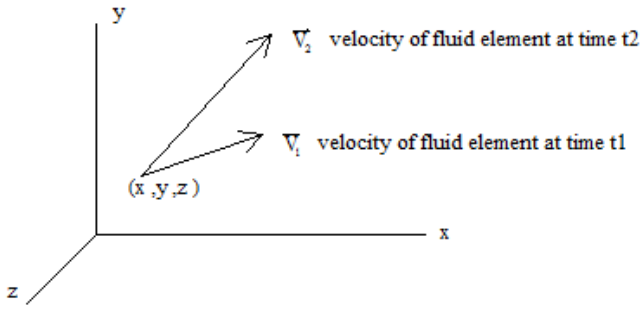


Figure 3.2 Eulerian approach to the movement of a fluid particle

Fundamental Concepts of Fluid Flow

1. Fluid Types

a) Ideal Fluid: Inviscid fluids are fluids in which the fluid viscosity is assumed to be zero.

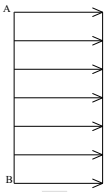


Figure 3.3 Velocity profile for ideal fluid flow

b) Viscous Fluid: Viscous fluids (real fluids) are fluids in which the fluid viscosity cannot be neglected.

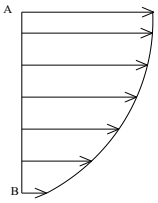


Figure 3.4 Velocity profile for viscous fluid flow

2. Flow Types;

The fluid flow property at a point in a flow is a function of its position as well as time. This can be expressed as:

$$E = E(x, y, z, t) \tag{3.3}$$

where E is any fluid property such as density, velocity, temperature, etc.

a) Steady and Unsteady Flow

If the properties at a point in the flow field do not change with time, the fluid flow is called steady flow.

$$\frac{\partial E}{\partial t} = 0 \quad \text{or} \quad E = E(x, y, z) \tag{3.4}$$

In certain types of flow (such as turbulent flow) there are always small fluctuations of velocity at a point. Under such conditions, the flow is said to be steady if the temporal mean velocity (time-averaged velocity) given by

DRAFT COPY

$$V_t = \frac{1}{t}\int_0^t V dt \qquad (3.5)$$

this does not change with time.

The flow, on the other hand, is unsteady if the properties do vary at point with respect to time.

$$\frac{\partial E}{\partial t} \neq 0 \qquad (3.6)$$

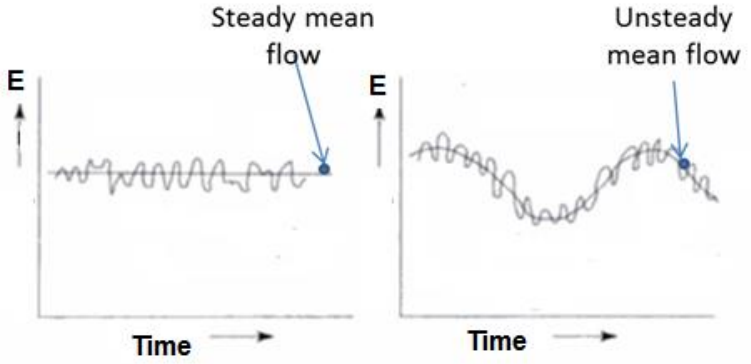
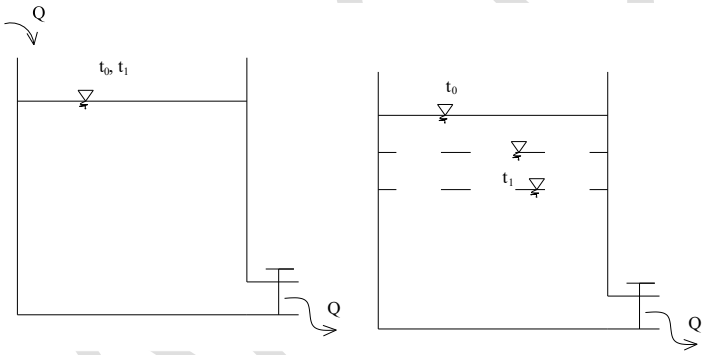


Figure 3.5 Variation of properties E for steady and unsteady flow



a) Steady Flow b) Unsteady flow

Figure 3.6 Steady and unsteady flow

b) Uniform and Non-uniform Flow;

Contrary to the steady flow which is defined with respect to time, uniform flow is defined with respect to space at any instant of time (Fig.3.5).

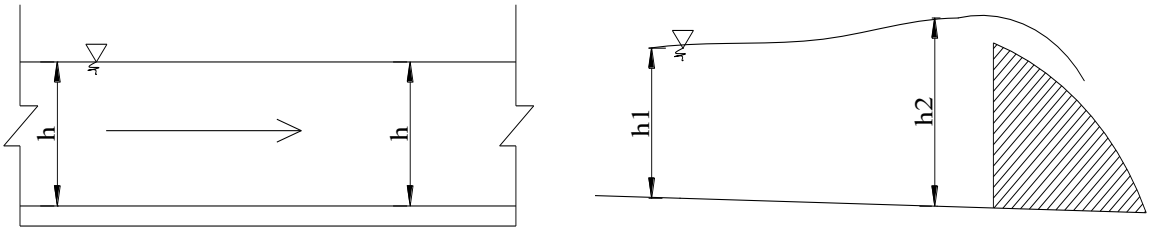
The flow of a fluid is said to be uniform if the fluid properties do not change with respect to space at any instant of time. However, the properties may vary from time to time.

$$\frac{\partial E}{\partial s} = 0 \qquad (3.7)$$

where s refers to the position in space.

On the other hand, a non-uniform flow is one in which properties vary from point to point at the same instant of time.

$$\frac{\partial E}{\partial s} \neq 0 \qquad (3.8)$$



a) Uniform flow b) Non- uniform flow

Figure 3.7 Uniform and non-uniform flow

3. Streamline

Streamlines are lines tangent to the velocity field or streamlines are lines drawn such that their tangents at any point are in the direction of the particle velocity. If streamlines are closer together, the flow field has higher velocity or vice versa.

The equation of streamlines may be described as follows;

$$\vec{V} = u\,\vec{i} + v\,\vec{j} \tag{3.9}$$

also, the displacement of the fluid particle at point P may be written as where δ_x and δ_y are the rectangular components of the displacement in the x and y directions respectively.

$$\vec{\delta}_s = \delta_x\,\vec{i} + \delta_y\,\vec{j} \tag{3.10}$$

Since velocity and displacement vectors are in the same direction we must have

$$\frac{\delta_x}{u} = \frac{\delta_y}{v} \tag{3.11}$$

or, in differential form

$$\frac{dx}{u} = \frac{dy}{v} \tag{3.12}$$

$$vdx - udy = 0 \tag{3.13}$$

this gives the equation of a streamline in a two-dimensional flow field. In three dimensions, the equation would become,

$$\frac{dx}{u} = \frac{dy}{v} = \frac{dz}{w} \tag{3.14}$$

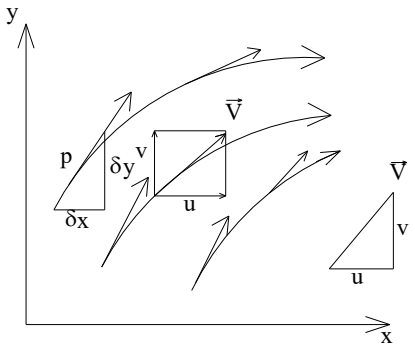


Figure 3.8 Streamlines and velocity vectors

DRAFT COPY

4. Streaklines

A streakline is an instantaneous picture of the line joining all particles which have emerged from the same source and which have passed through the same point in a flow field. If a dye is injected at a point in a flow field, the dye would follow the path of the fluid particles passing through that point. A photograph of flow taken at any instant of time would show a streak of dye which is a streakline. If the flow is steady, photographs taken at various times would be identical. But if the flow is unsteady, different photographs taken at different times would show different streaklines

5. Pathlines

A pathline is the line traced out by a given particle as it flows from one point to another. The pathline is a Lagrangian concept.

A streakline is an instantaneous picture of the line joining all particles or a streakline is the line joining all particles passing through the same fixed point in the fluid.

For steady flow, streamlines, streaklines, and pathlines are the same. The streamline is often used in analytical work while the streaklines and pathlines are often used in experimental work.

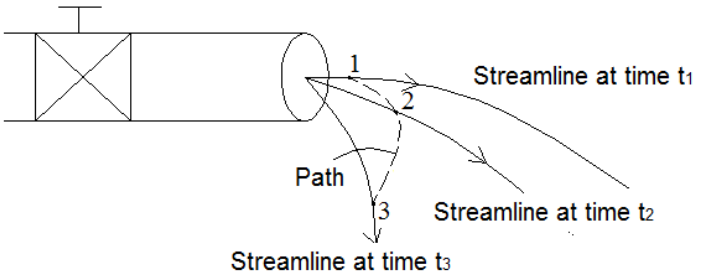


Figure 3.9 Streamlines and pathlines for unsteady flow

In a steady flow, since the direction of velocity is fixed at any point, the streamline is also fixed in space and coincides with the path of movement of a particle (Fig.3.9). In an unsteady flow, the direction of velocity is not fixed with time; therefore, a streamline may shift in space from time to time. A particle, then, may follow one streamline at one instant of time and another one at another instant of time. Since the velocity vector is tangent to a streamline, it must have no component normal to it. Therefore, there is no flow across a streamline.

6. Stream Tube

A stream-tube is a volume of fluid enclosed by a cylindrical surface formed by streamlines as shown in Figure 3.10 or another definition of a stream tube is the tube formed by an envelope of streamlines. Since the velocity component normal to a streamline is zero, there can be no flow across a stream-tube.

A stream-tube of very small cross-section with no velocity variation over it is sometimes called a stream filament.

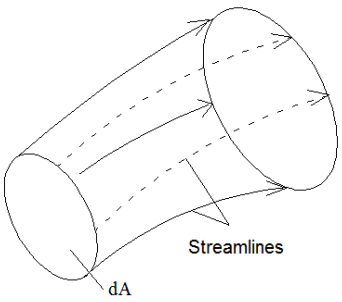


Figure 3.10 Stream tube

7. System and Control Volume

A system is a specific mass of fluid, which is isolated for experimentation. A system changes its shape or position but its mass always constant (Fig. 3.11).

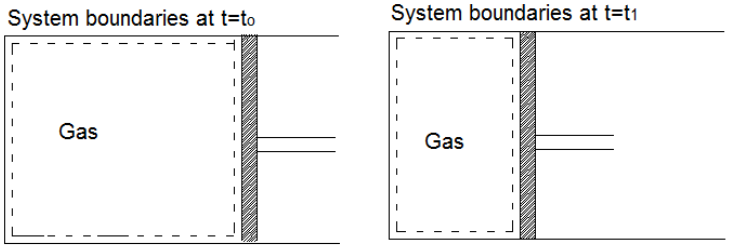


Figure 3.11 System

A control volume is a specific region of the flow system in space through which fluid flows and which has been isolated to study the effect of forces and other agents on the fluid. The physical boundary of the control volume is called the control surface. A section of a stream-tube may be used as a control volume. Fig. 3.12 shows a section of a pipe through which a fluid is flowing. A part of the pipe has been isolated as shown by the broken lines. This is a control volume and the broken lines are the control surface.

The concept of a system is useful when studying fluids from the Lagrangian point of view, while the concept of control volume is useful when studying fluids from the Eulerian point of view.

In general, the control volume approach to studying fluid behavior is preferred over the system approach. There are two basic reasons for this. First, it is always more convenient to study the behavior of a moving fluid as it flows past a fixed region than to study the behavior of a fluid as it goes from place to place. Second, in general, we are not interested in the motion of a given mass of fluid, but rather in the effect of such a motion on the surrounding objects.

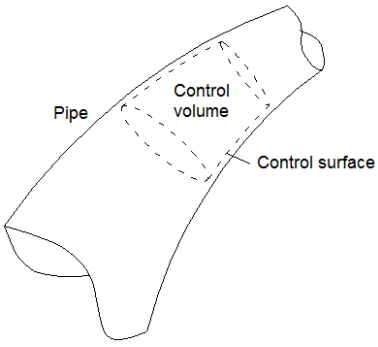
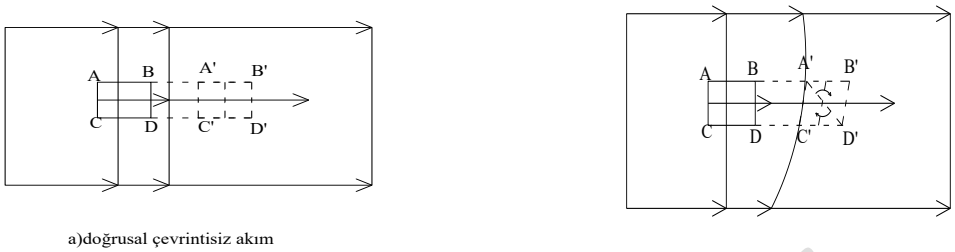


Figure 3.12 Control volume

8. Rotational and Irrotational Flows

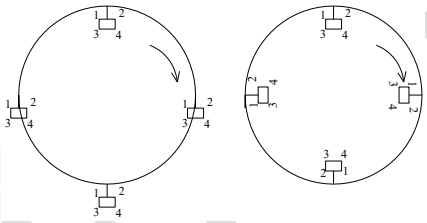
A fluid element is said to have been rotated if the sides of the element have rotated through an angle without any change in shape of the fluid element. Hence, there is no change in volume of the fluid element and no shear stresses are present in it. If the average rate of rotation is zero, then the flow is irrotational.



a)doğrusal çevrintisiz akım

Figure 3.13 Irrotational and rotational flow

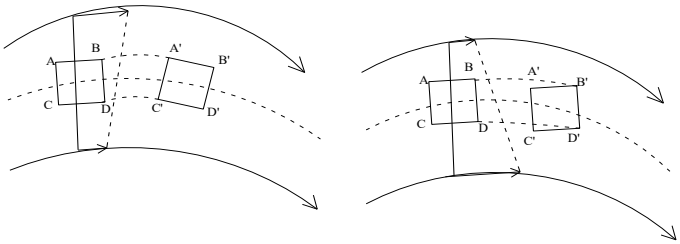
Rotation of a fluid particle can be caused only by a torque applied by shear forces on the sides of particle. Such forces, however, are impossible in an ideal or inviscid fluid and so the flow of an ideal fluid is necessarily irrotational. The concept of an ideal fluid is that of one which is inviscid and incompressible; the flow is also assumed to be irrotational. The analysis of ideal fluid flow is also referred to as "potential flow". Ferris wheel in a fun-fair may be given as an example to define potential flow, because every car don't rotate around their axis during the ferris wheel rotates around its axis. If the cars are fixed on the wheel, they will rotate which define the rotational flow (Fig.3.14).



a) Rotational flow b) Irrotational flow

Figure 3.14 An example of ferries wheel in fun fair to potential flow

Two examples of fluid flow are dedicated in Fig 3.15. A small fluid element with sides AB and CD is considered in a curvilinear flow. At (a) both sides rotate in the same direction and an equal amount clockwise rotation occurs as the element moves along a streamline; there is rotation but little distortion. At (b) the side AB rotates clockwise as the element moves along a streamline, CD rotates an equal amount with AB but counterclockwise so that the net rotation is zero. Although undergoing distortion, the element is thus not rotating; the flow is irrotational.



a) Rotational flow b) Irrotational flow

Figure 3.15 Rotational and irrotational flow

9. Laminar and Turbulent Flows

A real flow may be laminar, transitional or turbulent. In laminar flow the fluid flows in a well-ordered pattern, fluid layers are assumed to slide over one another. In this kind of flow, that

DRAFT COPY

occurs at the lower velocities, the particles of fluid are evidently moving entirely in straight lines even though the velocity with which particles move along one line is not necessarily the same as that along another line. If a transition is taking place from a previously well-ordered flow to an unstable flow, the flow is called a turbulent flow.

In laminar flow, the individual particles of a fluid move in a regular fashion, they do not cross each other's path and there is no mixing of the fluid. The fluid is thus considered to be moving in layers or laminas. Due to low velocity, the viscous forces dominate over the inertia forces.

In turbulent flow, the motion of individual fluid particles is highly erratic producing eddies and cross-currents. In the turbulent flow, the paths of individual particles of fluid are no longer everywhere straight but are sinuous, intertwining and crossing one another in a disorderly manner so that a thorough mixing of the fluid takes place. Turbulent flow, in short, is characterized by the fact that superimposed on the principal motion of the fluid are countless, irregular, haphazard secondary components and three dimensionality. Turbulent motion is essentially irregular motion on a small scale. Indeed, the fluctuations in the magnitude of the velocity of individual particles are often so small compared with the average magnitude.

When turbulent flow takes place in a cylindrical tube, for example, only the average motion of the fluid is parallel to the axis of the tube even though the flow has three-dimensionality.

Precise mathematical analysis of turbulent flow is difficult; therefore, experimentation is usually needed.

Whether the flow through a conduit is laminar or not, can be determined by calculating the Reynolds number after the work of Osborne Reynolds (1880). The apparatus used by Reynolds was basically as shown in Fig. 3.16. A straight circular glass tube was placed in a tank, which is full of water. The flow rate was controlled by a valve at the outlet end. A fine nozzle connected to a small reservoir of a liquid dye discharge a colored filament into the inlet of the glass tube. If the velocity of the water remained low in the tube, the filament of dye passes down the tube without mixing with the water. When the valve was opened further, the velocity of the water increased, and the stream of dye began to wave and fluctuated and mixed more or less completely with the water in the tube. Reynolds also found that it was not the velocity, but also the density and viscosity of the fluid and the diameter of the tube which determined the flow regime. An exact mathematical analysis of turbulent flow is impossible and experimental results are widely used in the analysis of turbulent flow problems

The flow is considered to be laminar if the Re number is less than, or equal to 2320.

$$\mathrm{Re} = \frac{V D}{\nu} \tag{3.15}$$

Re<2320	Laminar flow
Re≈2320	Transient flow
Re>2320	Turbulent flow

DRAFT COPY

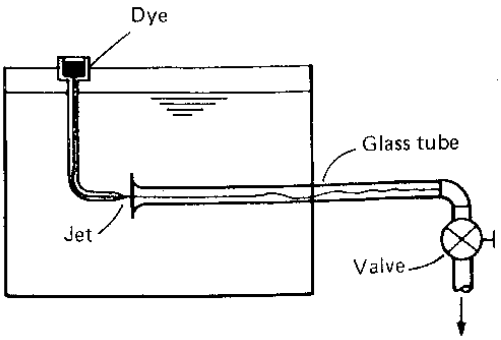
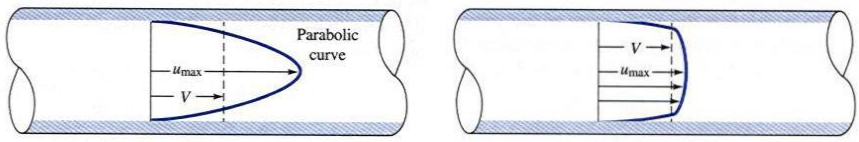


Figure 3.16 Reynolds experiment

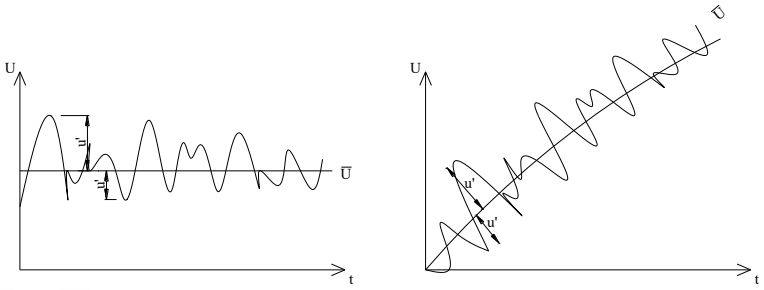
As shown in Fig. 3.17, velocity distribution in a pipe has a parabolic pattern for laminar flow, but the velocity distribution in turbulent flow is more uniform due to momentum interchange between fluid layers.



(a) Velocity profile for laminar flow (b) Velocity profile for turbulent flow

Figure 3.17 Velocity profiles

Fig.3.18 shows the traces of velocity at a fixed position for steady turbulent and unsteady turbulent flows. In turbulent flow, the instantaneous velocity (total) velocity has two components; these are fluctuated and time-averaged velocity.



a) Steady case b) Unsteady case

Figure 3.18 Steady and unsteady turbulent flow

Resultant instantaneous velocity

$$u(t) = \overline{u} \pm u'(t) \qquad (3.16)$$

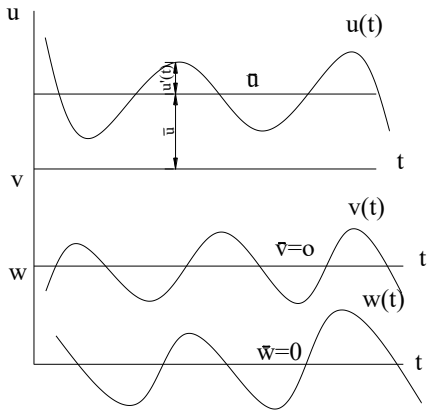


Figure 3.19 Turbulent velocities for one-dimensional flow

Time-averaged velocity

$$\bar{u}.T = \int_0^T u(t).dt \rightarrow \bar{u} = \frac{1}{T}\int_0^T u(t).dt \qquad (3.17)$$

For one directional flow $\bar{v} = \bar{w} = 0$

Time averaged fluctuated velocity in x direction equals to zero, because

$$\overline{u'} = \frac{1}{T}\int_0^T u'(t)\,dt = \frac{1}{T}\int_0^T (u(t) - \bar{u})\,dt = \bar{u} - \bar{u} = 0 \qquad (3.18)$$

Instantaneous velocities in three dimensions may be written as follows;

$$\left.\begin{aligned} u(t) &= \bar{u} \pm u'(t) \\ v(t) &= \bar{v} \pm v'(t) \\ w(t) &= \bar{w} \pm w'(t) \end{aligned}\right\} \qquad (3.19)$$

10. One-, Two-, and Three- Dimensional Flows

In almost any flow situation, the velocity field actually contains all three velocity components (u, v and w). In many situations the three-dimensional flow characteristics are important in terms of the physical effects they produce.

In many situations one of the velocity components may be small (in some sense) relative to the two other components. In situations of this kind it may be reasonable to neglect the smaller component and assumed two-dimensional flow.

It is sometimes possible to further simplify a flow analysis by assuming that two of the velocity components are negligible, leaving the velocity field to be approximated as a one dimensional flow field.

Motion of a Fluid Element

The motion of the small fluid element can take place in four different ways. They are translation, linear deformation, angular deformation and rotation (Fig.3.20).

DRAFT COPY

Translation

A fluid element is said to have been translated if it only shifts in position without undergoing any deformation. There are no shear stresses in the fluid element. There is no change in the volume of the element.

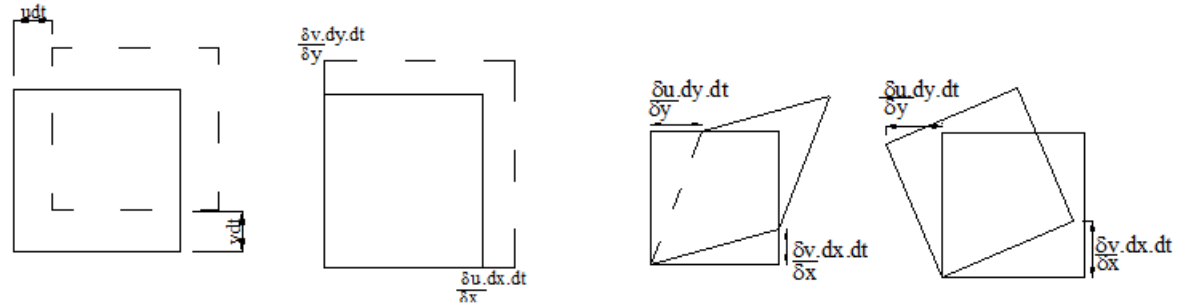


Figure 3.20 Motion of a fluid element

Linear Deformation

A fluid element is said to have been linearly deformed if it has expanded or contracted parallel to the original axes. In this case, there is a change in the volume of element. Shear stresses are present, due to existence of relative motion between fluid layers in the element. The relative change in volume with respect to the original volume is;

$$\frac{(dx+\frac{\partial u}{\partial x}dxdt)(dy+\frac{\partial v}{\partial y}dydt)(dz+\frac{\partial w}{\partial z}dzdt)-dxdydz}{dxdydz}$$

$$=\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}+\frac{\partial w}{\partial z}$$

$$=\nabla.\vec{V} \qquad (3.20)$$

Angular Deformation

A fluid element is said to have been angularly deformed if the sides of the element have deformed in the angular direction. In this case, there is a change in the shape of the fluid element. Shear stresses are presented in the element. Considering counterclockwise motion is positive, the angular deformation θ_{xy} is given by;

$$\theta_{xy}=\left(\frac{\frac{\partial v}{\partial x}dxdt}{dx}\right)-\left(\frac{-\frac{\partial u}{\partial y}dydt}{dy}\right)$$

$$\theta_{xy}=\left(\frac{\partial v}{\partial x}+\frac{\partial u}{\partial y}\right)dt \qquad (3.21)$$

Angular velocities or rate of angular deformations;

DRAFT COPY

$$\left.\begin{aligned}
\dot{\theta}_{xy} &= \left(\frac{\partial v}{\partial x}+\frac{\partial u}{\partial y}\right) \qquad x-y\,\text{plane}\\
\dot{\theta}_{yz} &= \left(\frac{\partial v}{\partial z}+\frac{\partial w}{\partial y}\right) \qquad y-z\,\text{plane}\\
\dot{\theta}_{xz} &= \left(\frac{\partial w}{\partial x}+\frac{\partial u}{\partial z}\right) \qquad x-z\,\text{plane}
\end{aligned}\right\} \tag{3.22}$$

Heat is generated and mechanical energy is dissipated through angular deformation, shear stresses are present.

Rotation

A fluid element is said to have been rotated if the sides of the element have rotated through an angle without any change in the shape of the fluid element. Hence, there is no change in volume of the fluid element and no shear stresses are present in the element. No heat is generated and no mechanical energy is dissipated through rotation.

The rate of rotation is the angular velocity.

$$\omega_z = \frac{1}{2}\left(\frac{\partial v}{\partial x}-\frac{\partial u}{\partial y}\right) \tag{3.23}$$

$$\omega_x = \frac{1}{2}\left(\frac{\partial w}{\partial y}-\frac{\partial v}{\partial z}\right) \tag{3.24}$$

$$\omega_y = \frac{1}{2}\left(\frac{\partial u}{\partial z}-\frac{\partial w}{\partial x}\right) \tag{3.25}$$

The resultant angular velocity ω of the fluid element is given by;

$$\vec{\omega} = \omega_x\vec{i}+\omega_y\vec{j}+\omega_z\vec{k}$$

$$\vec{\omega} = \frac{1}{2}\begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u & v & w \end{vmatrix} = \frac{1}{2}\,\text{curl}\vec{V} = \frac{1}{2}\,\text{rot}\vec{V} = \frac{1}{2}\nabla\times\vec{V} \tag{3.26}$$

A flow is said to be irrotational if the resultant angular velocity or rotation is zero so equation (3.27) may be obtained. Vorticity is defined as twice the angular velocity or rotation.

$$\left.\begin{aligned}
\frac{\partial v}{\partial x} &= \frac{\partial u}{\partial y} \quad \text{on} \quad x-y\,\text{plane}\\
\frac{\partial w}{\partial y} &= \frac{\partial v}{\partial z} \quad \text{on} \quad y-z\,\text{plane}\\
\frac{\partial u}{\partial z} &= \frac{\partial w}{\partial x} \quad \text{on} \quad x-z\,\text{plane}
\end{aligned}\right\} \tag{3.27}$$

Fluid Acceleration

i) In terms of a natural coordinate system

DRAFT COPY

Acceleration of a fluid particle is the rate of change of its velocity as it moves along a streamline. Let a particle of fluid move from a position s at time t to a position s+ds in time dt as shown Fig.3.21. Tangential acceleration;

$$a_S = \frac{dV(s,t)}{dt} \tag{3.28}$$

$$a_S = \frac{\partial V_S}{\partial s}\frac{\partial s}{\partial t} + \frac{\partial V_S}{\partial t}$$

where $\partial s/\partial t$ is local velocity

$$a_S = V_s\frac{\partial V_s}{\partial s} + \frac{\partial V_s}{\partial t} \tag{3.29}$$

The above expression (3.29) for acceleration contains (material acceleration) two types of terms. The first term describes change of velocity with respect to position while the second term describes the change of velocity with respect to time. The first term is known as the convective or spatial acceleration and the second term is known as the local or temporal acceleration.

Since a streamline is a curve, it is subjected to an additional acceleration in the direction normal to the streamline (Fig. 3.22). The acceleration towards the center of curvature of the path is called centrifugal acceleration.

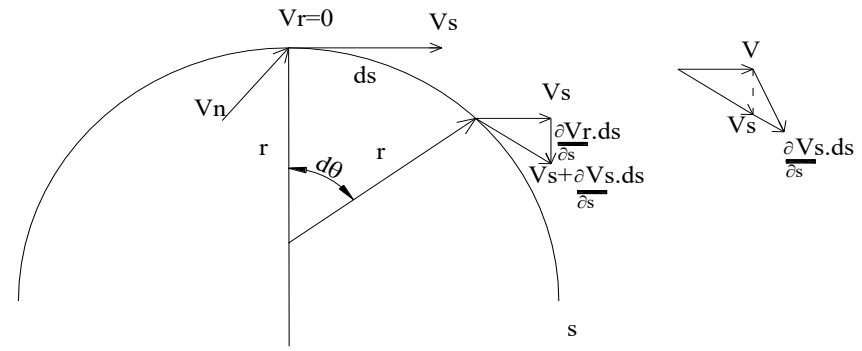


Figure 3.21 Velocity of a fluid element

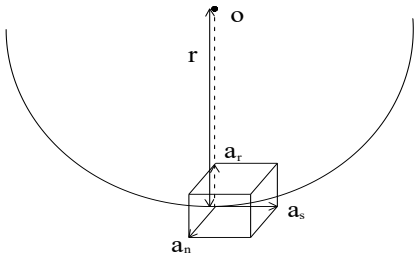


Figure 3.22 Acceleration components in natural coordinates

1) Tangential acceleration

$$a_s = \frac{dV_s}{dt} = \frac{\partial V_s}{\partial s}\frac{ds}{dt} + \frac{\partial V_s}{\partial t} = \frac{\partial V_s}{\partial t} + V_s\frac{\partial V_s}{\partial s} = \frac{\partial V_s}{\partial t} + \frac{1}{2}\frac{\partial (V_s^2)}{\partial s} \tag{3.30}$$

2) Centrifugal acceleration

$$a_r = \frac{dV_r}{dt} = \frac{\partial V_r}{\partial s}\frac{ds}{dt} + \frac{\partial V_r}{\partial t} \qquad (3.31)$$

$$\frac{\partial V_r}{\partial s} ds = (V_s + \frac{\partial V_s}{\partial s} ds) \sin d\theta$$

$d\theta$ is very small so $\sin d\theta = \frac{ds}{r}$ where r radius of curvatureof the streamline,

$$\frac{\partial V_r}{\partial s} ds = (V_S + \frac{\partial V_S}{\partial s} ds)\frac{ds}{r}$$

$$\frac{\partial V_r}{\partial s} ds = \frac{V_S}{r} ds + \frac{1}{r}\frac{\partial V_S}{\partial s}(ds)^2$$

ds^2 is second order term so it is very small,

$$\frac{\partial V_r}{\partial s} = \frac{V_S}{r}$$

if this equation is substituted in Eq.(3.31)

$$a_r = \frac{V_S^2}{r} + \frac{\partial V_r}{\partial t} \qquad (3.32)$$

3) Normal acceleration (it is in the plane)

$$a_n = \frac{dV_n}{dt} = \frac{\partial V_n}{\partial t} \qquad (3.33)$$

Streamline (tangential) and centrifugal components of acceleration occur even in steady flows.

ii) In terms of rectangular components, the acceleration vector may be written;

$$a = a_X \vec{i} + a_y \vec{j} + a_Z \vec{k}$$

In the Eulerian method a velocity vector at the location (x, y, z) is function of the local coordinates and time

$$\left.\begin{aligned} &\vec{V} = [u, v, w] \\ &u = u(x, y, z, t) \\ &v = v(x, y, z, t) \\ &w = w(x, y, z, t) \end{aligned}\right\}$$

The acceleration is the time rate of change of the velocity of a given particle; hence the material derivative may be used. Thus, application of the chain rule gives.

$$\Delta u = \frac{\partial u}{\partial x}\Delta x + \frac{\partial u}{\partial y}\Delta y + \frac{\partial u}{\partial z}\Delta z + \frac{\partial u}{\partial t}\Delta t$$

$$\frac{\Delta u}{\Delta t} = \frac{\Delta x}{\Delta t}\frac{\partial u}{\partial x} + \frac{\Delta y}{\Delta t}\frac{\partial u}{\partial y} + \frac{\Delta z}{\Delta t}\frac{\partial u}{\partial z} + \frac{\Delta t}{\Delta t}\frac{\partial u}{\partial t}$$

Δx, Δy, Δz and Δt goes to zero for limit case;

DRAFT COPY

$$\frac{\Delta u}{\Delta t} \to \frac{du}{dt}, \quad \frac{\Delta x}{\Delta t} \to u, \quad \frac{\Delta y}{\Delta t} \to v, \quad \frac{\Delta z}{\Delta t} \to w$$

$$\frac{du}{dt} = u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z} + \frac{\partial u}{\partial t} \qquad \text{in x direction} \tag{3.34}$$

when similar concepts are used to the other components of acceleration

$$\frac{dv}{dt} = u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial v}{\partial z} + \frac{\partial v}{\partial t} \qquad \text{in y direction} \tag{3.35}$$

$$\underset{\text{(Total,material)}}{\frac{dw}{dt}} = \underbrace{u\frac{\partial w}{\partial x} + v\frac{\partial w}{\partial y} + w\frac{\partial w}{\partial z}}_{\text{(convective)}} + \underset{\text{(Local)}}{\frac{\partial w}{\partial t}} \qquad \text{in z direction} \tag{3.36}$$

Resultant acceleration,

$$a = a_X\vec{i} + a_y\vec{j} + a_Z\vec{k}$$

$$\vec{a} = \frac{D\vec{V}}{Dt}$$

In cylindrical coordinates(r, θ, z) with velocity components v_r, v_θ, v_z, the acceleration coordinates are:

$$\left.\begin{aligned} a_r &= v_r\frac{\partial v_r}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_r}{\partial \theta} + v_z\frac{\partial v_r}{\partial z} - \frac{v_\theta^2}{r} + \frac{\partial v_r}{\partial t} \\ a_\theta &= v_r\frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_\theta}{\partial \theta} + v_z\frac{\partial v_\theta}{\partial z} + \frac{v_r v_\theta}{r} + \frac{\partial v_\theta}{\partial t} \\ a_z &= v_r\frac{\partial v_z}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_z}{\partial \theta} + v_z\frac{\partial v_z}{\partial z} + \frac{\partial v_z}{\partial t} \end{aligned}\right\} \tag{3.37}$$

DRAFT COPY

4
Fluid Dynamics

4.1. Introduction

In general, the velocity of a fluid particle varies from point to point in a flow field. Since force is required to change velocity, the pressure must vary from point to point. The relationship between the changes in pressure and velocity in a flow field is described by Euler equation, named after the Swiss mathematician Leonard Euler (1707-83).

In this chapter, we will provide firstly an introduction to the differential equations that describe (in detail) the motion of fluids. Unfortunately, we will also find that these equations are rather complicated, partial differential equations that cannot be solved exactly except in a few cases, at least without making some simplifying assumptions. There are some exact solutions for laminar flow. In addition, by making some simplifying assumptions many other analytical solutions can be obtained. For example, in some circumstances it may be reasonable to assume that the effect of viscosity is small and can be neglected which is called inviscid flow.

It is known that for certain types of flows, the flow field can be conceptually divided into two regions; a very thin region near the boundaries of the system in which viscous effects are important, and a region away from the boundaries in which the flow is essentially inviscid.

Second introduction is that the finite control volume analysis describes the fluid motions. Many fluid mechanics problems can be solved using control volume analysis. The finite control volume formulas are easy to interpret physically and thus are not difficult to use.

4.2. Forces

i) *Surface Forces*; which act on the surface of a differential element. There are two components, which are normal force and shear (tangential) force. Stress is force per unit area. Stress at a point can be described by defining two vectors; an elementary area around the point and the resultant force acting across that area.

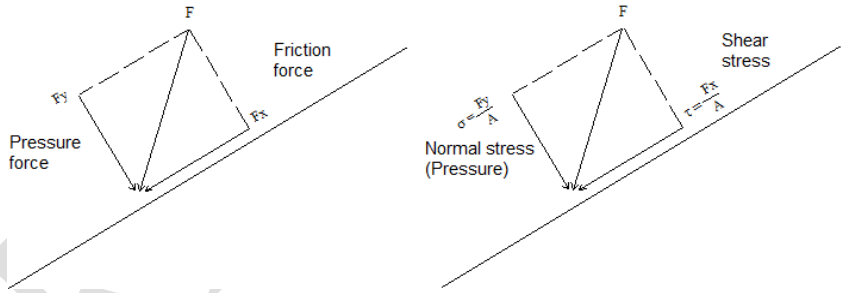


Figure 4.1 Surface forces

ii) *Body Forces*; these forces are distributed through the volume of the element, they are proportional to either its mass or volume such as gravitational and electromagnetic forces, and their resultant acts through the center of the element.

iii) *Inertia Force*; It is the force developed due to the movement of the fluid element.

iv) *Elastic Force*; It is important for compressible fluids and also unsteady flows.

4.2.1. Fundamental Principles

Basis of the analysis method is some fundamental principles of physics, namely,

i) Conservation of mass gives *the Continuity Equation.*

ii) Conservation of momentum gives *the Dynamic Equations or Equations of Motions* obtained in the coordinate directions. If they are written on the orthogonal planes or in the Cartesian (or rectangular) coordinate directions, three equations are found. They are classified into three different categories according to type of fluid flows;

- Euler equations are valid for inviscid fluid flow or frictionless flow,
- Navier-Stokes Equations are valid for viscous fluid flow (Newtonian fluids),
- Reynolds Equations are valid for turbulent flow.

iii) Conservation of energy (first law of thermodynamics) gives *the energy equation*.

In this chapter, first we will study the motion of a particle of an inviscid (non-viscous) fluid leading to the development of Euler and Bernoulli equations. Then the energy equation for a real fluid will be developed.

DRAFT COPY

4.3. Inviscid Fluid Dynamics

4.3.1. Continuity Equation

The Law of Conservation of Mass states that in the absence of a nuclear reaction, matter can neither be created nor destroyed. When applied to a control volume, it would mean that;

$$\left\{\begin{array}{l}\text{Rate of mass flowing (i.e., mass flux)}\\ \text{into the control volume}\end{array}\right\}-\left\{\begin{array}{l}\text{Rate of mass flowing}\\ \text{out of the control volume}\end{array}\right\}=\left\{\begin{array}{l}\text{Rate of mass accumulating}\\ \text{in the control volume}\end{array}\right\} \qquad (4.1)$$

The continuity equation is a mathematical statement of the Law of Conservation of Mass.

a) The Continuity equation is now derived for an elementary rectangular control volume of a fluid shown in Fig. 4.2. Conservation of the fluid mass requires that the mass flux entering the control volume minus the mass flux leaving the control volume must be equal to the rate of change of mass inside the control volume. Figure 4.2 shows the mass flux across the six faces of the control volume.

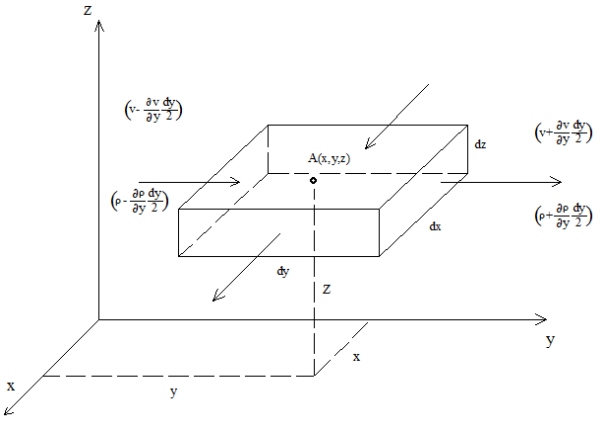


Figure 4.2 Control volume

According to a point A (x, y, z) in the control volume, mass flux in y direction is

i) The mass flux entering the CV in unit time;

$$\left(v-\frac{\partial v}{\partial y}\frac{dy}{2}\right)\left(\rho-\frac{\partial \rho}{\partial y}\frac{dy}{2}\right)dxdz = v\rho dxdz - v\frac{\partial \rho}{\partial y}\frac{dy}{2}dxdz - \rho\frac{\partial v}{\partial y}\frac{dy}{2}dxdz$$

ii) The mass flux leaving the CV in unit time;

$$\left(v+\frac{\partial v}{\partial y}\frac{dy}{2}\right)\left(\rho+\frac{\partial \rho}{\partial y}\frac{dy}{2}\right)dxdz = v\rho dxdz + v\frac{\partial \rho}{\partial y}\frac{dy}{2}dxdz + \rho\frac{\partial v}{\partial y}\frac{dy}{2}dxdz$$

iii) The net mass flux in the y-direction is the difference of ((i)-(ii))

$$-v\frac{\partial \rho}{\partial y}dxdydz - \rho\frac{\partial v}{\partial y}dxdydz = -\left(v\frac{\partial \rho}{\partial y}+\rho\frac{\partial v}{\partial y}\right)dxdydz = -\frac{\partial(v\rho)}{\partial y}dxdydz$$

Similarly;

$$-\frac{\partial(\rho u)}{\partial x}dxdydz \qquad \text{in x direction}$$

$$-\frac{\partial(\rho w)}{\partial z}dxdydz \qquad \text{in z direction}$$

iv) Total net mass flux in dt time;

$$-\left(\frac{\partial(\rho u)}{\partial x}+\frac{\partial(\rho v)}{\partial y}+\frac{\partial(\rho w)}{\partial z}\right)dxdydzdt$$

v) The rate of change of mass in dt time inside the control volume;

$$\frac{\partial \rho}{\partial t}dxdydzdt = \frac{\partial m}{\partial t}dt$$

According to the Law of Conservation of Mass: (iv) = (v)

DRAFT COPY

$$-\left(\frac{\partial(\rho u)}{\partial x}+\frac{\partial(\rho v)}{\partial y}+\frac{\partial(\rho w)}{\partial z}\right)dxdydzdt=\frac{\partial \rho}{\partial t}dxdydzdt$$

$$\boxed{\frac{\partial \rho}{\partial t}+\frac{\partial(\rho u)}{\partial x}+\frac{\partial(\rho v)}{\partial y}+\frac{\partial(\rho w)}{\partial z}=0} \qquad (4.2)$$

This equation is commonly referred to as the Continuity Equation. If the Continuity Equation is written in vector notation;

$$\frac{d\rho}{dt}+\text{div}(\rho\vec{V})=0 \qquad (4.3)$$

This is the general form of the continuity equation applicable to compressible and incompressible fluids under steady and unsteady flow conditions. For an incompressible fluid, the density may be assumed constant. Also, for steady flow $\partial\rho/\partial t=0$. Therefore, the continuity equation for the steady flow of an incompressible fluid may be written as

$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}+\frac{\partial w}{\partial z}=0 \quad \text{or} \quad \text{div}(\vec{V})=0 \qquad (4.4)$$

and in a steady flow $\dfrac{\partial \rho}{\partial t}=0$

$$\frac{\partial(\rho u)}{\partial x}+\frac{\partial(\rho v)}{\partial y}+\frac{\partial(\rho w)}{\partial z}=0 \text{ or } \text{div}(\rho\vec{V})=0 \qquad (4.5)$$

b) To define the continuity Equation in natural coordinate systems for steady flow, the equation for a section of pipe or a duct may be derived by setting up a control volume as shown in Fig. 4.3. Assume that the fluid enters the control volume through section 1 and leaves it through section 2. There is no flow across the control volume.

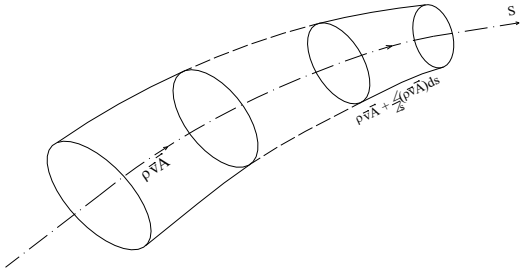


Figure 4.3 Stream tube in a pipe

According to the Law of Conservation of Mass:

$$\rho\vec{V}\vec{A}-\left[\rho\vec{V}\vec{A}+\frac{\partial}{\partial s}(\rho\vec{V}\vec{A})ds\right]=\frac{\partial}{\partial t}(\rho\vec{A})ds$$

$$\frac{\partial(\rho\vec{A})}{\partial t}+\frac{\partial}{\partial s}(\rho\vec{V}\vec{A})=0 \qquad (4.6)$$

For steady flow

$$\frac{\partial(\rho\vec{A}\vec{V})}{\partial s}=0 \qquad (4.7)$$

Hence,

Assuming that the density and the velocity at a section are constant, the above equation becomes

$m=\rho\vec{A}\vec{V}$ continuity equation for one dimensional incompressible fluid flow

$$m=\rho\vec{V}\vec{A}=\int\rho u dA=\rho V_1A_1=\rho V_2A_2=\text{constant} \qquad (4.8)$$

where V is the average velocity at cross-sectional area A. For an incompressible fluid and steady flow, the density is constant and

DRAFT COPY

$$Q = \vec{V}\vec{A} = \int udA = V_1A_1 = V_2A_2 = \text{constant} \tag{4.9}$$

where Q is called volumetric discharge or shortly discharge or volumetric flow rate (m^3/s).

4.3.2 Equations of Motion (Euler and Bernoulli Equations)

The equation of motion, or the dynamic equations, are a statement of Newton's second law of motion F=ma in differential form in terms of the properties of the motion at a point, that is using an Eulerian description.

a) Equation of Motion in Cartesian Coordinates

Consider a rectangular differential volume and the net forces acting on the differential volume are given in Fig. 4.5. Forces acting on the volume may be classified as body forces and surface forces. The body forces in the Cartesian coordinate directions per unit of mass are X, Y, Z and their units are given in acceleration. Surface forces include only normal forces; it is pressure, since Euler's equations are obtained for a frictionless flow. In frictionless flow, shear stresses don't present the surface forces are due to pressure.

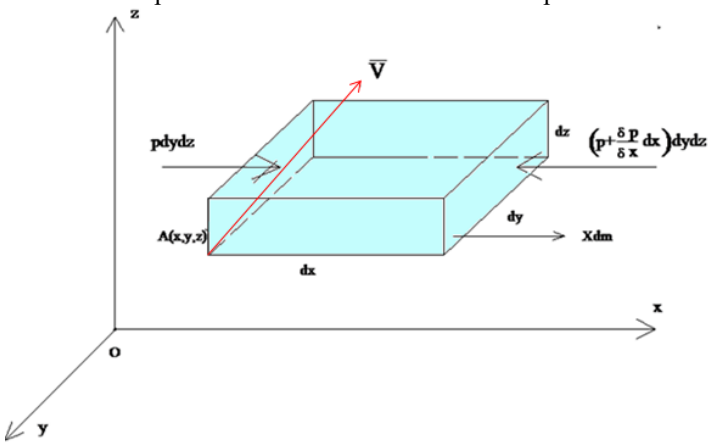


Figure 4.5 Control volume

Forces acting in the x-direction may be written using Newton's second law as follows.

$$pdydz - \left(p + \frac{\partial p}{\partial x}dx\right)dydz + X\,dm = dm\,a_x$$

$$pdydz - pdydz - \frac{\partial p}{\partial x}dxdydz + X\rho dxdydz = \rho dxdydz\frac{du}{dt}$$

$$\frac{du}{dt} = X - \frac{1}{\rho}\frac{\partial p}{\partial x} \tag{4.10}$$

Similarly
in the y-direction and the z-direction, respectively

$$\frac{dv}{dt} = Y - \frac{1}{\rho}\frac{\partial p}{\partial y} \tag{4.11}$$

$$\frac{dw}{dt} = Z - \frac{1}{\rho}\frac{\partial p}{\partial z} \tag{4.12}$$

$$\left.\begin{aligned}
\frac{1}{\rho}\frac{\partial p}{\partial x} &= X - \left(\frac{\partial u}{\partial t} + \frac{\partial u}{\partial x}u + \frac{\partial u}{\partial y}v + \frac{\partial u}{\partial z}w\right)\\
\frac{1}{\rho}\frac{\partial p}{\partial y} &= Y - \left(\frac{\partial v}{\partial t} + \frac{\partial v}{\partial x}u + \frac{\partial v}{\partial y}v + \frac{\partial v}{\partial z}w\right)\\
\frac{1}{\rho}\frac{\partial p}{\partial z} &= Z - \left(\frac{\partial w}{\partial t} + \frac{\partial w}{\partial x}u + \frac{\partial w}{\partial y}v + \frac{\partial w}{\partial z}w\right)
\end{aligned}\right\} \tag{4.13}$$

DRAFT COPY

These equations are called Euler's Equations of motion. Multiplying Euler's Equations in each direction by dx, dy and dz, and then they are summed, we may get the general equation of hydraulics.

$$\frac{du}{dt}dx = Xdx - \frac{1}{\rho}\frac{\partial p}{\partial x}dx$$

$$\frac{dv}{dt}dy = Ydy - \frac{1}{\rho}\frac{\partial p}{\partial y}dy$$

$$\frac{dw}{dt}dz = Zdz - \frac{1}{\rho}\frac{\partial p}{\partial z}dz$$

$$\frac{du}{dt}dx + \frac{dv}{dt}dy + \frac{dw}{dt}dz = Xdx + Ydy + Zdz - \frac{1}{\rho}\left(\frac{\partial p}{\partial x}dx + \frac{\partial p}{\partial y}dy + \frac{\partial p}{\partial z}dz\right)$$

$$udu + vdv + wdw = Xdx + Ydy + Zdz - \frac{1}{\rho}dp$$

If the resultant velocity vector is written as follows;

$$\vec{V} = V(u, v, w)$$

$$V^2 = u^2 + v^2 + w^2$$

$$2VdV = 2udu + 2vdv + 2wdw$$

$$2VdV = 2(udu + vdv + wdw)$$

$$VdV = (udu + vdv + wdw)$$

and instead of velocity components, resultant velocity vector subsitutes in the above equation,

$$VdV = Xdx + Ydy + Zdz - \frac{1}{\rho}dp \qquad \text{or}$$

$$\boxed{\frac{1}{\rho}dp = Xdx + Ydy + Zdz - VdV} \qquad (4.14)$$

This equation is called "*the general equation of hydraulics*".

By integrating Euler's equation or general equation of hydraulics (4.14), we get the Bernoulli equation. The Bernoulli equation describes the motion of a fluid particle along a streamline when the flow is steady and the fluid is incompressible and inviscid (commonly called ideal fluids). It can be obtained from the Eq.(4.14) by substituting the conditions for a steady flow an incompressible fluid.

X=0, Y=0, Z=-g

$$\frac{1}{\rho}dp = -gdz - VdV$$

$$\frac{1}{\rho}p = -gz - \frac{1}{2}V^2 + C$$

$$\frac{1}{2}V^2 + gz + \frac{1}{\rho}p = C$$

$$\frac{V^2}{2g} + z + \frac{p}{\gamma} = \frac{C}{g} = C'$$

$$\frac{V^2}{2g} + \frac{p}{\gamma} + z = \text{constant} \qquad (4.15)$$

This is commonly known as the *Bernoulli equation*. In this equation, each term has units of length. They may be also interpreted as energy per unit weight of the fluid. Then, the first term $V^2/2g$ is the energy per unit weight of the fluid due to its velocity, and is called the *velocity head*. The second term p/γ is the energy per unit weight of the fluid due to its pressure and is called the *pressure head*. The third term z is the energy per unit weight of the fluid due to its elevation and is called the *elevation head*.

The sum of these heads remain constant (equal to H) along the streamline, but it may vary from the streamline to streamline.

DRAFT COPY

The assumptions by which the Bernoulli equation holds are;

a) Steady flow,

b) Incompressible flow,

c) Frictionless flow,

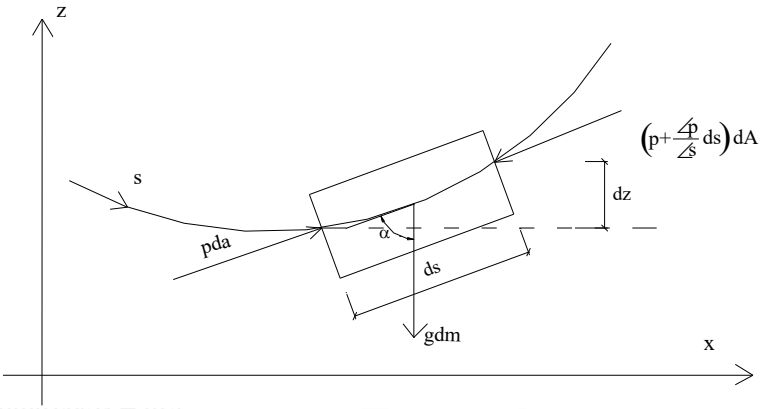
d) Flow along a streamline,

e) It can be applied throughout a flow field provided that the flow is irrotational. In other words, the Bernoulli equation remains constant from streamline to streamline in an irrotational flow field. Thus, for irrotational flow it can be written any two points in the flow area as below:

$$\frac{V_1^2}{2g}+\frac{p_1}{\gamma}+z_1=\frac{V_2^2}{2g}+\frac{p_2}{\gamma}+z_2$$

b) Equation of Motion in Natural Coordinates

In s direction;



$$pdA-\left(p+\frac{\partial p}{\partial s}ds\right)dA-dmg\cos\alpha=ma_s$$

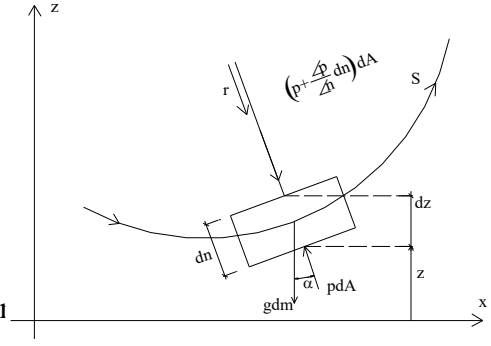
$$-\frac{\partial p}{\partial s}ds\,dA-\rho g\,ds\,dA\cos\alpha=\rho ds dA\left(\frac{\partial v_s}{\partial t}+\frac{1}{2}\frac{\partial\left(v_s^2\right)}{\partial s}\right)$$

$\frac{\partial v_s}{\partial t}=0$ and $\cos\alpha=\frac{dz}{ds}$ for steady flow

$$\frac{1}{\rho}\frac{dp}{ds}-g\frac{dz}{ds}=\frac{1}{2}\frac{d\left(v_s^2\right)}{ds}$$

$$\frac{1}{2}\frac{d\left(v_s^2\right)}{ds}+\frac{1}{\rho}\frac{dp}{ds}+g\frac{dz}{ds}=0 \qquad (4.16)$$

in n direction



$$pdA-\left(p+\frac{\partial p}{\partial n}dn\right)dA-gdı$$

$$-\frac{\partial p}{\partial n}dndA-\rho gdndA\cos\alpha=\left(\frac{\partial v_r}{\partial t}+\frac{v_s^2}{r}\right)\rho dndA$$

DRAFT

$$\frac{\partial v_r}{\partial t} = 0 \text{ and } \cos\alpha = \frac{dz}{dn} \qquad \text{for steady flow}$$

$$-\frac{1}{\rho}\frac{dp}{dn} - g\frac{dz}{dn} = \frac{v_s^{\,2}}{r}$$

$$\frac{v_s^{\,2}}{r} + \frac{1}{\rho}\frac{dp}{dn} + g\frac{dz}{dn} = 0 \qquad (4.17)$$

In a curvilinear flow, pressure distribution deviates from the hydrostatic pressure law. Fig.4.9 shows the pressure distributions in concave and convex flows.

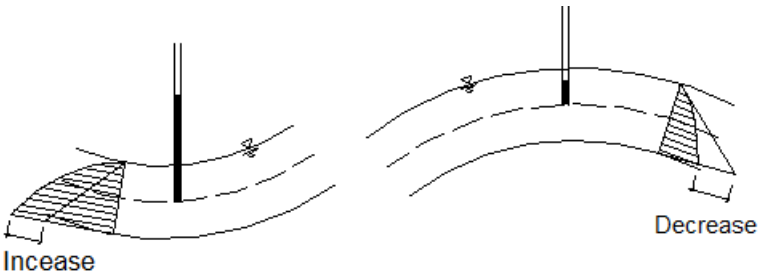


Figure 4.9 Pressure distributions for concave and convex flows

4.3.3 Energy Equation

The total energy of a mass of fluid consists of:

1. *Kinetic energy* or energy due to the motion of the fluid.
2. *Potential energy* or energy due to the position of the fluid, usually gravitational energy but it may also be electrical or magnetic.
3. *Internal energy* or energy due to the kinetic and potential energies of the molecules of the fluid. It is a function of the internal state of the fluid system only and does not depend on the position and the velocity of the system as a whole.

Flow work is the work done by pressure forces as the system moves through space. Let us consider the basic figure 4.10 depicting the system and control volume to get a better understanding of flow work.

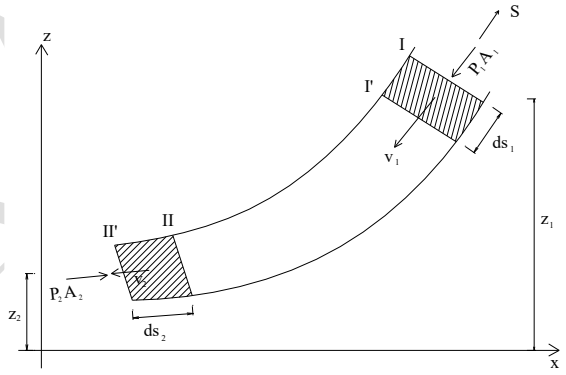


Figure 4.10 Forces on control volume

Consider the area A_2, which is the left end of the fluid system. The force acting on surrounding fluid will be p_2A_2, and the distance traveled by the area in the time Δt will be $\Delta s_2 = V_2\Delta t$. The work done on the surrounding fluid because of this force in time Δt will be the product of the component of the force in the direction of motion (p_2A_2) and the distance by the area ($V_2\Delta t$).

(Work done by pressure force) + (Work done by gravity force) = (Energy due to the motion of the fluid) (4.18)

Energy due to pressure force : $E_b = p_1A_1ds_1 - p_2A_2ds_2 = \forall(p_1 - p_2)$

Energy due to position (or elevation) : $E_p = mg(z_1 - z_2) = \rho\forall g(z_1 - z_2) = \gamma\forall(z_1 - z_2)$

DRAFT COPY

Energy due to the kinetic : $E_k = m\left(\frac{V_2^2 - V_1^2}{2}\right) = \rho\forall\frac{V_2^2 - V_1^2}{2} = \gamma\forall\frac{V_2^2 - V_1^2}{2g}$

If the energy definitions substitute in the principle (4.18)

$$E_k = E_b + E_p$$

where $\forall$ is volume and it is constant

$$\gamma\forall\frac{V_2^2 - V_1^2}{2g} = \forall(p_1 - p_2) + \gamma\forall(z_1 - z_2)$$

$$\frac{V_2^2 - V_1^2}{2g} = z_1 - z_2 + \frac{p_1 - p_2}{\gamma}$$

$$\frac{V_1^2}{2g} + \frac{p_1}{\gamma} + z_1 = \frac{V_2^2}{2g} + \frac{p_2}{\gamma} + z_2 \qquad (4.19)$$

This result is known as "*Bernoulli's equation*". The first term is the kinetic energy per unit weight of fluid called velocity head, the second term represents the work done in pushing a body of fluid by fluid pressure and is known as pressure energy per unit weight of fluid called pressure head, the third term is the elevation or potential energy per unit weight of fluid with respect to an arbitrary datum called elevation or potential head. So Bernoulli's equation is a summation of the energy terms.

Kinetic energy + pressure energy + potential (elevation) energy=constant (4.20)

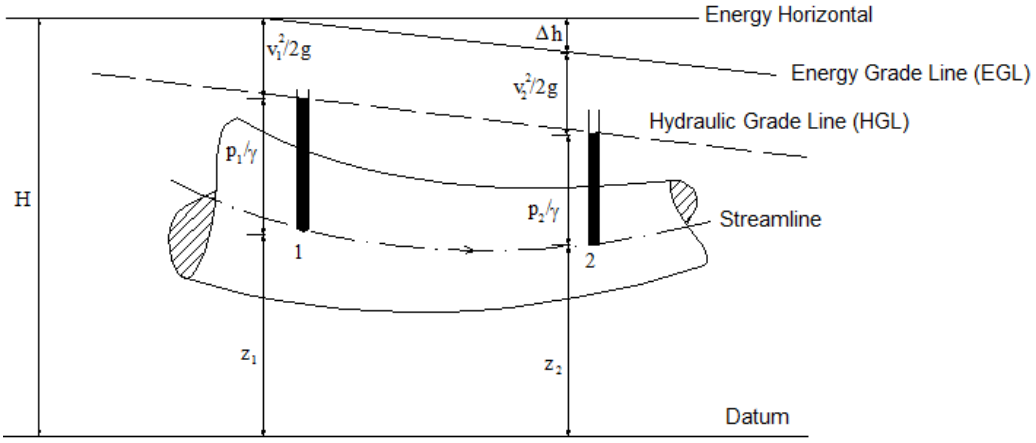


Figure 4.11 Graphical description of Bernoulli's equation

4.3.4 Practical Applications of Bernoulli's Equations

Various flow meters are governed by the Bernoulli and Continuity equations.

Venturimeter

The goal of venturimeter is placing of some type of restriction within the pipe as shown in Fig. 4.12 and to measure the pressure difference between the low velocity, high-pressure upstream section (1), and the high-velocity, low pressure downstream section (2).

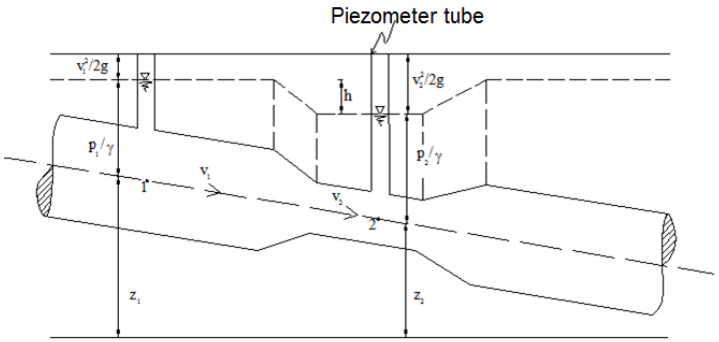


Figure 4.12 Venturimeter

The Bernoulli's equation becomes;

$$\frac{p_1}{\gamma}+\frac{V_1^2}{2g}+z_1=\frac{p_2}{\gamma}+\frac{V_2^2}{2g}+z_2$$

$$\frac{p_1-p_2}{\gamma}+(z_1-z_2)=\frac{V_2^2-V_1^2}{2g}$$

the left hand side of above equation shows the difference of piezometric elevations, h.

$$h=\frac{V_2^2-V_1^2}{2g}$$

If the continuity equation substitutes in the above equation or combination of these two equations, results in the following theoretical flow rate.

$$h=\frac{Q^2}{2g}\left(\frac{1}{A_2^2}-\frac{1}{A_1^2}\right)=\frac{Q^2}{2gA_1^2}\left(\left(\frac{A_1}{A_2}\right)^2-1\right)$$

$$Q=A_1\sqrt{\frac{2gh}{\left(\frac{A_1}{A_2}\right)^2-1}} \qquad (4.21)$$

This equation gives theoretical discharge, in practice the discharge is less than above Eq. So a coefficient must be inserted to Eq. (4.21) to get discharge for real fluid flow which is due to the head loss between sections (1) and (2) in venturimeter.

$$Q=C_dA_1\sqrt{\frac{2gh}{\left(\frac{A_1}{A_2}\right)^2-1}} \qquad (4.22)$$

where C_d is discharge coefficients. The flow rate is a function of the pressure difference across the flow meter.

Pitot tube

Pitot tube is used to measure fluid flow velocity. A pitot tube in its simplest form is an L-shaped tube held against the flow as shown in Fig.4.13, creating a stagnation point in the flow. Inner tube gives total pressure and outer tube gives static pressure and hence the difference between the water levels in the tubes gives dynamic pressure.

DRAFT COPY

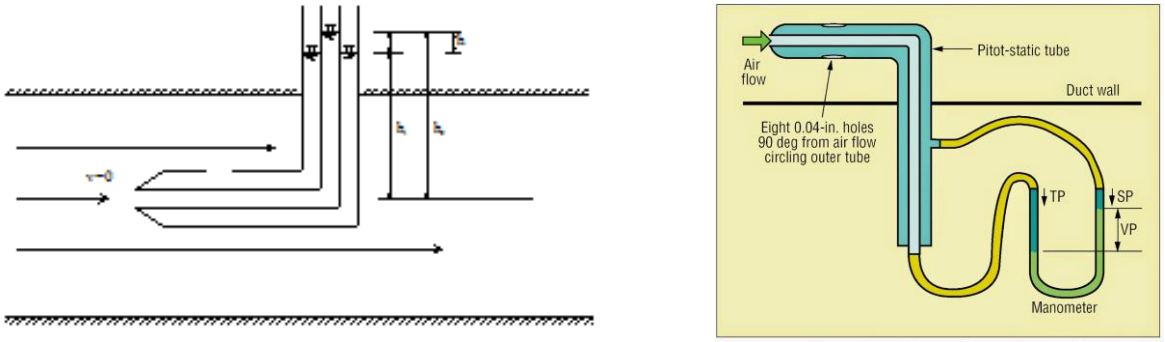


Figure 4.13 Pitot tube

Static pressure $p_s=\gamma h_1$ (outer tube) (4.23)

Total pressure $p_t=\gamma h_2$ (inner tube) (4.24)

Dynamic pressure $p_d=p_t-p_s=\gamma h_2-\gamma h_{1=\gamma}h$ (4.25)

Since the dynamic pressure head equals to velocity head, Eq. (4.25) may be written in the following form.

$$h=\frac{V^2}{2g} \tag{4.26}$$

so flow velocity may be found from Eq. (4.26)

$$V=\sqrt{2gh} \tag{4.27}$$

so pitot tube is a device which measures the fluid flow velocity.
The stagnation pressure may be also defined by considering a bridge pier in Fig. 4.14, velocity is zero at the stagnation point where velocity head is compensated by rising the water level which gives total pressure at point 2 equals static and dynamic pressure.

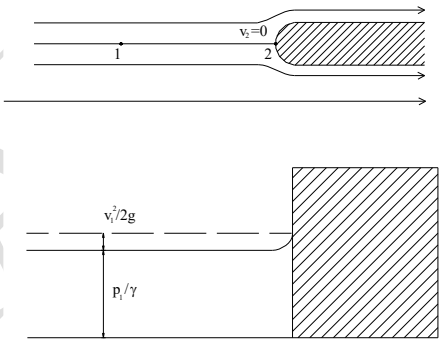


Figure 4.14 Stagnation point and dynamic pressure

If Bernoulli equation is written between the points (1) and (2):

$$\frac{p_1}{\gamma}+\frac{V_1^2}{2g}=\frac{p_2}{\gamma}$$

$$p_1+\rho\frac{V_1^2}{2}=p_2$$

Since conservation of energy along the streamline due to ideal fluid, p_2 presents the total pressure which is the sum of static and dynamic pressure.

4.4. Impulse-Momentum Equation

In the previous chapters we applied Newton's second law of motion to a particle of fluid moving along a streamline and derived the Euler and Bernoulli equations. In this chapter, we will apply Newton's second law of motion to a mass of particles of fluid to obtain the momentum equation. The momentum equation may be used to analyze forces exerted on a solid surface. Although in developing the Euler equation, we used only one small fluid system (differential fluid volume), we now define the fluid system to include all of the fluid in a specified

DRAFT COPY

control volume at a particular time. Although shear stress is not explicitly included in the development, the equation we derive will apply equally well to real fluids as well as ideal fluids. However some adjustments must be made for real fluids due to velocity profiles.

4.4.1 Linear Impulse-Momentum Equation

A flying baseball can simply be caught with a glove. A moving automobile however is difficult to stop in a short time. Therefore, the velocity is not sufficient to study the effects of bodily motion, but the product, mV, of the mass m and the velocity V can be used as an indicator of the consequences of motion. This is called the linear momentum.

Momentum of a body is the product of its mass and velocity and Newton's second law of motion states that the resultant external force acting on anybody in any direction equal to the rate of change of momentum of the body in that direction.

$$\Sigma\vec{F} = \frac{d\left(m\vec{V}\right)}{dt} \tag{4.28}$$

where (mV) is linear momentum, and (ΣF)dt is the impulse in the time dt from the sum of all external forces acting on the control volume. Eq.(4.28) is known as impulse-momentum equation.

$$\Sigma\vec{F}dt = m\int_{V_1}^{V_2} d\vec{V}$$

$$\Sigma\vec{F}dt = m\vec{V_2} - m\vec{V_1}$$

$$\boxed{\Sigma\vec{F}dt + m\vec{V_1} = m\vec{V_2}} \tag{4.29}$$

In solid mechanics, the impulse applied to a body is equal to the change of momentum for given time interval. The directly analogous situation for fluid flow is case where liquid in a pipe is accelerated by means of a pressure differential along the pipe. Both cases may be termed simple unsteady-state cases, for which the basic equation of linear momentum is applicable. When we have non-uniform flow, which may be either steady or unsteady, the process becomes more complicated, and we turn to basic control volume approach to develop the momentum equations.

Linear momentum equations may be used to calculate the force exerted on the moving fluid. According to these equations, in a steady flow, the net force on the fluid contained within the control volume is equal to the net momentum flux leaving the control volume. Of course, the force and the momentum must have the same direction. It may be noticed that only the conditions at the inlet and the outlet to the control volume enter into the calculations. Whatever occurs inside the control volume does not concern us.

$m_{inflow} = \rho Q_1.dt = \rho V_1 A_1 dt$

$m_{outflow} = \rho\, Q_2.dt = \rho V_2 A_2 dt$

$$\Sigma\vec{F}dt = \rho Q_2\vec{V_2}dt - \rho Q_1\vec{V_1}dt$$

$$\Sigma\vec{F} + \rho Q_1\vec{V_1} = \rho Q_2\vec{V_2} \tag{4.30}$$

It is often convenient to use three scalar equations in the x, y, and z coordinate directions in rectangular coordinate system instead of a single vector equation. Thus the momentum equation in vectorial form modified for the case of uniform velocity across flow sections, is written as follows in these directions:

$$\Sigma F_x + (\rho Q_1 V_1)_x = (\rho Q_2 V_2)_x \tag{4.31}$$

$$\Sigma F_y + (\rho Q_1 V_1)_y = (\rho Q_2 V_2)_y \tag{4.32}$$

Steps of Application of Impulse-Momentum Equations

1. Draw a control surface involves the changing of velocity vector i.e., magnitude and/or direction of velocity.
2. Define the external forces on the control volume. Pressure and impulse forces will act in the direction of fluid flow at the inlet and these forces act in the opposite direction of fluid flow at the outlet. Gravity force acts on the CV in downward, however if the system is on the horizontal plane, gravity force may be neglected. Reaction force will act on CV in any direction.
3. Select a coordinate system, (x, y).
4. Impulse-Momentum equations will be set in the coordinate directions.

DRAFT COPY

5. Check the knowns and unknowns in the equations.
6. If there are unknowns such as Q, V, p, z, D in the momentum equations the continuity and Bernoulli's equations are used.
7. If there is a jet flow, pressure force may be neglected because diameters of the jets are generally small.
8. Remember that the forces at the end sections of the control volume act towards the sections.

Example:
The 2m pipe carrying 6.4 m^3/s of water is connected to a 1 m pipe by a standard reducer fitting as is shown in Fig. 4.15. Pressure is measured at section I-I 24 N/cm^2. What force is exerted by the water on the reducer, neglecting any lost head? The pipe is on the horizontal plane.

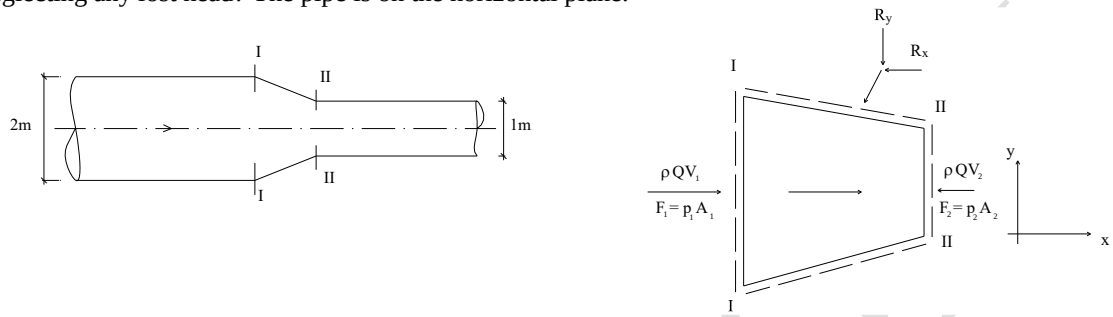


Figure 4.15

$\sum F_x = 0$ $\rho QV_1+p_1A_1-\rho QV_2- p_2A_2-R_x=0$

$\sum F_y = 0$ $R_y=0$

Unknowns are V_1, V_2, p_2, R_x

Continuity Equation is $Q=V_1A_1= V_2A_2$

$$V_1 = \frac{Q}{A_1} = \frac{6.4}{\frac{\pi \times 2^2}{4}} = 2.04 m/s \quad V_2 = \frac{Q}{A_2} = \frac{6.4}{\frac{\pi \times 1^2}{4}} = 8.16 m/s$$

Bernoulli Equation is

$$\frac{p_1}{\gamma} + \frac{V_1^2}{2g} + \cancel{z_1} = \frac{p_2}{\gamma} + \frac{V_2^2}{2g} + \cancel{z_2}$$

$$\frac{p_2}{\gamma} = \frac{V_1^2}{2g} - \frac{V_2^2}{2g} + \frac{p_1}{\gamma}$$

$$\frac{p_2}{\gamma} = \frac{2.04^2}{19.62} - \frac{8.16^2}{19.62} + \frac{24\times10^4}{10^4} \qquad \frac{p_2}{\gamma} = 21 mwc$$

$$1000\times6.4\times2.04 + 240000\times\frac{\pi}{4}\times(2)^2 - 1000\times6.4\times8.16 - 210000\times\frac{\pi}{4}\times1^2 - R_x = 0$$

$$R_x = 56\times10^4 N \qquad R = R_x = 56\times10^4 N$$

An important application of the momentum equation is the deflectors (or vanes) of pumps, turbines, or compressors. The applications involve both stationary deflectors and moving deflectors. The following assumptions are made for:

1. The frictional force between the fluid and the deflector is negligible.
2. The pressure is assumed to be constant as the fluid moves over the deflector.
3. The body force is assumed to be negligible.
4. The effect of the lateral spreading of the fluid stream is neglected

Force on a Stationary Vertical Flat Plate

If a jet of fluid strikes normally at a flat plate, the plate would experience a force which can be calculated using the momentum equation.

DRAFT COPY

Consider a jet of fluid striking a plate as shown in Fig.4.15a. Establish a control volume as shown in figure. Applying the momentum equation in the x-direction:
Since the jet is horizontal and the plate is vertical

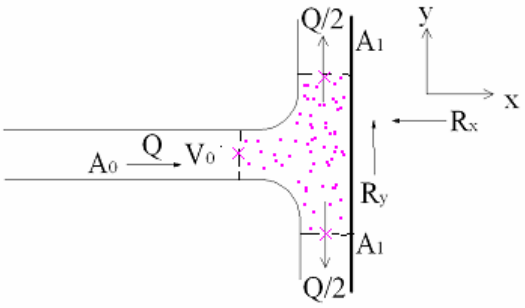


Figure 4.15a

$$\frac{P_0}{\gamma}+\frac{V_0^2}{2g}+z_0=\frac{P_1}{\gamma}+\frac{V_1^2}{2g}+z_1$$

$V_0=V_1$

Inflow discharge=Q, Outflow discharges =Q/2

$$R_x=\rho Q V_0$$

Force on a Stationary Inclined Flat Plat

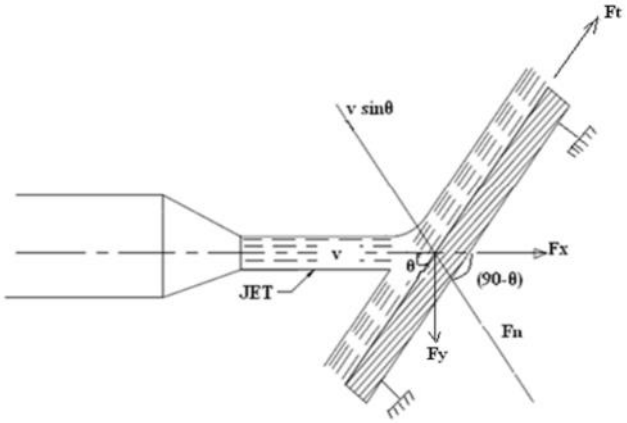


Figure.2.

Force on a Stationary Inclined Curved Plate

When a free jet impinges onto a smooth vane that is curved as fig. 4.15c, the jet is deflected, its momentum is changed, and a force is exerted on the vane. The jet is assumed to flow onto the vane in a tangential direction, without shock; and furthermore the frictional resistance between jet and vane is neglected. The velocity is assumed to be uniform throughout the jet upstream and downstream from the vane. Since the jet is open to the air, it has the same pressure at each end of the vane. When the small change in elevation between ends, is neglected, application of Bernoulli's equation shows that the magnitude of the velocity is unchanged for fixed vanes.

PRO COPY

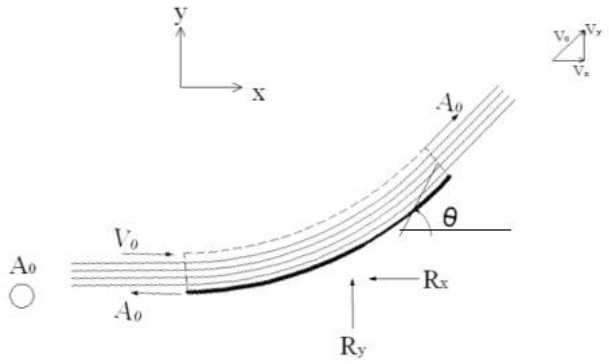


Figure 4.15c

$-R_x+\rho QV-\rho QV\cos\theta = 0$

$R_x=\rho Q(V-V\cos\theta)= \rho QV(1-\cos\theta)$

$R_y= \rho QV\sin\theta$

Jet Deflected by a Moving Plate or a Vane

If a jet of fluid strikes a plate, which is moving, the strike velocity will be the relative velocity between the jet and plate. Hence, if the plate is moving in the same direction as the jet, the strike velocity will be the difference between the jet and the plate velocities (Fig. 4.16), and if the plate is moving in the opposite direction of the jet, the strike velocity will be the sum of the jet and the plate velocities.

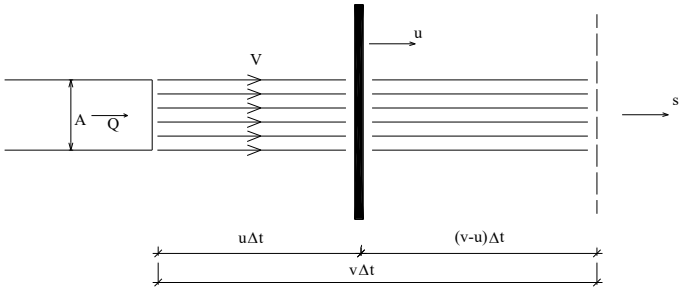


Figure 4.16 Moving plate

If we consider a curved vane as shown in Fig.4.17, a control volume should be drawn and identify control surface between sections 1 and 2. Applying the momentum equations in the x and y-direction, we have

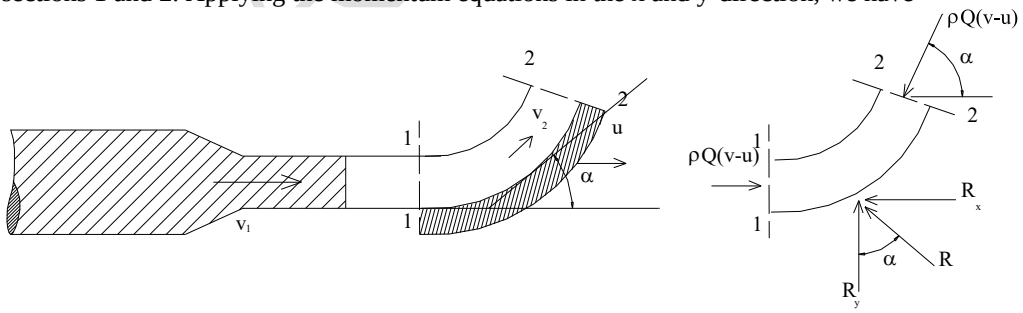


Fig 4.17 Moving vane

$$R_x = \rho Q(V-u)(1-\cos\alpha)$$

$$R_y = \rho Q(V-u)\sin\alpha$$

A practical example may be a wheel, with buckets fixed radially around the circumference, moving under the force of a jet. A bucket is shown in Fig. 4.18.

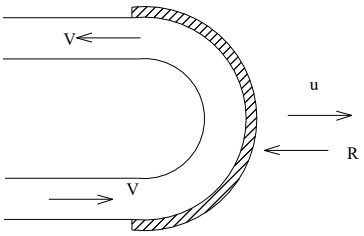


Figure 4.18 Moving bucket

Applying the momentum equation in the x-direction, we have
$R=2\rho Q(V-u)$
Work done per second on the bucket is
$N=u.R$
$N=2\rho Q(V-u)u$
The maximum efficiency can be found by differentiating N with respect to u and equating to zero, or

$$\frac{\partial N}{\partial u}=0$$

$$2\rho Q(V-u)-2\rho Qu=0$$

and u=V/2 is obtained.
hence, maximum transferred force from jet to the bucket will be only 50% of jet velocity, then

$$N_{max}=\rho Q\frac{V^2}{2}=\gamma Q\frac{V^2}{2g}$$

However, the wheel such as an impulse turbine rotates with a velocity u and water jet strikes at the center of the buckets. After a rotation of the wheel, the discharge will be constant on the wheel because remainder water mass always will strike on the subsequence buckets.
While consider a plate (in Fig. 4.16) or a curved vane (in Fig. 4.17) moves in the same direction as the jet flow at velocity u, fluid mass on the moving body will be different because fluid mass is a function of velocity ($\rho dsdA=\rho VdtdA$).The relative velocity of the jet flow compared with the plate is (V-u). Hence the discharge strike the moving plate is,

$$Q'=(V-u)A$$

$$R=\rho Q'(V-u)$$

or
$R=\rho A(V-u)^2$
If the plate has an angle with the horizontal plane, the force which is perpendicular to the plate is
$F=\rho Q'(V-u)\sin\theta=\rho Q((V-u)^2/V)\sin\theta$
Force F_x acting in the direction of the jet is
$F_x=F\sin\theta$
and force F_y acting in the direction at right angles to the jet is
$F_y=F\cos\theta$

4.4.2 Moment of Momentum Equation (Angular Momentum)

The angular momentum in the case where a body of mass m is rotating at radius r and rotational velocity V is given by
Angular Momentum= Moment of Inertia × Angular Velocity

$$=mr^2\frac{V}{r}=mrV$$

The torque (rotational couple) on this body is given by
Torque=Change of Angular Momentum (i.e. respect to time)
=Moment of Inertia×Angular Acceleration
This is equivalent to Newton's second law of motion, and expressed as the law of conservation of angular momentum.
Let's we consider an ice skater. Whenever the skater revolves with the same angular momentum, if she spreads out her arms and stretches out one of her legs to enlarge the moment of inertia, she will slow down.

DRAFT COPY

The total torque that acts on the control volume is equal to the rate of efflux of angular momentum from the control volume plus the rate of change of angular momentum inside the control volume. The general vector equation of torque is

$$\Sigma \vec{Fr} = \frac{d\left(m\vec{V}\vec{r}\right)}{dt} \tag{4.33}$$

$$\Sigma M + \left(\rho Q_1 \overrightarrow{V_1}\right)\vec{r_1} - \left(\rho Q_2 \overrightarrow{V_2}\right)\vec{r_2} = 0 \tag{4.34}$$

This is the general moment of momentum equation for a control volume. It has a great value in the analysis of certain flow problems, e.g., in turbomachinery where torques are more significant than forces.

4.5. Flow of Viscous Fluid

All fluids are viscous. In the case where the viscous effect is minimal, the flow can be treated as an ideal fluid; otherwise the fluid must be treated as a viscous fluid. For example; it is necessary to treat a fluid as a viscous fluid in order to analyse the pressure loss due to a flow, the drag acting on a body in a flow and the phenomenon where flow separates from a body.

4.5.1 Continuity Equation

The continuity equation is independent of whether the fluid is viscous or not, the same is applicable to an ideal fluid. Since we have obtained the continuity equation in last chapter we will not obtain it again. Viscosity is the fluid property that causes shear stress in a moving fluid; it is also one means by which irreversibilities or losses are developed. Without viscosity in a fluid there is no fluid resistance. Although the velocity is uniformly distributed in ideal flow, in viscous flow the velocity distribution is not uniform; it is zero on boundary and increases by diverging from the boundary.



(a) Ideal fluid (b) Viscous fluid

Figure 4.21 Velocity distributions between two plates

For a one dimensional viscous flow the discharge;

$$Q = \int_A u dA \tag{4.35}$$

The average velocity over the section (Figure 4.22):

$$V = \frac{Q}{A} = \frac{1}{A}\int_A u dA \tag{4.36}$$

If the discharge is written in terms of average velocity

$$Q = V.A \tag{4.37}$$

This equation is similar to the continuity equation obtained for the ideal fluid.

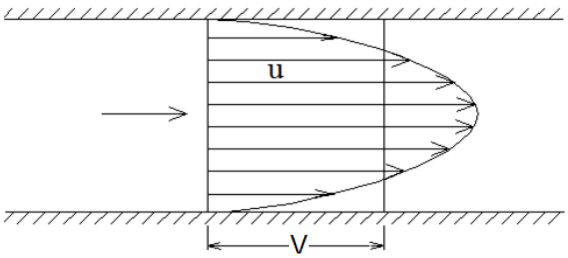


Figure 4.22 Velocity distribution and average velocity in viscous flow

DRAFT COPY

4.5.2 Momentum Equations

4.5.2.1 Navier-Stokes Equations

The equations of motion for a real fluid can be developed from consideration of the forces acting on a small element of the fluid, including the shear stresses generated by fluid motion and viscosity. Consider the element of fluid of dx, dy, and dz in Fig. 4.23 and shows the surface stresses on the element. The first subscript of the shear stress is the normal direction of the surface over which the stress component is acting. The second subscript is the direction of the stress component. Let's calculate the forces acting on the fluid element (Fig 4.24)

There are 18 stress components. Since the volume is very small numbers of components are reduced to nine. These nine quantities form a stress tensor. These nine components of the stress tensor called the stress matrix. The stress tensor and corresponding matrix are symmetric, which means that two shearing stresses with subscripts, which differ only in their order, are equal. Owing to the fact that

$\tau_{xy}=\tau_{yx}$, $\tau_{zy}=\tau_{yz}$, $\tau_{xz}=\tau_{zx}$

the stress matrix contains only six different stress components and becomes symmetrical with respect to the principal diagonal.

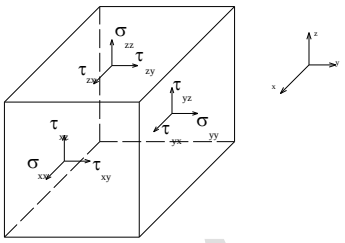


Figure 4.23 Stress components of fluid element

The stress tensor (stress matrix);

$$\begin{bmatrix} \sigma_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_{zz} \end{bmatrix} \qquad (4.38)$$

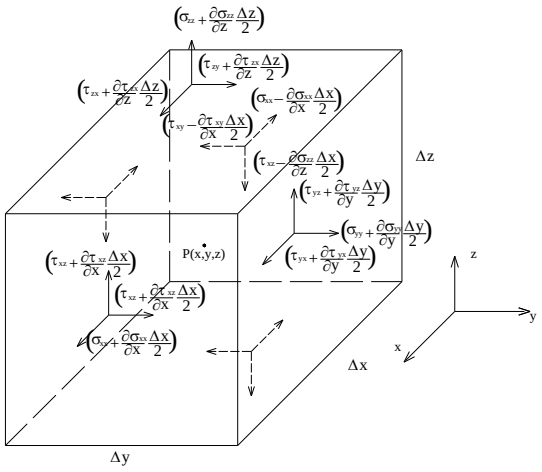


Figure 4.24 Stress components

The resultant force acting on a unit volume in the x-direction

$$F_x = \left(\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} \right) \Delta x \, \Delta y \, \Delta z \qquad (4.39)$$

The resultant force acting on a unit volume in the y-direction

DRAFT COPY

$$F_y = \left(\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \tau_{zy}}{\partial z} \right) \Delta x \, \Delta y \, \Delta z \qquad (4.40)$$

The resultant force acting on a unit volume in the z-direction

$$F_z = \left(\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} \right) \Delta x \, \Delta y \, \Delta z \qquad (4.41)$$

If Eqs. (4.39, 4.40, 4.41) are used with the Euler Equations, the equations for the real flow are obtained.

$$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} + \rho X = \rho \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} + \frac{\partial u}{\partial t} \right) \qquad (4.42)$$

$$\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \tau_{zy}}{\partial z} + \rho Y = \rho \left(u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} + \frac{\partial v}{\partial t} \right) \qquad (4.43)$$

$$\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} + \rho Z = \rho \left(u \frac{\partial w}{\partial y} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} + \frac{\partial w}{\partial t} \right) \qquad (4.44)$$

Since the stresses are related with the movement, by the help of relations between stresses and deformation we obtain new equations;

$$\rho \frac{du}{dt} = \rho X - \frac{\partial p}{\partial x} - \frac{2}{3} \mu \frac{\partial}{\partial x} \left(\operatorname{div} \vec{V} \right) + \mu \frac{\partial}{\partial x} \left(\operatorname{div} \vec{V} \right) + \mu \nabla^2 u \qquad (4.45)$$

For an incompressible flow since ρ= constant and div $\vec{V}$ =0 (continuity condition) than Eq.(4.45) becomes:

$$\underbrace{\rho \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} + \frac{\partial u}{\partial t} \right)}_{\text{inertia term}} = \underbrace{\rho X}_{\text{body force term}} - \underbrace{\frac{\partial p}{\partial x}}_{\text{pressure term}} + \underbrace{\mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right)}_{\text{viscous term}} \qquad (4.46)$$

If the same procedures are done for all directions:

$$\left.\begin{aligned} &\text{for x direction} \quad \rho \frac{du}{dt} = \rho X - \frac{\partial p}{\partial x} + \mu \nabla^2 u \\ &\text{for y direction} \quad \rho \frac{dv}{dt} = \rho Y - \frac{\partial p}{\partial y} + \mu \nabla^2 v \\ &\text{for z direction} \quad \rho \frac{dw}{dt} = \rho Z - \frac{\partial p}{\partial z} + \mu \nabla^2 w \end{aligned}\right\} \qquad (4.47)$$

are obtained.

These equations are called the Navier-Stokes equations. *The Navier-Stokes equations are the basic differential equations describing the flow of incompressible Newtonian fluids.*

4.5.3 Basic Equations of Turbulent Flow

4.5.3.1 Continuity Equation

For a three-dimensional flow, the velocity components are decomposed into their time averages and fluctuation terms and expressed as follows;

$$u = \bar{u} + u'$$
$$v = \bar{v} + v'$$
$$w = \bar{w} + w'$$

where u, v, and w are instantaneous velocities, $\bar{u}$, $\bar{v}$, and $\bar{w}$ are time wise mean velocities (time averaged, mean) and u', v', and w' are the fluctuating velocities. Now, the velocities are substituted into incompressible turbulent flow. The continuity equation is obtained as:

$$\frac{\partial}{\partial x} \left(\bar{u} + u' \right) + \frac{\partial}{\partial y} \left(\bar{v} + v' \right) + \frac{\partial}{\partial z} \left(\bar{w} + w' \right) = 0 \qquad (4.48)$$

Each term is averaged for Δt time interval

DRAFT COPY

$$\overline{\frac{\partial}{\partial x}(\bar{u}+u')}=\frac{1}{\Delta t}\int_{t_0}^{t_0+\Delta t}\left[\frac{\partial}{\partial x}(\bar{u}+u')\right]dt$$

The term can be either written as:

$$\overline{\frac{\partial}{\partial}(\bar{u}+u')}=\frac{\partial}{\partial x}\left[\frac{1}{\Delta t}\int_{t_0}^{t_0+\Delta t}(\bar{u}+u')dt\right]$$

Since the mean (time averaged) of the fluctuation is zero, the time averaged continuity equation for the incompressible turbulent flow can be written as:

$$\frac{\partial\bar{u}}{\partial x}+\frac{\partial\bar{v}}{\partial y}+\frac{\partial\bar{w}}{\partial z}=0 \tag{4.49}$$

If this equation is subtracted from Eq. (4.48)

$$\frac{\partial u'}{\partial x}+\frac{\partial v'}{\partial y}+\frac{\partial w'}{\partial z}=0 \tag{4.50}$$

is obtained. It is seen that the time-averaged velocity components and the fluctuating components each satisfies the incompressible equation of continuity.

4.5.3.2 Momentum Equations

4.5.3.2.1 Turbulent Shear Stress

Before obtaining momentum equations for turbulent flow let's examine the turbulent shearing stress only. As shown in Figure 4.25 the fluid, which passes in unit time in the y direction through minute area dA parallel to the x-axis, is:

$$\rho\,dx\,dz\,v\,dt$$

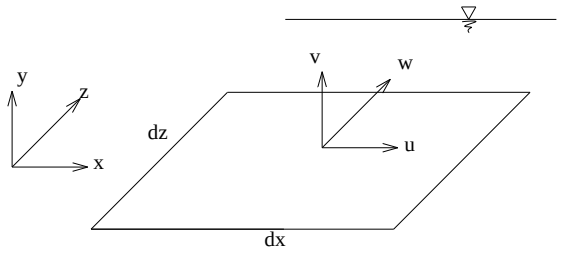


Figure 4.25 Velocities in dxdz plane

Since this fluid is at relative velocity u, the momentum is;

$$(\rho\,dx\,dz\,v\,dt)u$$

and

$$\frac{d(mu)}{dt}=F_x=\rho\,dx\,dz\,u\,v$$

F_x is the force, which causes change of momentum. The force created by the fluid;

$$K_x=-F_x=-\rho\,dx\,dz\,u\,v$$

force in the unit area or stress is obtained as

$$\tau=\frac{K_x}{dx\,dz}=-\rho\,u\,v \tag{4.51}$$

Therefore, a shearing stress develops on face (dxdz). In other words it is found that the shearing stress due to the turbulent flow is proportional to $\rho\,\overline{u'v'}$

The time average of this shearing stress in sufficient Δt time

$$\bar{\tau}=-\frac{\rho}{\Delta t}\int_{t}^{t+\Delta t}u(t)v(t)dt \tag{4.52}$$

DRAFT COPY

If the velocity components are expressed for turbulent flow

$$\bar{\tau} = -\frac{\rho}{\Delta t}\int_{t}^{t+\Delta t}\left(\bar{u}+u'\right)\left(\bar{v}+v'\right)dt \tag{4.53}$$

If it is written for a one dimensional (x-direction) flow; $\bar{v}=0$

$$\bar{\tau} = -\frac{\rho}{\Delta t}\int_{t}^{t+\Delta t}\bar{u}\,v'dt - \frac{\rho}{\Delta t}\int_{t}^{t+\Delta t}u'v'dt$$

or

$$\bar{\tau} = -\frac{\rho}{\Delta t}\bar{u}\cancelto{0}{\int_{t}^{t+\Delta t}v'dt} - \frac{\rho}{\Delta t}\int_{t}^{t+\Delta t}u'v'dt$$

Since the time wise mean of the fluctuating velocities are zero

$$\bar{\tau} = -\frac{\rho}{\Delta t}\int_{t}^{t+\Delta t}u'v'dt \tag{4.54}$$

$$\overline{u'v'} = \frac{1}{\Delta t}\int_{t}^{t+\Delta t}u'v'dt$$

is obtained.

$$\bar{\tau} = -\rho\overline{u'v'} \tag{4.55}$$

This is the turbulent shear stress or eddy stress. The negative sign is a consequence of a plus y-direction flux giving a deficiency in u′, or a negative u′, in the new environment. Fluctuations in the negative y-direction will give a positive u′. The rate of change of momentum represent by Eq. 4.55 is an effective resistance to motion and an effective shearing stress. The total shear stress is:

$\tau = \tau_l + \tau_t$

The shearing stress τ of a turbulent flow is now the sum of laminar flow shearing stress (viscous friction stress), which is the frictional force acting between the two layers at different velocities, and so-called the turbulent shearing stress τ_t, where numerous rotating molecular groups (eddies) mix with each other. τ_t is the stress developed by the turbulent flow, which is called the Reynolds stress.

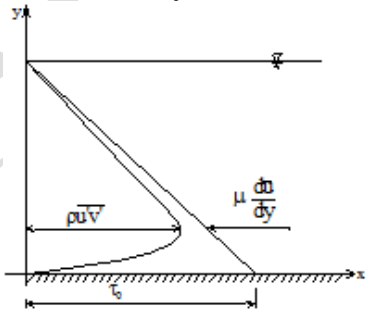


Figure 4.26 Shear stress distribution in a flow with free surface

$$\tau = \mu\frac{du}{dy} + \left(-\rho\overline{u'v'}\right) \tag{4.56}$$

If the flow is laminar u′=v′=0, so that $\overline{u'v'}=0$ and Eq. 4.56 reduces to a customary random molecule-motion-induced laminar shear stress, $\tau = \mu\frac{du}{dy}$. For turbulent flow it is found that the turbulent shear stress, $\bar{\tau} = -\rho\overline{u'v'}$, is positive. In a very narrow region near the wall the laminar shear stress is dominant. Away from the wall the turbulent portion of the shear stress is dominant. The transition between these two regions occurs in the overlap layer.

DRAFT COPY

4.5.3.2.2 Reynolds Equations

Now, to obtain momentum equations for turbulent flow we are going to use Navier-Stokes equations.

Navier-Stokes equation in x-direction:

$$\rho\left(u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}+w\frac{\partial u}{\partial z}+\frac{\partial u}{\partial t}\right)=\rho X-\frac{\partial p}{\partial x}+\mu\nabla^2 u$$

For u; $(\bar{u}+u')$, for v; $(\bar{v}+v')$, for w; $\bar{w}+w'$ are substituted to the equation:

$$\rho(\bar{u}+u')\frac{\partial}{\partial x}(\bar{u}+u')+\rho(\bar{v}+v')\frac{\partial}{\partial y}(\bar{u}+u')+\rho(\bar{w}+w')\frac{\partial}{\partial z}(\bar{u}+u')+\rho\frac{\partial}{\partial t}(\bar{u}+u')$$

$$=\rho X-\frac{\partial \bar{p}}{\partial x}+\mu\nabla^2(\bar{u}+u')$$

If each term is expanded, for example the first term:

$$\rho(\bar{u}+u')\frac{\partial}{\partial x}(\bar{u}+u')=\rho\bar{u}\frac{\partial\bar{u}}{\partial x}+\rho\bar{u}\frac{\partial\overline{u'}}{\partial x}+\rho\overline{u'}\frac{\partial\bar{u}}{\partial x}+\rho\overline{u'}\frac{\partial\overline{u'}}{\partial x}$$

If average of each term $\left(\bar{u}=\frac{1}{\Delta t}\int_{t_0}^{t_0+\Delta t}u\,dt\right)$; $\rho\bar{u}\frac{\partial\bar{u}}{\partial x}$ the first term doesn't change with time because it is time-averaged term, the time average of $\partial u'/\partial x$ is zero. The first term doesn't change with time because it is time averaged term, the second term $\left(\overline{u'}=\frac{1}{\Delta t}\int_{t_0}^{t_0+\Delta t}u'dt=\frac{1}{\Delta t}\int_{t_0}^{t_0+\Delta t}(u-\bar{u})dt=\bar{u}-\bar{u}=0\right)$, the third term $\rho u'$ is zero and last term $\rho u'\partial u'/\partial x$ is the multiplication of two fluctuating component, cannot be zero by time averaging. After the expansion of all terms with the same manner:

$$\rho\left(\bar{u}\frac{\partial\bar{u}}{\partial x}+\bar{v}\frac{\partial\bar{u}}{\partial y}+\bar{w}\frac{\partial\bar{u}}{\partial z}+\frac{\partial\bar{u}}{\partial t}\right)=\rho X-\frac{\partial\bar{p}}{\partial x}+\mu\nabla^2\bar{u}-\left(\rho\overline{u'\frac{\partial u'}{\partial x}}+\rho\overline{v'\frac{\partial u'}{\partial y}}+\rho\overline{w'\frac{\partial u'}{\partial z}}\right)\qquad(4.57)$$

is obtained.

The last term of the above equation can be written as:

$$-\left(\rho\overline{u'\frac{\partial u'}{\partial x}}+\rho\overline{v'\frac{\partial u'}{\partial y}}+\rho\overline{w'\frac{\partial u'}{\partial z}}\right)=\frac{\partial}{\partial x}\left(-\rho\overline{u'^2}\right)+\frac{\partial}{\partial y}\left(-\rho\overline{u'v'}\right)+\frac{\partial}{\partial z}\left(-\rho\overline{u'w'}\right)$$

The equations of turbulent flow of incompressible viscous fluid in x-direction, y-direction and z-direction (Eqs.) 4.58, 4.59, 4.60 are given below respectively:

$$\rho\left(\bar{u}\frac{\partial\bar{u}}{\partial x}+\bar{v}\frac{\partial\bar{u}}{\partial y}+\bar{w}\frac{\partial\bar{u}}{\partial z}+\frac{\partial\bar{u}}{\partial t}\right)=\rho X-\frac{\partial\bar{p}}{\partial x}+\mu\nabla^2\bar{u}+\frac{\partial}{\partial x}\left(-\rho\overline{u'^2}\right)+\frac{\partial}{\partial y}\left(-\rho\overline{u'v'}\right)+\frac{\partial}{\partial z}\left(-\rho\overline{u'w'}\right)\qquad(4.58)$$

$$\rho\left(\bar{u}\frac{\partial\bar{v}}{\partial x}+\bar{v}\frac{\partial\bar{v}}{\partial y}+\bar{w}\frac{\partial\bar{v}}{\partial z}+\frac{\partial\bar{v}}{\partial t}\right)=\rho Y-\frac{\partial\bar{p}}{\partial y}+\mu\nabla^2\bar{v}+\frac{\partial}{\partial x}\left(-\rho\overline{v'u'}\right)+\frac{\partial}{\partial y}\left(-\rho\overline{v'^2}\right)+\frac{\partial}{\partial z}\left(-\rho\overline{v'w'}\right)\qquad(4.59)$$

$$\rho\left(\bar{u}\frac{\partial\bar{w}}{\partial x}+\bar{v}\frac{\partial\bar{w}}{\partial y}+\bar{w}\frac{\partial\bar{w}}{\partial z}+\frac{\partial\bar{w}}{\partial t}\right)=\rho Z-\frac{\partial\bar{p}}{\partial z}+\mu\nabla^2\bar{w}+\frac{\partial}{\partial x}\left(-\rho\overline{w'u'}\right)+\frac{\partial}{\partial y}\left(-\rho\overline{w'v'}\right)+\frac{\partial}{\partial z}\left(-\rho\overline{w'^2}\right)\qquad(4.60)$$

These equations are known as Reynolds equations. The additional stress components on the right hand side of the Eqs. (4.58, 4.59, 4.60) are known as Reynolds or Eddy stresses.

4.5.4 Modified Energy Equation for Real Fluid Flows

Bernoulli's equation can be modified in the case of real incompressible fluid flow;

By introducing a loss term in the Eq.(4.15) which would take into account the energy expended in overcoming the frictional resistances caused by viscous and turbulent shear stresses and other resistances due to changes of section, valves, fittings, etc. and by correcting the velocity term for true velocity distribution.

The frictional losses depend upon the type of flow; in a laminar pipe flow they vary directly with the viscosity, the length and the velocity and inversely with the square of the diameter whereas in turbulent flow they vary

DRAFT COPY

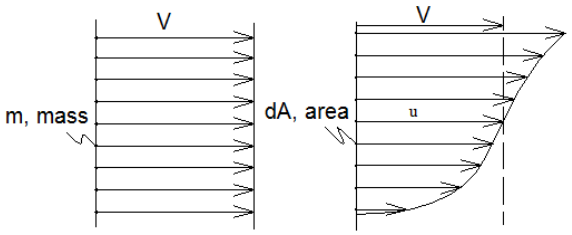
directly with the length, square of the velocity and inversely with the diameter. The turbulent losses also depend upon the roughness of the interior surface of the pipe wall and the fluid properties of density and viscosity. Therefore, for real incompressible fluid flow, we can write:

$$\frac{V_1^2}{2g}+\frac{P_1}{\gamma}+z_1=\frac{V_2^2}{2g}+\frac{P_2}{\gamma}+z_2+\text{losses} \tag{4.61a}$$

$$z_1+\frac{P_1}{\gamma}+\alpha_1\frac{V_1^2}{2g}=z_2+\frac{P_2}{\gamma}+\alpha_2\frac{V_2^2}{2g}+\Delta H \tag{4.61b}$$

where α is the velocity (kinetic energy) correction factor.

Note a) A general energy equation from the principles of conservation of energy can be derived for a fluid flow taking into account the mass, momentum and heat transfer and the thermal energy due to friction in real fluid.



(a) (b)

Figure 4.27 Velocity (kinetic energy) correction factor

$$z_1+\frac{P_1}{\gamma}+\alpha_1\frac{V_1^2}{2g}\pm H_m=z_2+\frac{P_2}{\gamma}+\alpha_2\frac{V_2^2}{2g}+\Delta H \tag{4.62}$$

or

$$H_1\pm H_m=H_2+\Delta H$$

here H_m is the external energy supplied by some machine.

b) Velocity (energy energy) correction factor, α:

Total kinetic energy at the section = ½ (m+m+m+…..) V^2

= ½ (G/g) V^2

= $V^2/2g$ per unit weight of the fluid

In the case of non-uniform velocity distribution (Fig 4.27), the particles move with different velocities.

Mass of individual elements passing through an elementary area dA = ρdAu

Kinetic energy of individual mass element = ½ ρdAuu^2

And hence total kinetic energy over the section $=\int_A\frac{1}{2}\rho u^3 dA$

$=\alpha\frac{1}{2}\rho AVV^2$ or $\alpha\frac{1}{2}\rho AV^3$, V being the average velocity at the section

$$\alpha=\frac{1}{A}\int_A(u/V)^3 dA \tag{4.63}$$

α is commonly referred to as *Coriolis coefficient.* For turbulent flows α lies between 1.03 and 1.3 and laminar flows α is 2.0.

4.5.5 Impulse-Momentum Equation

Momentum of a body is the product of its mass and velocity. Newton's second law of motion states that the resultant external force acting on anybody in any direction is equal to the rate of change of momentum of the body in that direction

$$\sum\vec{F}+\rho\beta_1Q\vec{V}_1+\left(-\rho\beta_2Q\vec{V}_2\right)=0 \tag{4.64}$$

its components;

DRAFT COPY

$$\Sigma F_x + (\rho\beta_1 Q V_1)_x - (\rho\beta_2 Q V_2)_x = 0$$
$$\Sigma F_y + (\rho\beta_1 Q V_1)_y - (\rho\beta_2 Q V_2)_y = 0$$

In the case of non-uniform velocity distribution (Fig.4.27b), the particles move with different velocities across a section of the flow field.

$$\int_A \rho\, dA\, u\, u = \beta \rho A V V$$

where V is the average velocity at the section

$$\beta = \frac{1}{A}\int_A \left(\frac{u}{v}\right)^2 dA \qquad (4.65)$$

β is commonly referred to as the *Boussinesq coefficient.*

For turbulent flows β is seldom greater than 1.1 and for laminar flows β is 1.33.

4.5 Boundary Layer Concept

In 1904 Prandtl developed the concept of the boundary layer. It provides an important link between ideal fluid flow and real fluid-flow. For fluids having relatively small viscosity, the effect of internal friction in a fluid is appreciable only in a narrow region surrounding the fluid boundaries. From this hypothesis, the flow can be divided into two regions. They are:

a) The region near the wall where the movement of flow is controlled by the frictional resistance. Relations within boundary layer region can be computed from the general equations for viscous fluids; and

b) The flow outside the narrow region near the solid boundaries may be considered as ideal flow or potential flow because this region is not affected by the friction.

The former is called the boundary layer and the latter the main flow.

Development of Boundary Layer

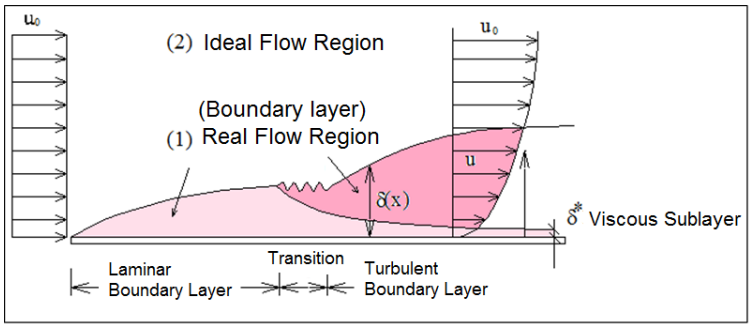
When flow is started in a fluid having very small viscosity, the flow is essentially irrotational in the first instants. Since the fluid at the boundaries has zero velocity relative to the boundaries, there is a steep velocity gradient from the boundary into the flow. This velocity gradient in a real fluid sets up near the boundary shear forces that reduce the flow relative to the boundary. That fluid layer which has had its velocity affected by the boundary shear is called the *boundary layer*. The velocity in the boundary layer approaches the velocity in the main flow asymptotically. The boundary layer is very thin at the upstream end of a streamlined body at rest in an otherwise uniform flow. As this layer moves along the body, the continual action of shear stress tends to slow down additional fluid particles, causing the thickness of the boundary layer to increase with distance from the upstream point. The fluid in the layer is also subjected to a pressure gradient, determined from the potential flow, that increases the momentum of the layer if the pressure decreases downstream and decreases its momentum of the layer if the pressure increases downstream (adverse pressure gradient). The flow outside the boundary layer may also bring momentum into the layer.

For smooth upstream boundaries the boundary layer starts out as a *laminar boundary layer* in which the fluid particles move in smooth layers. As the laminar boundary layer increases in thickness, it becomes unstable and finally transforms into a *turbulent boundary layer* in which the fluid particles move in haphazard paths, although their velocity has been reduced by the action of viscosity at the boundary. Thickness of the layer increases more rapidly. These changes occur over a relatively short distance called the transition region. When the boundary layer has become turbulent, there is still a very thin layer close to the boundary that has a laminar motion. It is called the *viscous sublaye*r.

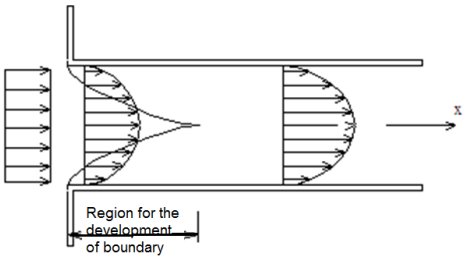
It is difficult to distinguish the boundary layer thickness; the distance from the body surface when the velocity reaches 99% of the main flow velocity is defined as the *boundary layer thickness, δ.* The boundary layer continuously thickens with the distance over which it flows. This thickness is less than a few millimeters on the frontal part of a high-speed aero plane, but reaches as much as 50 cm on the rear part of an airship.

Consider the case where a flat plate is placed in a uniform flow. The flow velocity is zero on the plate surface. Since the shearing stress due to viscosity acts between this layer and the layer immediately outside it, the velocity of the outside layer is reduced. Such a reduction extends to a further outside and thus the boundary layer increases its thickness in succession, beginning from the front end of the plate shown in Fig (4.28a).

DRAFT COPY



a)Boundary layer on a flat board surface



b) Development of the boundary layer in a pipe

Figure 4.28

Separation of Boundary Layer; and Wake

Along a flat plate, the boundary layer continues to grow in the downstream direction, regardless of the length of the plate, when the pressure gradient remains zero. In a flow where pressure decreases in the direction of the flow, the fluid is accelerated and the boundary layer reduces in thickness. In a contraction flow, the pressure has such a negative (favorable) gradient that the flow stabilizes while the turbulence gradually decreases.

In contrast, things are quite different in a flow with a positive (adverse) pressure gradient where the pressure increases in the flow direction, such as a divergent flow or flow on a curved wall as shown on Fig. 4.29. The adverse gradient and the boundary shear decrease the momentum in the boundary layer; and if both act over a sufficient distance, they cause the boundary layer to come to rest. This means that the fluid is no longer able to follow the contour of the surface and separates from it. Separation actually starts at a point where the velocity gradient du/dy at the surface becomes zero. The reverse flow near the surface, after separation has occurred produces eddies, resulting in loss of energy. This causes the pressure to remain constant after separation and not let it rise which would have been the case in the absence of separation. This phenomenon is called *separation* (Fig.4.30). The boundary streamline must leave the boundary at the separation point, and downstream from this point the adverse pressure gradient causes backflow near the wall. This region downstream from the streamline that separates from the boundary is known as the *wake*. In a wake lots of eddies are present and where pressure is constant. The effect of separation is to decrease the net amount of flow work that can be done by fluid element on the surrounding fluid at the expense of its kinetic energy, with the net result that pressure recovery is incomplete and flow losses (drag) increase. In this separation zone, more vortices develop than in the ordinary boundary layer, and the flow becomes more turbulent. For this reason the energy loss increases. Therefore, an expansion flow is readily destabilized with a large loss of energy.

Streamlined bodies are so designed that the separation point occurs as far downstream line along the body as possible. If separation can be avoided, the boundary layer remains thin and the pressure is almost recovered downstream along the body. The only loss or drag is due to shear stress in the boundary layer, called skin friction. In the wake, the pressure is not recovered, and a pressure drag results. Reduction of wake size reduces the pressure drag on a body. In general, the drag is caused by both skin friction and pressure drag.

DRAFT COPY

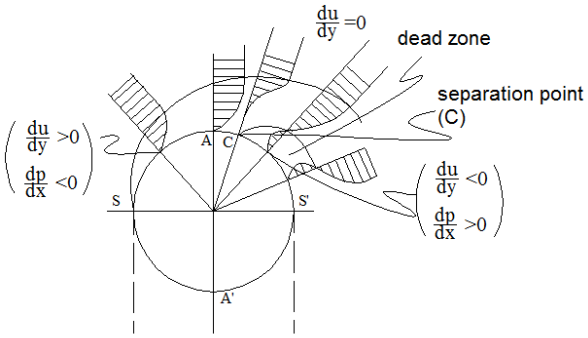


Figure 4.29 Separation from a cylindrical body

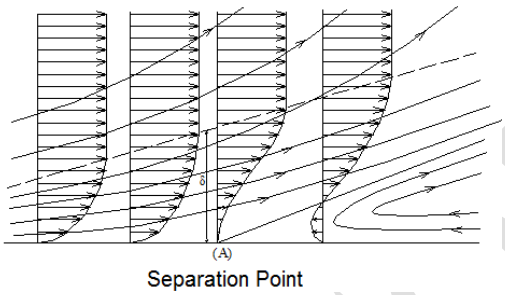


Figure 4.30 Effect of adverse pressure gradient on boundary layer (separation)

4.6 Hydrodynamics of Submerged Bodies

4.6.1 Hydrodynamic Forces

A body in the flow experiences a fluid force when there is a relative motion between the fluid and body. Relative motion means that there exist a velocity difference between the body and fluid; i.e. the body may be at rest or moving at a velocity lower or higher than that of the flow.

The fluid force acting in the direction of the motion of the body is called the *drag* force. The fluid force acting normal (or perpendicular) to the direction of motion of the body is called the *lift* force.

Figure 4.31 shows the drag and lift forces on a sediment particle moving close to the bottom. The particle velocity is assumed to be smaller than the local fluid velocity. The relative velocity is V_r.

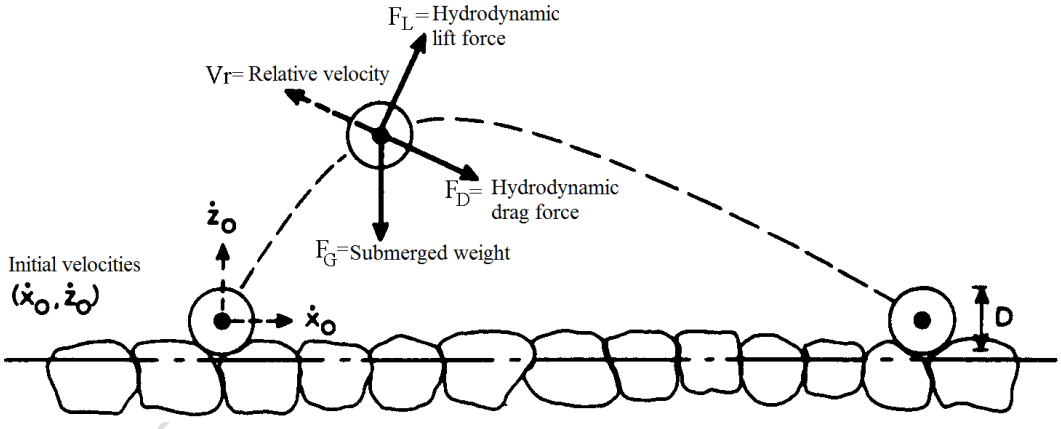


Figure 4.31 Lift and drag forces on a particle

Theoretical determination of the fluid forces is not yet possible. The usual approach is to determine the fluid forces via the introduction of drag and lift coefficients (C_D and C_L), which are determined experimentally. Since the flow around a body may be in the laminar, turbulent or transitional regime, the fluid forces (or better the drag and lift coefficient) are strongly related to the type of flow that is characterized by the Reynolds' number. Thus, C_D and C_L are functions of Re.

Figure 4.32 shows the flow around a horizontal cylinder at different Re numbers. Flow separation can be observed in both cases, but the flow separation or wake regions have different dimensions yielding different C_D coefficients. Point S is called the stagnation point and point L is called the separation point.

FREE COPY

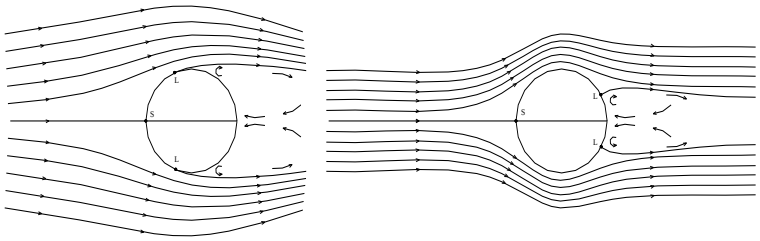


Figure 4.32 Turbulent flow around a cylinder

4.6.1.1 Drag Force

Basically, the drag force acting on the body surface consists of a pressure component and a (viscous) friction component. The pressure component is directed normal to the body surface; the friction component is directed parallel to the body surface.

Usually, the drag force is expressed in terms of the pressure component only, because this latter component is dominant for small bodies; the friction effects are taken into account via the drag coefficient. Applying the Bernoulli equation along the streamline through the stagnation point S (Fig. 4.32) yields;

$$\frac{U_0^2}{2g}+\frac{p_0}{\gamma}+z_0=\frac{U_S^2}{2g}+\frac{p_S}{\gamma}+z_S \qquad (4.66)$$

Since $z_0 = z_S$ and $U_S = 0$, it follows that:

$$p_S - p_0 = \frac{1}{2}\rho U_0^2 \qquad (4.67)$$

Eq. (4.67) express an over pressure in point S equal to $1/2\rho U_0^2$. Downstream of point L there is under pressure. Based on this, the drag force

$$F_D = C_D \frac{1}{2}\rho U_0^2 A \qquad (4.68)$$

in which: C_D is drag coefficient due to pressure and friction effects (-), U_0 is fluid velocity of undisturbed approaching flow relative to the velocity of the body (m/s), A is cross-sectional area of body normal to flow direction (m^2), ρ is fluid density (kg/m^3).

If the body is moving at a velocity U_1 with $U_1 < U_0$, then the relative velocity $V=U_0- U_1$ has to be applied in Eq. (4.68).

The C_D coefficient depends on:

The shape of the body,

The roughness of the body surface,

The type of flow characterized by the Reynolds' number

The Reynolds' number is expressed as,

$$Re = U_0 D/\nu \qquad (4.69)$$

in which the length scales D is the smallest body size normal to the flow direction. Fig. 4.33 shows the (experimentally) determined C_D coefficient as a function of Re for smooth bodies of different shapes.

1-In case of laminar flow ($Re < 10^2$), the influence of the shape and roughness of the body is relatively small.

2-In case of a plate normal to the flow the shape effect is dominant.

3-In case of a well-streamlined body the C_D coefficient is extremely small ($C_D<0.1$).

4-Usually, the C_D coefficients are known for relatively small bodies (spheres, plates, cubes). In case of a very long body the friction component should be taken into account separately.

5-Detailed information of C_D coefficients is given by Schlichting, 1968.

If a fluid is inviscid (an ideal fluid) and if the flow of such a fluid is initially incompressible and irrotational, then the flow will be irrotational throughout the entire flow field. Then, if the flow is also steady, Bernoulli's equation will have been satisfied. Because the flow pattern is symmetrical with either the vertical or the horizontal axis through the center of the cylinder, the pressure distribution on the surface of the cylinder, obtained by application of Bernoulli's equation, is also symmetrical.

DRAFT COPY

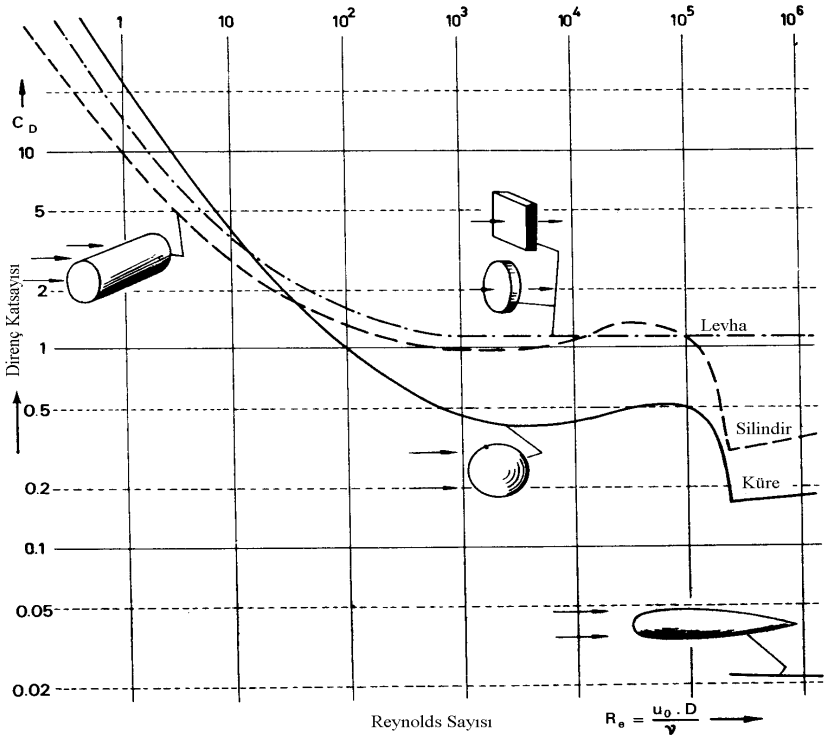


Figure 4.33 Drag coefficient

In Fig.4.34 (a) the relative pressure C_p is plotted outward (negative) or inward (positive) from the surface of the cylinder, depending on the sign of the relative pressure and on a line normal to the surface of the cylinder.
Now, consider the flow of a real (viscous) fluid past a circular cylinder, as shown in Figs. 4.34 a and b. The flow pattern upstream of the midsection of the cylinder is quite similar to the flow pattern of irrotational flow about a cylinder, except very close to the boundary surface. Here, because of the viscous resistance, a thin layer of fluid has its velocity reduced from that predicted by irrotational theory. The normal tendency is for the layer of reduced velocity, which is called "the boundary layer" to grow in thickness in the direction of flow. When separation occurs from the surface of the cylinder, the flow pattern is changed from that of irrotational flow. Therefore, one would expect corresponding changes in the pressure distribution, which do, in fact, occur. For flow past a blunt body, the slight change (due to viscosity) of the flow pattern next to the forward part of the body changes the pressure distribution only slightly. However, in the zone of separation, marked changes in pressure result.

DRAFT COPY

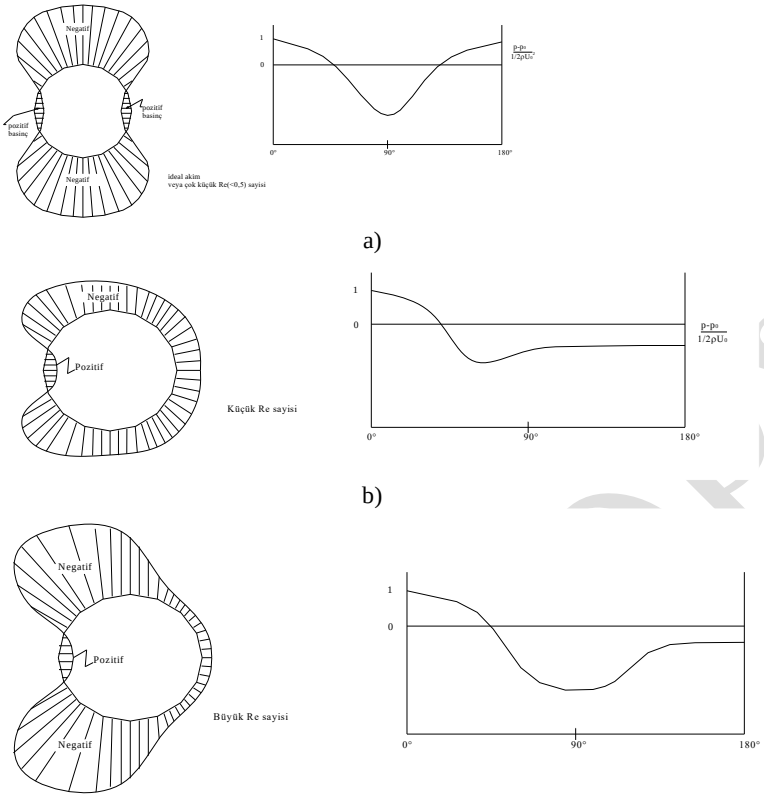


Figure 4.34 Pressure distribution around cylinder

4.6.1.2 Fall Velocity of a Sphere (or Terminal Velocity)

A sphere with diameter D falling in a still fluid obtain a constant fall velocity w_s because the upward fluid drag force will be equal to the downward gravity force on the body. Terminal velocity may be defined as final velocity of falling particles of a specified sediment grade in still water at a specified temperature.

The fluid force is

$$F_D = C_D\left(\frac{1}{2}\rho w_S^2\right)\left(\frac{1}{4}\pi D^2\right) \tag{4.70}$$

The gravity force is

$$F_G = (\rho_S - \rho)\left(\frac{1}{6}\pi D^3\right)g \tag{4.71}$$

From $F_D=F_G$ it follows that:

$$w_S = \left(\frac{4}{3C_D}\frac{\rho_S - \rho}{\rho} g D\right)^{1/2} \tag{4.72}$$

The C_D coefficient is strongly dependent on the Reynolds' number ($Re=w_S D/\nu$ in this case).

Two extreme conditions can be identified:

1-Stokes range ($Re<0.5$) theoretically, it can be derived that $C_D=24/Re$, giving

$$w_S = \frac{1}{18}\frac{\rho_S - \rho}{\rho} g \frac{D^2}{\nu} \tag{4.73}$$

2-Newtons range ($Re<10^3$) experimentally, it can been found that; $C_D = 0.4$, giving

$$w_S = \left(3\frac{\rho_S - \rho}{\rho} gD \right)^{1/2} \tag{4.74}$$

4.6.1.3 Lift Force

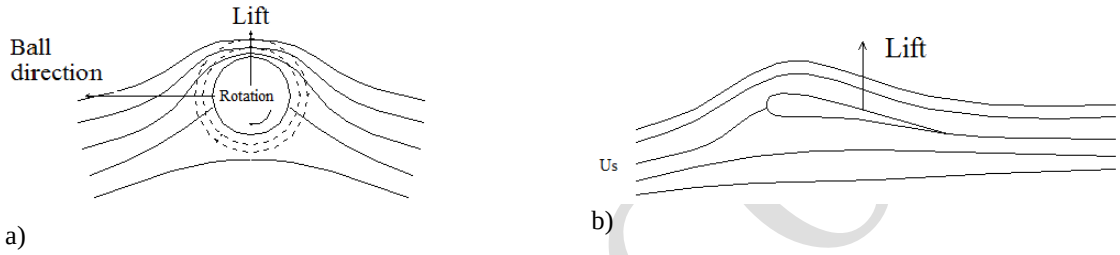
Lift forces on a body can be generated in two ways:

1- By the spinning motion of a body, this is known as the Magnus effect. For example, a ping-pong ball experiences a lift force when it spins during its path through the air. Top spin generates a downward force, while back spin generates an upward force (Fig. 4.35).

2- By the presence of fluid velocity difference on both sides of a body. For example, the shape of an airfoil is designed in such a way that the fluid velocity above the airfoil is larger than that under the airfoil giving a lower pressure above and a higher pressure under the airfoil. The resulting force is an upward lift force (Fig. 4.35). The lift force can be expressed as:

$$F_L = C_L \frac{1}{2} \rho U_0^2 A \tag{4.75}$$

where C_L is the lift coefficient.



a) b)

Figure 4.35(a) Spinning ping-pong ball (b) airfoil

DRAFT COPY

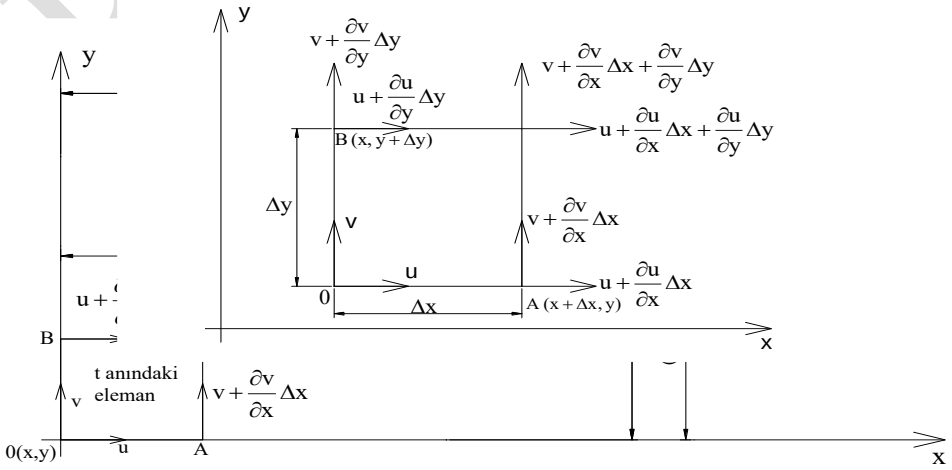
5
Introduction to Potential Flow

5.1. Introduction

The analysis of ideal fluid flow is also referred to as "potential flow". The concept of an ideal fluid is that of one which is inviscid and incompressible; the flow is also assumed to be irrotational. Since flow in boundary layers is rotational and strongly influenced by viscosity the analytical techniques of ideal fluid flow cannot be applied in such circumstances. However in many situations the flow of real fluids outside the boundary layer, where viscous effects are small, approximates closely to that of an ideal fluid.
The object of the study of ideal fluid flow is to obtain the flow pattern and pressure distribution in the fluid flow around prescribed boundaries.

5.2 Rotation

In this section it is known that the assumption of irrotational flow leads to the existence of a velocity potential. By use of these relations and assumption of a conservative body force, the Euler equations can be integrated.
The individual particles of a frictionless incompressible fluid initially at rest cannot be caused to rotate. This can be visualized by considering a small free body of fluid in the shape of a sphere. Surface forces act normal to its surface, since the fluid is frictionless, and therefore act through the center of the sphere. Similarly, the body force acts at the mass center. Hence no torque can be exerted on the sphere, and it remains without rotation. Likewise, once an ideal fluid has rotation, there is no way of altering it, as no torque can be exerted on an elementary sphere of the fluid.
An analytical expression for fluid rotation of a particle about an axis parallel to the z-axis during its translation is developed. The velocity components at O(x,y) are u, v and the velocity components at points (x, y+Δy) and (x+Δx, y) are $(u+\frac{\partial u}{\partial y}\Delta y)$ and $(v+\frac{\partial v}{\partial x}\Delta x)$ respectively. Now, assume that element rotate about z-axis during translation. The element which is initially at O moves to O′ in time dt. During this movement corner A and B of the element moves $(u+\frac{\partial u}{\partial y}\Delta y)dt$ and $(v+\frac{\partial v}{\partial x}\Delta x)dt$ respectively. At point O′ sides OA and OB will rotate through the angles dβ and dα to the new position OA′ and OB′ as shown in Fig.5.1.
a) The corner velocities of a two dimensional fluid element



b)
Figure 5.1 Rotation of a particle during its translation

DRAFT COPY

In this case the net displacement of A relative to O in y-direction

$$\left(v+\frac{\partial v}{\partial x}\Delta x\right)dt - vdt = \frac{\partial v}{\partial x}\Delta x dt$$

net displacement of B relative to O in x-direction

$$\left(u+\frac{\partial u}{\partial y}\Delta y\right)dt - udt = -\frac{\partial u}{\partial y}\Delta y dt$$

Since the displacement is in negative x-direction, the net displacement is negative.
The angular velocity of OA is

$$\omega_{OA} = \frac{d\beta}{dt} = \frac{\frac{\partial v}{\partial x}\Delta x dt}{\Delta x}\frac{1}{dt} = \frac{v}{r} = \frac{\partial v}{\partial x} \text{ rad/sec} \tag{5.1}$$

For small angles since tan dβ=dβ

$$d\beta = \frac{\frac{\partial v}{\partial x}\Delta x dt}{\Delta x} \tag{5.2}$$

The angular velocity of OB is

$$\omega_{OB} = \frac{d\alpha}{dt} = -\frac{\frac{\partial u}{\partial y}\Delta y dt}{\Delta y}\frac{1}{dt} = -\frac{\frac{\partial u}{\partial y}\Delta y}{\Delta y} = -\frac{\partial u}{\partial y} \text{ rad/sec} \tag{5.3}$$

$$d\alpha = \frac{\frac{\partial u}{\partial y}\Delta y dt}{\Delta y} \tag{5.4}$$

If counter clockwise rotation is considered positive, the rotation in the x-y plane is the average of the rotation of sides δx and δy. Also, the rotation in the x-y plane takes place about an axis parallel to the z-axis.

$$\omega_z = \frac{\omega_{OA}+\omega_{OB}}{2} = \frac{1}{2}\left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right) \tag{5.5}$$

Rotation of the element about other two coordinate axes can be obtained in a similar manner with the result that for rotation the x- axis.

$$\omega_x = \frac{1}{2}\left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}\right) \tag{5.6}$$

and for rotation about the y- axis

$$\omega_y = \frac{1}{2}\left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}\right) \tag{5.7}$$

The examination of this result reveals that ω is equal to the curl of velocity vector. That is

$$\vec{\omega} = \frac{1}{2}\text{curl}\vec{V} = \frac{1}{2}\text{rot}\vec{V} = \frac{1}{2}(\nabla \times \vec{V}) \tag{5.8}$$

$$\vec{\omega} = \frac{1}{2}\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u & v & w \end{vmatrix} \tag{5.9}$$

DRAFT COPY

$$\vec{\omega} = \frac{1}{2}\begin{vmatrix} \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v & w \end{vmatrix}\vec{i} - \frac{1}{2}\begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial z} \\ u & w \end{vmatrix}\vec{j} + \frac{1}{2}\begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} \\ u & v \end{vmatrix}\vec{k} \tag{5.10}$$

$$\vec{\omega} = \frac{1}{2}\left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}\right)\vec{i} + \frac{1}{2}\left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}\right)\vec{j} + \frac{1}{2}\left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right)\vec{k} \tag{5.11}$$

$$\omega_x = \frac{1}{2}\left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}\right), \quad \omega_y = \frac{1}{2}\left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}\right), \quad \omega_z = \frac{1}{2}\left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right) \tag{5.12}$$

The three components ω_x, ω_y, ω_z can be combined to give the rotation vector, ω, in the form

$$\vec{\omega} = \omega_x\vec{i} + \omega_y\vec{j} + \omega_z\vec{k} \tag{5.13}$$

The vorticity, ξ, is defined as a vector that is twice the rotation vector that is,

$$\xi = 2\vec{\omega} = \nabla x\vec{V} \tag{5.14}$$

Vorticity is a function of both time and space.

If there is no rotation in a fluid flow $\vec{\omega} = 0$ or $\text{curl}\vec{V} = \text{rot}\vec{V} = \nabla x\vec{V} = 2\omega = 0$. If angular velocity vector is different from zero the fluid is said to be rotational. If each component of velocity vector is zero the fluid is irrotational or potential, that is

$$\left.\begin{aligned} \frac{1}{2}\left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}\right) &= 0 \\ \frac{1}{2}\left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}\right) &= 0 \\ \frac{1}{2}\left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right) &= 0 \end{aligned}\right\} \tag{5.15}$$

or

$$\text{curl}\vec{V} = \text{rot}\vec{V} = 0 \tag{5.16}$$

5.3 Potential and Stream Function

Potential Function

In mathematics if there is a scalar function, ϕ, which can be differentiated

$$\text{curl grad}\phi = \text{rot grad}\phi = 0 \tag{5.17}$$

Since $\text{rot}\vec{V} = 0$ in irrotational flow, the velocity vector, $\vec{V}$, should be equal to gradient of a scalar function, ϕ. It can also be demonstrated for three-dimensional flow. In vector form,

$$\vec{V} = -\text{grad}\phi = -\nabla\phi \tag{5.18}$$

The velocity components in Cartesian coordinates are negative. The partial derivatives of the function ϕ (x, y, z, t)

$$u = -\frac{\partial\phi}{\partial x}, \quad v = -\frac{\partial\phi}{\partial y}, \quad w = -\frac{\partial\phi}{\partial z} \tag{5.19}$$

The function ϕ is known as the *velocity potential* or the *potential function*. The minus sign in the above equations appears as a matter of convention to indicate that the velocity potential decreases in the direction of flow. The flow described by the potential flow is irrotational. It is to be noted, however, that the velocity potential can be defined for a general three-dimensional flow, whereas the stream function is restricted to two-dimensional flows.

1-Continuity Equation

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \tag{5.20}$$

DRAFT COPY

2- Potential function equation; this is the condition of irrotational motion

$$\frac{\partial v}{\partial x}-\frac{\partial u}{\partial y}=0 \tag{5.21}$$

Conversely if a flow is irrotational, function ϕ as in the following equation must exist for u and v, $\phi(x,y)$

$$d\phi=\frac{\partial \phi}{\partial x}dx+\frac{\partial \phi}{\partial y}dy=udx+vdy \tag{5.22}$$

Consequently, when the function ϕ has been obtained, velocities u and v can also be obtained by differentiation, and thus the flow pattern is found. This function ϕ is called velocity potential, and such a flow is called *potential or irrotational flow*. In other words, *the velocity potential is a function whose gradient is equal to the velocity vector*.

3- For the potential flow of an incompressible fluid, when the velocity components in terms of velocity potential are substituted into Eq. (5.20) or continuity equation, the following relationship is obtained

$$\frac{\partial^2 \phi}{\partial x^2}+\frac{\partial^2 \phi}{\partial y^2}=0 \tag{5.23}$$

In vector form Eq (5.23)

$$\nabla.\nabla\phi=\nabla^2\phi=0 \tag{5.24}$$

here $\nabla.\nabla(\quad)=\nabla^2(\quad)=0$ is the *Laplacian operator*.

Eq. 5.24 is called *Laplace's equation*. Thus, inviscid, incompressible, irrotational flow fields are governed by Laplace's equation.

Stream Function

Steady, incompressible, plane, two-dimensional flow represents one of the simplest types of flow of practical importance. By plane, two-dimensional flow means that there are only two velocity components, such as u and v, when the flow is considered to be in the x-y plane. For this flow the continuity equation reduces to

$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}=0 \tag{5.25}$$

We still have two variables, u and v, to deal with, but they must be related in a special way as indicated by continuity equation. This equation suggests that if we define a function $\psi(x, y)$, called the *stream function*, which relates the velocities as

$$u=\frac{\partial \psi}{\partial y},\ v=-\frac{\partial \psi}{\partial x} \tag{5.26}$$

then the continuity is identically satisfied. The conclusion can be verified by simply substituting the expression for u and v into continuity equation so that

$$\frac{\partial}{\partial x}\left(\frac{\partial \psi}{\partial y}\right)+\frac{\partial}{\partial y}\left(-\frac{\partial \psi}{\partial x}\right)=\frac{\partial^2 \psi}{\partial x \partial y}-\frac{\partial^2 \psi}{\partial x \partial y}=0 \tag{5.27}$$

Thus, whenever the velocity components are defined in terms of the stream function we know that conservation of mass will be satisfied. Of course, we still do not know what $\psi(x, y)$ is for a particular problem, but at least we have simplified the analysis by having to determine only one unknown, $\psi(x, y)$, rather than the two functions u(x, y) and v(x, y).

Another particular advantage of using the stream function is related to the fact that *lines along ψ which is constant, are streamlines*. Streamlines are lines in the flow field that are everywhere tangent to the velocities, as is illustrated in Fig. 5.2. It follows from the definition of the streamline that the slope at any point along a streamline is given by

$$\frac{dy}{dx}=\frac{v}{u} \tag{5.28}$$

$$udy-vdx=0 \tag{5.29}$$

DRAFT COPY

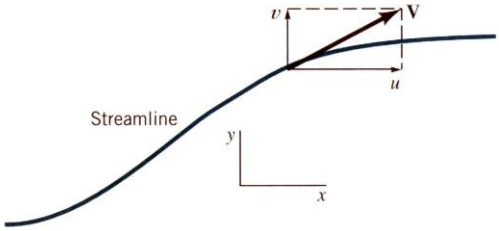


Figure 5.2 Velocity and velocity components along a streamline

The change in the value of ψ as we move from one point (x, y) to a nearby point (x+dx, y+dy) is given by the relationship:

$$d\psi = \frac{\partial \psi}{\partial x}dx + \frac{\partial \psi}{\partial y}dy = -vdx + udy \tag{5.30}$$

Along a line of constant ψ we have dψ=0 so that

-vdx+udy=0

and, therefore, along a line of constant ψ

$$\frac{dy}{dx} = \frac{v}{u}$$

This is the defining equation for a streamline. Thus, if we know the function ψ(x, y) we can plot lines of constant ψ to provide the family of streamlines that are helpful in visualizing the pattern of flow. There are an infinite number of streamlines that make up a particular flow field, since for each constant value assigned to ψ a streamline can be drawn.

The actual numerical value associated with a particular streamline is not of particular significance, but the change in the value of ψ is related to the volume rate of flow. Consider two closely spaced streamlines, as are shown in Fig. 5.3. The lower streamline is designated ψ and the upper one ψ+dψ. Let dq represent the volume rate of flow (per unit width perpendicular to the x-y plane) passing between the two streamlines. Note that flow never crosses streamlines, since by definition the velocity is tangent to the streamline. From conservation of mass we know that the inflow, dq, crossing the arbitrary surface AC of Fig. 5.3 must equal the net outflow through surfaces AB and BC. Thus,

$$dq = udy - vdx \tag{5.31}$$

$$dq = \frac{\partial \psi}{\partial y}dy + \frac{\partial \psi}{\partial x}dx \tag{5.32}$$

The right-hand side Eq. 5.32 is equal to dψ so that

$$dq = d\psi \tag{5.33}$$

Thus, the volume rate of flow, q, between two streamlines such as ψ_1 and ψ_2 of Fig.5.3b can be determined by integrating Eq. 5.33 to yield

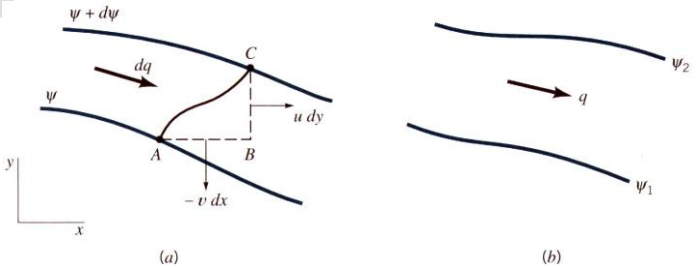


Figure 5.3 Flow between two streamlines

$$q = \int_{\psi_1}^{\psi_2} d\psi = \psi_2 - \psi_1 \tag{5.34}$$

If the upper streamline, ψ_2 has a value greater than the lower streamline, ψ_1 than q is positive, which indicates that the flow is from left to right. For $\psi_1 > \psi_2$ the flow is from right to left.

If stream function is substituted in vorticity vector

DRAFT COPY

$$\omega_z = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 0 = \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} \tag{5.35}$$

Laplace equation for the irrotational flow is obtained.

$$\nabla^2 \psi = 0 \tag{5.36}$$

If potential and stream functions are thought all together

$$u = -\frac{\partial \phi}{\partial x} = -\frac{\partial \psi}{\partial y} \tag{5.37}$$

$$v = -\frac{\partial \phi}{\partial y} = \frac{\partial \psi}{\partial x} \tag{5.38}$$

"*Cauchy-Riemann*" differential equations are obtained.

The Laplace differential equations are known as basic equations of potential flow.

$$\boxed{\begin{aligned} \nabla^2 \phi &= 0 \\ \nabla^2 \psi &= 0 \end{aligned}} \tag{5.39}$$

5.4 Bernoulli Equation in Potential Flow

Euler's equation in x-direction

$$u\frac{\partial u}{\partial x}dx + v\frac{\partial u}{\partial y}dx = -\frac{1}{\rho}\frac{\partial p}{\partial x}dx \tag{5.40}$$

The irrotationality condition for potential flow

$$\frac{\partial u}{\partial y} - \frac{\partial v}{\partial x} = 0 \quad \text{or} \quad \frac{\partial u}{\partial y} = \frac{\partial v}{\partial x}$$

$$u\frac{\partial u}{\partial x}dx + v\frac{\partial v}{\partial x}dx = -\frac{1}{\rho}\frac{\partial p}{\partial x}dx$$

$$\frac{\partial\left(\frac{u^2}{2}\right)}{\partial x}dx + \frac{\partial\left(\frac{v^2}{2}\right)}{\partial x}dx = -\frac{1}{\rho}\frac{\partial p}{\partial x}dx$$

Since V=V(u,v)

$$\frac{\partial\left(\frac{V^2}{2}\right)}{\partial x}dx = -\frac{1}{\rho}\frac{\partial p}{\partial x}dx \tag{5.41}$$

Euler's equation in y-direction

$$u\frac{\partial v}{\partial x}dy + v\frac{\partial v}{\partial y}dy = -\frac{1}{\rho}\frac{\partial p}{\partial y}dy - g\,dy \tag{5.42}$$

$$u\frac{\partial u}{\partial y}dy + v\frac{\partial v}{\partial y}dy = -\frac{1}{\rho}\frac{\partial p}{\partial y}dy - g\,dy$$

$$\frac{\partial(\frac{u^2}{2})}{\partial y}dy + \frac{\partial(\frac{v^2}{2})}{\partial y}dy = -\frac{1}{\rho}\frac{\partial p}{\partial y}dy - g\,dy$$

$$\frac{\partial(\frac{V^2}{2})}{\partial y}dy = -\frac{1}{\rho}\frac{\partial p}{\partial y}dy - g dy \tag{5.43}$$

DRAFT COPY

If equations 5.41 and 5.43 are summed and rearranged

$$\frac{\partial(\frac{V^2}{2})}{\partial x}dx+\frac{\partial(\frac{V^2}{2})}{\partial y}dy=-\frac{1}{\rho}(\frac{\partial p}{\partial x}dx+\frac{\partial p}{\partial y}dy)-gdy \tag{5.44}$$

$$d(\frac{V^2}{2})=-\frac{dp}{\rho}-gdy \tag{5.45}$$

If this equation is integrated between any two points in flow and arranged

$$\int_1^2 d(\frac{V^2}{2})=-\int_1^2\frac{dp}{\rho}-g\int_1^2 dy \tag{5.46}$$

$$\frac{V_2^2}{2}+\frac{p_2}{\rho}+gy_2=\frac{V_1^2}{2}+\frac{p_1}{\rho}+gy_1 \tag{5.47}$$

or

$$\frac{V_2^2}{2g}+\frac{p_2}{\rho}+y_2=\frac{V_1^2}{2g}+\frac{p_1}{\rho}+y_1=\text{constant} \tag{5.48}$$

The available energy is everywhere constant through the fluid. This is Bernoulli's equation for an irrotational fluid. It should be emphasized that the Bernoulli equation, as expressed by Eq. 5.48, is restricted to the following:
Inviscid flow
Steady flow
Incompressible flow
Although Bernoulli equation is valid any two points on the same streamline for rotational flow, for irrotational or potential flow, it will be satisfied in all points of flow area.

5.5 Flow Net

A graphical representation of streamlines of ψ=constant and equipotential lines of ϕ=constant within a two-dimensional flow field is called a flow net. A portion of a flow net is shown in Fig.5.4. In this figure Δn denotes the distance between the two adjacent streamlines and Δs denotes the distance between two adjacent equipotential lines. In practice Δn is made equal to Δs, then in the limits when Δs=Δn→0, the region A (Fig.5.4) becomes a perfect square. A flow net may be drawn through trial and error by first sketching a number of streamlines and then drawing equipotential lines in such a way that approximate squares are drawn.
Flow nets can provide valuable information about the flow patterns in a flow net.

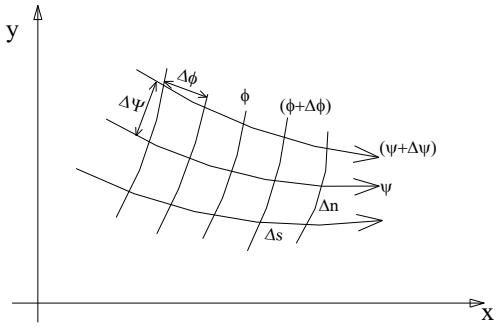


Figure 5.4 Elements of a flow net

The change in ϕ as we move from one point (x, y) to a nearby point (x+dx, y+dy) is given by the relationship

$$d\phi=\frac{\partial\phi}{\partial x}dx+\frac{\partial\phi}{\partial y}dy=udx+vdy \tag{5.49}$$

Along a line of constant ϕ we have dϕ=0 so that

$$\left.\frac{dy}{dx}\right|_\phi=-\frac{u}{v} \tag{5.50}$$

DRAFT COPY

We have previously shown that lines of constant ψ are streamlines; that is

$$d\psi = \frac{\partial \psi}{\partial x} dx + \frac{\partial \psi}{\partial y} dy = 0 \qquad (5.51)$$

$$\left.\frac{dy}{dx}\right|_{\psi} = \frac{v}{u} \qquad (5.52)$$

$$\left.\frac{dy}{dx}\right|_{\psi} . \left.\frac{dy}{dx}\right|_{\phi} = \frac{v}{u} . (-\frac{u}{v}) = -1 \qquad (5.53)$$

It is evident that the product of the two slopes given by Eqs.(5.50) and (5.52) is equal to negative one. This means that the streamlines (lines along which ψ is constant) and the equipotential lines (lines along which ϕ is constant) cross each other orthogonally. In a flow field, the network formed by the streamlines and equipotential lines is known as a flow net.

It may be pointed out here that a stream function ψ can be defined for a rotational or an irrotational flow, but a velocity potential ϕ exists only for an irrotational flow

If the directions of streamline and equipotential lines are shown as s and n respectively,

$$\frac{\Delta \phi}{\Delta s} = \frac{\Delta \psi}{\Delta n} \qquad (5.54)$$

can be written and the discharge in a stream tube

$$\Delta q = \Delta \psi = V . \Delta n \qquad (5.55)$$

is obtained. Velocity at a point is approximately

$$V \cong \frac{\Delta \psi}{\Delta n} \cong \frac{\Delta \phi}{\Delta s} \qquad (5.56)$$

From the Eq. (5.56) it is said that the velocity increases as the streamlines converges and decreases when the streamlines diverges.

Darchy's Law can be written in the generalized form

$$V_f = K\, J \qquad (5.57)$$

where

J = slope of the piezometric line

K= coefficient of permeability

$$J = -\frac{d(\frac{P}{\gamma} + z)}{ds} \qquad (5.58)$$

$$V_f = -K \frac{d(\frac{P}{\gamma} + z)}{ds} \qquad (5.59)$$

or

$$V_f = -\frac{d(Kh)}{ds} \qquad (5.60)$$

or

$$V_f = -\frac{d\phi}{ds} \qquad (5.61)$$

and the potential function

$$\phi = K(\frac{P}{\gamma} + z) = Kh \qquad (5.62)$$

DRAFT COPY

As it is shown ground water flow is a potential flow. This conclusion has a contradiction with the explanations made before. Because flow near solid boundaries doesn't obey potential flow, but groundwater flows in minor voids. If it is examined in macroscopic scale, the vortices will eliminate each other and the average velocity will have potential.

DRAFT COPY

6

Dimensional Analysis and Modeling

6.1 Introduction

The method of dimensional analysis is used in every field of engineering especially in such fields as fluid dynamics and thermodynamics where problems with many variables are handled. Engineers are most often interested in knowing the relationship between various physical quantities, which affect a fluid phenomenon. Most physical quantities have dimensions; for example distance has a dimension of length and velocity has dimension of length/time. Some physical quantities have no dimensions; for example, Angular displacement θ is dimensionless. Dimensional analysis is a technique whereby physical quantities may be arranged in dimensionless parameters. Dimensionless parameters play an important role in understanding the flow of fluid in the laboratory- i.e., under experimental conditions.

Hydraulic engineering structures or machines can be designed using;

(i) pure theory

(ii) empirical methods

(iii) semi-empirical methods which are mathematical formulations based on theoretical concepts supported by suitably designed experiments or

(iv) physical models

(v) mathematical models which is also called computational fluid dynamics (CFD).

The purely theoretical approach in hydraulic engineering is limited to a few cases of laminar flow, for example the Hagen Poisseuille equation for the hydraulic gradient in the laminar flow of an incompressible fluid in a circular pipeline.

Empirical methods are based on correlations between observed variables affecting a particular physical system. Such relationships should only be used under similar circumstances to those under which the data were collected. Due to the inability to express the physical interaction of the parameters involved in mathematical terms some such methods are still in use. One well-known example is in the relationship between wave height, fetch, wind speed and duration for the forecasting of ocean wave characteristics.

Semi-empirical methods were obtained from theoretical concepts and experiments designed on the basis of dimensional analysis; it is universally applicable to all Newtonian fluids. A good example of a semi-empirical relationship is the Colebrook-White equation for the friction factors in turbulent flow in pipes.

Dimensional analysis also forms the basis for the design and operation of physical scale models, which are used to predict the behavior of physical scale models that are used to

DRAFT COPY

predict the behavior of their full-sized counterparts called "prototypes". Such models, which are generally geometrically similar to the prototype, are used in the design of aircraft, ships, submarines, pumps, turbines, harbors, breakwaters, river and estuary engineering works, spillways, etc.

While mathematical modeling techniques have progressed rapidly due to the advent of high-speed computers, enabling the equations of motion coupled with semi-empirical relationships to be solved for complex flow situations such as pipe network analysis, pressure transient in pipeline, unsteady flows in rivers and estuaries, etc., there are many cases, particularly where localized flow patterns cannot be mathematically modeled, when physical models are still needed.

6.2 Dimensional Analysis

In a laboratory a considerable amount of work and finance would be required to establish a relationship between the dependent and independent variables. The basis of dimensional analysis is to condense the number of separate variables involved in a particular type of physical system into a smaller number of non-dimensional groups of the variables. So, low expense of time and money can be obtained by using dimensionless parameters. Dimensional parameters play an important role in understanding the flow of fluid in laboratory, i.e. the experimental conditions. Using dimensionless parameters, limited experimental results can be applied to physical conditions involving much larger dimensions, and often different fluid properties. Also, the concept of dimensional analysis coupled with the understanding of the mechanism of flow, may be used to generalize the experimental data. This would mean that limited and selective experimental results could be applied to a broad category of larger physical problems.

The arrangement of the variables in the groups is generally chosen so that each group has a physical significance.

All physical parameters can be expressed in terms of a number of basic dimensions; in engineering the basic dimensions, mass (M), length (L) and time (T) are sufficient for this purpose.

Dimensional Analysis is used for:

1. To control of the equations,
2. To control of the units,
3. Derivation of the equations,
4. One of the basic methods for the similarity,
5. Systematic way to perform an experiment.

6.2.1 Physical Significance of Non-Dimensional Groups

The main components of force, which may act on a fluid element, are those due to viscosity, gravity, pressure, surface tension and elasticity. The resultant of these components is called the inertial force and the ratio this force to each of the force components indicates the relative importance of the force types in a particular flow system.

For example the ratio of inertial force to viscous force,

$$\frac{F_i}{F_\mu} = \rho \frac{L^3 L T^{-2}}{\tau L^2}$$

DRAFT COPY

$$\tau = \mu \frac{dV}{dy} = \mu L T^{-1} L^{-1}$$

$$\frac{F_i}{F_\mu} = \rho \frac{L^2 T^{-1}}{\mu} = \rho \frac{L V}{\mu} = \rho \frac{LV}{\mu} = \frac{L V}{\nu}$$

where L is a typical length dimension of the particular system.

The dimensionless term $\rho \frac{L V}{\mu}$ is in the form of the Reynolds number. Low Reynolds numbers indicate a significant dominance of viscous forces in the system which explains why this non-dimensional parameter may be used to identify the type of flow, i.e. whether laminar or turbulent.

Similarly it can be shown that the Froude number is the ratio of inertial force to gravity force in the form

$$Fr = \frac{V^2}{g L} \text{ (usually expressed as } Fr = \frac{V}{\sqrt{g L}} \text{)}$$

The most popular dimensionless numbers in hydraulics;

1) Reynolds Number $= \dfrac{\text{Inertia Force}}{\text{Viscous Force}} = \dfrac{V L}{\nu}$ (6.1)

2) Froude Number $= \dfrac{\text{Inertia Force}}{\text{Gravitational Force}} = \dfrac{V}{\sqrt{g L}}$ (6.2)

3) Weber Number $= \dfrac{\text{Inertia Force}}{\text{Surface Tension Force}} = \rho \dfrac{V^2 L}{\sigma}$ (6.3)

4) Euler Number $= \dfrac{\text{Pressure Force}}{\text{Inertia Force}} = \dfrac{p}{\rho V^2}$ (6.4)

5) Cauchy Number $= \dfrac{\text{Inertia Force}}{\text{Compressibility Force}} = \dfrac{\rho V^2}{E}$ (6.5)

6) Mach Number $= \sqrt{\text{Cauchy Number}}$ (6.6)

7) Strouhal Number $= \dfrac{\text{Inertia(local)Force}}{\text{Inertia(convective)Force}} = \dfrac{fL}{V}$ (6.7)

DRAFT COPY

Table 6.1 Symbols and dimensions of some basic quantities

Physical Quantity	**Symbol**	**Dimensional Form**
Length	l	L
Time	t	T
Mass	m	M (or $KL^{-1}T^{2}$)
Velocity	V	LT^{-1}
Acceleration	a	LT^{-2}
Discharge	Q	$L^{3}T^{-1}$
Force	F, K	MLT^{-2} (or K)
Pressure	P	$ML^{-1}T^{-2}$ (or KL^{-2})
Power	P, N	$ML^{2}T^{-3}$
Density	ρ	ML^{-3} (or $KL^{-4}T^{2}$)
Dynamic viscosity	μ	$ML^{-1}T^{-1}$ (or $KL^{-2}T$)
Kinematic viscosity	ν	$L^{2}T^{-1}$
Surface tension	σ	MT^{-2}
Bulk modulus of elasticity	K, E	$ML^{-1}T^{-2}$

6.2.3 The Buckingham π Theorem

Buckingham, in 1915, presented a theorem based on model experiments. A fundamental question we must answer is how many dimensionless products are required to replace the original list of variables? The answer to this question is supplied by the basic theorem of dimensional analysis that states the following:

If an equation involving n variables is dimensionally homogeneous. It can be reduced to a relationship among m=n-r independent dimensionless products, where r is the minimum number of reference dimensions required to describe the variables.

The dimensionless products are frequently referred to as "pi terms" and the theorem is called the Buckingham π theorem. Buckingham used the symbol π to represent a dimensionless product, and this notation is commonly used.

The pi theorem is based on the idea of dimensional homogeneity. Let A1, A2, A3,…., A_n be the n quantities involved, such that

$$f(A_1, A_2, A_3, ..., A_n) = 0 \tag{6.8}$$

Then these quantities can be arranged into (n-r) independent parameters π_1, π_2, π_3,…,π_{n-r} ,such as a way that

$$F(\Pi_1, \Pi_2, \Pi_3, ..., \Pi_{n-m}) = 0 \tag{6.9}$$

In fluid mechanics r=3 and therefore each π term basically contains four of the quantity terms.

Π Parameters ($M^0 L^0 T^0$ or $K^0 L^0 T^0$);

$\Pi_1 = A_1^{x_1} A_2^{y_1} A_3^{z_1} A_4$

DRAFT COPY

$$\Pi_2 = A_1^{x_2}\, A_2^{y_2}\, A_3^{z_3}\, A_5$$

$$\Pi_{n-m} = A_1^{x_{n-m}}\, A_2^{y_{n-m}}\, A_3^{z_{n-m}}\, A_n \qquad (6.10)$$

one of the π parameters can be written in function of the others,

$$\Pi_1 = \phi(\Pi_2, \Pi_3, \Pi_4, \ldots, \Pi_{n-m}) \qquad (6.11)$$

A dimensional analysis can be performed using a series of distinct steps as follows:

Step 1- List all the variables that are involved in the problem.

Step 2- Express each of the variables in terms of basic dimensions.

Step 3- Determine the required number of pi terms.

Step 4- Select a number of repeating variables, where the number required is equal to the number of reference dimensions (usually the same as the number of basic dimensions).

Step 5-From a π term by multiplying one of the non-repeating variables by the product of repeating variables each raised to an exponent that will make the combination dimensionless.

Step 6- Repeat Step 5 for each of the remaining repeating variables.

Step 7- Check all the resulting pi terms to make sure they are dimensionless.

Step 8- Express the final form as a relationship among the pi terms and think about what it means.

One of the most important and difficult steps in dimensional analysis is the selection of variables. It is often helpful to classify variables into three groups- geometry, material properties, and external effects.

Typically, in fluid mechanics, the required number of reference dimensions is three, but in some problems only one or two are required.

There is not a unique set of pi terms for a given problem. Once a correct set of pi terms is obtained, any other set can be obtained by manipulation of the original set. It is permissible to exponentiate any π group, e.g. π^{-1}, $\pi^{1/2}$, π^2, etc. to form a new group this will not alter the functional form. It should be emphasized, however, that the required number of pi terms cannot be reduced by this manipulation; only the form of the pi terms is altered.

Which form for the pi terms is best? Usually our only guideline is to keep the pi terms as simple as possible. The final choice remains an arbitrary one and generally will depend on the background and experience of the investigator.

A useful physical interpretation can often be given to dimensionless groups. Special names along with physical interpretations are given to the most common dimensionless groups.

Dimensional analysis greatly facilitates the efficient handling, interpretation, and correlation of experimental data. If only one pi term is involved in a problem, it must be equal to a constant, π_1=c. For problems involving only two pi terms, results of an experiment can be conveniently presented in a simple graph, $\pi_1=f(\pi_2)$. For problems involving more than two or three pi terms, it is often necessary to use a model to predict specific characteristics, $\pi_1=f(\pi_2, \pi_3)$.

6.4 Physical Modeling

A physical model is a scaled representation of a hydraulic flow situation. A hydraulic model may help the decision-makers to visualize and to picture the flow field, before selecting a

DRAFT COPY

"suitable" design. In civil engineering applications, a physical hydraulic model is usually a smaller size representation of the prototype (i.e. the full-scale structure). The physical system for which the predictions are to be made is called the *prototype*.

In a physical model, the flow conditions are said to be similar to those in the prototype if the model displays similarity of form (*geometric similarity*), similarity of motion (*kinematic similarity*) and similarity of forces (*dynamic similarity*).

The ratio of a model variable to the corresponding prototype variable is called the *scale* (r) for that variable.

"Geometric Similarity"; Geometric similarity requires that the ratio of any length in the model L_m to the corresponding length in the prototype L_p should be the same everywhere. This ratio is usually known as scale factor (r_L).

$$r_L = \frac{(L_r)_m}{(L_r)_p} = \frac{(L_2)_m}{(L_2)_p} \tag{6.12}$$

"Kinematic Similarity"; The term kinematic similarity means that at corresponding points in the prototype and in the model, the velocities (V_p ,V_m respectively) and accelerations should have a constant ratio.

$$r_V = \frac{V_m}{V_p} = \frac{L_m}{L_p}\frac{T_p}{T_m} = \frac{r_l}{r_t} \tag{6.13}$$

Where r_v ; velocity scale, r_t ;timescale. For acceleration;

$$r_a = \frac{a_m}{a_p} = \frac{\frac{L_m}{T_m^2}}{\frac{L_p}{T_p^2}} = \frac{L_m}{L_p}\frac{T_p^2}{T_m^2} = \frac{r_l}{r_t^2} \tag{6.14}$$

r_a ;acceleration scale

"Dynamic Similarity"; Dynamic similarity term means that at corresponding points in the model and prototype, the forces should have a constant ratio. A fluid may be subjected to various types of forces. Some possible types of forces are;

1- Pressure force, caused by the differences of pressure between various points in the fluid.

2- Viscous force, caused by the viscosity of the fluid.

3- Elastic force, caused by the compressibility of the fluid.

4- Surface tension force.

5- Gravity force.

6- Inertia force. From Newton's second law of motion, inertia forces are equal –ma-i.e., a hypothetical force (equal to-ma) is required to bring the acceleration of the fluid particle to zero.

$$r_F = \frac{F_m}{F_p} = \frac{(ma)_m}{(ma)_p} = \frac{\left(\rho L^3 \frac{L}{T^2}\right)_m}{\left(\rho L^3 \frac{L}{T^2}\right)_p} = r_\rho \frac{r_L^4}{r_t^2} = r_\rho\, r_l^3\, r_a \tag{6.15}$$

DRAFT COPY

where r_F ; force scale

The points corresponding to both in the model and prototype are called *homologue points*.

Models for which one or more similarity requirements are not satisfied are called *distorted models*. A distorted model is a physical model in which the geometric scale is different between each main direction. For example, river models are usually designed with a larger scaling ratio in the horizontal directions than in the vertical directions. The scale distortion does not disturb seriously the flow pattern and it usually gives good results.

With a distorted scale model, it is possible to select small physical models. In addition to the economical and practical benefits, distorted models also have the following advantages compared with non-distorted models;

1-the flow velocities and turbulence in the model are larger.

2-the time scale is reduced.

3-the model Reynolds number is larger, improving the prototype-to-model dynamic similarity, and

4-the larger vertical scale allows a greater accuracy on the flow depth measurements.

Scale effects may be defined as the distortion introduced by effects (e.g. viscosity, surface tension) other than the dominant one (e.g. gravity in the free-surface flows). They take place when one or more dimensionless parameters differ between model and prototype. Scale effects are often small but they are not always negligible altogether.

The procedures for designing models should be made so that the model and prototype will behave in a similar fashion. It will be provided by model design conditions, also called similarity requirements or modeling laws. Model similitude can be established by a number of different methods

1–Similitude by Dynamic Analysis: The dynamic scale ratios are equal in model and prototype.

2–Similitude by Dimensional Analysis: The similarity requirements for a model can be readily obtained with the aid of dimensional analysis. Similarity between a model and a prototype is achieved by equating pi terms.

3–Similitude by Differential Equations: Similarity laws can be directly developed from the equations governing the phenomenon of interest. Governing equations expressed in terms of dimensionless variables lead to the appropriate dimensionless groups.

6.4.1 Dynamic Similarity

The force ratios must be similar for model and prototype;

$$\frac{(F_a)_m}{(F_a)_p} = \frac{(F_g)_m}{(F_g)_p} = \frac{(F_\mu)_m}{(F_\mu)_p} = ...$$

Where; F_a; inertia force, F_g; gravity force, F_μ viscous force. From the above equation it follows that if any two of the force ratios are equal, the third force ratio must also be equal to these force ratios. Therefore, if only three forces are involved, the dynamic similarity would be achieved when any two of the forces are in direct proportion. For example, in a flow phenomenon, presume inertia and viscous forces are involved; then for dynamic similarity;

DRAFT COPY

$$\frac{(F_\mu)_m}{(F_a)_m} = \frac{(F_\mu)_p}{(F_\mu)_m}$$

There is no need to consider the third force, because the similarity condition for this force is automatically satisfied.

The acceleration of particles and, hence, the inertia forces play an important role in all fluid flow phenomena, and for this reason it has become customary to include inertia force as one of the two forces considered for dynamic similarity. The other force may be one of the pressure, viscous, elastic, surface tension or gravity forces. For example;

$$\frac{(F_g)_m}{(F_a)_m} = \frac{(F_g)_p}{(F_a)_p}$$

A specific number represents the ratio of inertia force to one of the other forces; for example, the ratio of the inertia force to the gravity force is represented by the Froude number. Therefore, it follows that dynamic similarity can be achieved by making the Mach, Reynolds, Froude or Weber number the same (as the case may be) in both the model and the prototype. The scale ratios are obtained from these numbers.

6.4.1.1 Froude Model

If the inertia forces and gravity forces are dominant forces in a physical system, the Froude number will satisfy dynamic similarity between model and prototype.

$$\frac{(F_g)_m}{(F_a)_m} = \frac{(F_g)_p}{(F_a)_p}$$

or;

$$\frac{(\rho g L^3)_m}{(\rho L^3 L T^{-2})_m} = \frac{(\rho g L^3)_p}{(\rho L^3 L T^{-2})_p}$$

Assuming; $\rho_m=\rho_p$

$$\frac{(L^3)_m}{(L^4 T^{-2})_m} = \frac{(L^3)_p}{(L^4 T^{-2})_p}$$

$$L/T = V$$

$$\frac{V_m^2}{g L_m} = \frac{V_p^2}{g L_p}$$

or

$$Fr_m = Fr_p$$

Therefore, Froude number must be same for the model and the prototype at the corresponding points.

$$r_{Fr} = \frac{Fr_m}{Fr_p} = 1 \qquad (6.16)$$

The scale ratios;

DRAFT COPY

$$r_{Fr} = \frac{V_m^2 L_p}{V_p^2 L_m} = 1$$

$$r_V = \sqrt{r_l} \tag{6.17}$$

For discharge

$$r_Q = \frac{Q_m}{Q_p} = \frac{A_m V_m}{A_p V_p}$$

$$r_Q = r_V \frac{L_m^2}{L_p^2}$$

$$r_Q = r_V \, r_l^2 \tag{6.18}$$

or using $\left(r_V = \sqrt{r_l}\right)$;

$$r_Q = r_l^{5/2} \tag{6.19}$$

6.4.1.2 Reynolds Model

If the inertia forces and viscous forces are dominant forces in a physical system, the Reynolds number will satisfy dynamic similarity between model and prototype.

$$\frac{\left(F_\mu\right)_m}{\left(F_a\right)_m} = \frac{\left(F_\mu\right)_p}{\left(F_a\right)_p}$$

or

$$\frac{\left(\mu L^2 L T^{-1} L^{-1}\right)_m}{\left(\rho L^3 L T^{-2}\right)_m} = \frac{\left(\mu L^2 L T^{-1} L^{-1}\right)_p}{\left(\rho L^3 L T^{-2}\right)_p}$$

$$L/T = \nu$$

$$\frac{V_m L_m}{\nu_m} = \frac{V_p L_p}{\nu_p}$$

or

$$Re_m = Re_p$$

Therefore, Reynolds number must be same for the model and the prototype at the corresponding points.

$$r_{Re} = \frac{Re_m}{Re_p} = 1 \tag{6.20}$$

The scale ratios;

$$r_{Re} = \frac{V_m L_m \nu_p}{V_p L_p \nu_m}$$

$$r_V = r_l^{-1} \, r \tag{6.21}$$

DRAFT COPY

Assuming $r_v = 1$

$$r_V = r_l^{-1} \tag{6.22}$$

For discharge

$$r_Q = r_V\, r_l^2 \tag{6.23}$$

$$r_Q = r_l \tag{6.24}$$

Stating that in order for the results of model testing to be valid for predicting the behavior of the prototype, the following conditions should be met may sum up the above discussion on similarity;

1-The model should be geometrically similar to the prototype.

2-Dynamic similarities must be present.

The condition of kinematic similarity is automatically met if the above two conditions are present.

Note: Both Froude and Reynolds similitude are not satisfied at the corresponding points between models and prototypes. Because scale ratios for velocity at the corresponding points in order to satisfy the both numbers;

$$\sqrt{r_l} = r_l^{-1}$$

$$r_l = r_l^{-2}$$

$$r_l = 1 \quad ; \quad L_m = L_p$$

which means that model equals to prototype. Therefore the dominant forces must be considered to find the scale ratios. Table 6.2 summarizes the scaling ratios for the Froude and Reynolds similitude. In general case, the engineer must choose a proper geometric scale. The selection procedure is an iterative process.

Step 1- Select the smallest geometric scale ratio to fit within the constraints of the laboratory.

Step 2- For r_L, and for the similitude criterion (e.g. Froude or Reynolds), check if the discharge can be scaled properly in the model, based upon the maximum model discharge.

Step 3- Check if the flow resistance scaling is achievable in the model.

Step 4- Check the model Reynolds number for the smallest test flow rate.

Step 5-Choose the convenient scale.

DRAFT COPY

Table 6.2 Scale ratios for Reynolds and Froude similarity

			Scales	
Parameter	Symbol	Unit	Froude	Reynolds
Length	L	L	r_l	r_l
Area	A	L^2	r_l^2	r_l^2
Volume	V	L^3	r_l^3	r_l^3
Time	T	T	$r_l^{1/2}$	$r_l^2 r_\gamma^{-1}$
Velocity	V	$L T^{-1}$	$r_l^{1/2}$	$r_l^{-1} r_\gamma$
Acceleration	A	$L T^{-2}$	1	$r_l^{-3} r_\gamma^2$
Angular velocity	ω	T^{-1}	$r_l^{-1/2}$	$r_l^{-2} r_\gamma$
Discharge	Q	$L^3 T^{-1}$	$r_l^{5/2}$	$r_l\ r_\gamma$
Unit discharge	Q	$L^2 T^{-1}$	$r_l^{3/2}$	r_γ
Mass	M	M	$r_l^3 r_\rho$	$r_l^3 r_\rho$
Force	P, K	$M L T^{-2}$	$r_l^3 r_\gamma$	$r_\mu^2 r_\rho^{-1}$
Pressure	P	$M L^{-1} T^{-2}$	$r_l r_\gamma$	$r_l^{-2} r_\mu^2 r_\rho^{-1}$
Impulse and Momentum	I	$M L T^{-1}$	$r_l^{7/2} r_\rho$	$r_l^2 r_\mu$
Energy and Work	E	$M L^2 T^{-2}$	$r_l^4 r_\gamma$	$r_l r_\mu^2 r_\rho^{-1}$
Power	N	$M L^2 T^{-3}$	$r_l^{7/2} r_\gamma$	$r_l^{-1} r_\mu^3 r_\rho^{-2}$

DRAFT COPY